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Nonlinear Auto-Oscillator Theory of Microwave Generation by Spin-Polarized Current

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This paper formulates a general analytic approach to the theory of microwave generation in magnetic nano-structures driven by spin-polarized current and reviews analytic results obtained in this theory. The proposed approach is based on the universal model of an auto-oscillator with negative damping and nonlinear frequency shift. It is demonstrated that this universal model, when applied to the case of a spin-torque oscillator (STO) based on a current-driven magnetic nano-pillar or nano-contact, gives adequate description of most of the experimentally observed properties of STO. In particular, the model describes the power and frequency of the generated microwave signal as functions of the bias current and magnetic field, predicts the magnitude and properties of the generation linewidth, and explains the STO behavior under the influence of periodic and stochastic external signals: frequency modulation, phase-locking to external signals, mutual phase-locking in an array of STO, broadening of the generation linewidth near the generation threshold, etc. The proposed nonlinear auto-oscillator theory is rather general and can be used not only for the development of practical nano-sized STO, but, also, for the description of nonlinear auto-oscillating systems of any physical nature.

***Index Terms*—Auto-oscillator, phase locking, spin-transfer torque, spin wave, thermal noise.**

I. Introduction

THE discovery of the spin-transfer torque effect by Slonczewski [1] and Berger [2] opened a possibility for a new method of generation of microwave oscillations that does not involve any semiconductor materials or devices. It turned out, that electric direct current passing through a magnetized magnetic layered structure becomes spin-polarized and, if the current density is sufficiently high, this spin-polarized current can transfer enough spin angular momentum between the magnetic layers to destabilize the static equilibrium orientation of magnetization in the thinner ("free") magnetic layer of the multilayered structure. Depending on the actual geometry and properties of the magnetic structure and the magnitude of the external bias magnetic field, this current-induced destabilization can lead either to the switching of the magnetization direction in the "free" layer or to the stable precession of the "free" layer magnetization. The frequency of this current-induced precession is close to the frequency of the most unstable spin wave mode of the "free" magnetic layer, depends on the current magnitude, and, typically, lies in the microwave range. The generated microwave signal can be registered as oscillations of the multilayer resistance through to the effects of giant magnetoresistance (GMR) [3] or tunneling magnetoresistance (TMR) [4] due to the fact that in the course of precession the orientation of the magnetization of the "free" layer relative to the static magnetization of the "fixed" layer oscillates with microwave frequency.

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Microwave spin-torque auto-oscillators based on either fully metallic GMR spin valves or magnetic tunnel junctions (MTJ), having a thin dielectric spacer and employing TMR effect, are very attractive for potential applications in active nano-sized devices in microwave spintronics, but a clear theoretical understanding of the current-induced microwave generation processes in magnetic nano-structures is necessary for the successful design and optimization of such devices.

In this paper, we develop a simple analytic theory of current-induced microwave generation in magnetic nano-structures based on the universal model (5) of a nonlinear auto-oscillator with negative damping and a nonlinear frequency shift. The outline of the paper is as follows.

Introduction is given in the Section I. In Section II, we give a short description of the possible geometries of magnetic nano-structures that can be used in spin-torque oscillators (STO) and present a brief review of the contemporary literature on the current-induced microwave generation in magnetic nano-structures. In Section III the equation of motion for the magnetization vector of the "free" layer of a magnetic nano-structure in the presence of the spin-torque effect and magnetic dissipation is introduced, and the torques acting on this magnetization vector are defined. In the Section IV we introduce a universal model of the nonlinear auto-oscillator with negative damping in the form of a nonlinear ordinary differential equation of the first order [see (5)], and show how the classical van der Pol auto-oscillator equation and the dynamic equation describing a current-driven normally magnetized magnetic nano-pillar can be reduced to this model. The autonomous dynamics of a nonlinear auto-oscillator in the framework of the model (5) is described in Section V. Section VI is devoted to the forced dynamics of a nonlinear auto-oscillator, and describes such phenomena as phase-locking of a nonlinear auto-oscillator to an external periodic signal and

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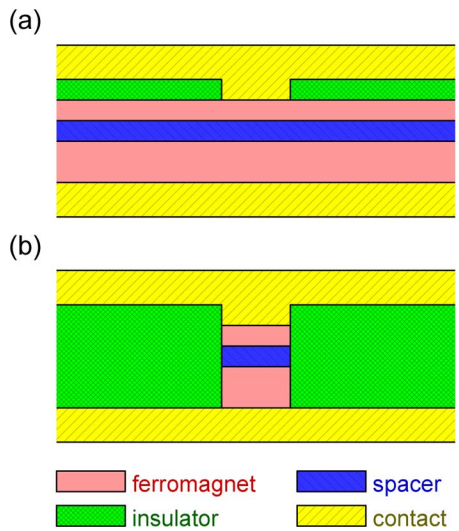

Fig. 1. Two possible geometries of spin-torque oscillator: (a) nano-contact geometry and (b) nano-pillar geometry. The bias current (for positive current polarity) flows from the bottom to the top of the structures. The top (thinner) ferromagnetic layer is the "free" magnetic layer, excited by the current. The bottom (thicker) ferromagnetic layer is the "fixed" magnetic layer. Magnetization in the "fixed" layer determines the direction of the spin polarization of the current. The spacer could be either a nonmagnetic metal (e.g., Cu) or a thin dielectric (e.g., MgO).

mutual phase-locking in a pair of oscillators or in an oscillator array. The stochastic dynamics of a nonlinear auto-oscillator under the influence of thermal noise is discussed in Section VII, and the output power and generation linewidth of such an auto-oscillator are calculated in this section. Conclusions are given in Section VIII. The expressions for the coefficients of the universal model (5) in the case of a spin-torque auto-oscillator based on an anisotropic magnetic nano-structure magnetized in an arbitrary direction are given in Appendix I, while the possible spatially nonuniform spin wave modes that might be excited by spin-polarized current in magnetic nano-structures are discussed in Appendix II.

II. Oscillator Geometries and Brief Review of Literature

Magnetic multilayered structures used for the development of STO, typically, consist of a relatively thick "fixed" magnetic layer, the direction of static magnetization of which determines the direction of spin-polarization of the traversing electric current, a nonmagnetic spacer, and a relatively thin magnetic "free" layer, the magnetization of which experiences microwave oscillation due to the spin transfer torque effect (see Fig. 1). The nonmagnetic spacer is either a nonmagnetic metal (e.g., copper) or a thin dielectric (e.g., MgO). In the first case the layered structure is, usually, called a giant magnetoresistance (GMR) spin valve, while in the second case it is called a magnetic tunnel junction (MTJ) and its performance is based on the effect of the tunneling magnetoresistance (TMR).

Two most common geometries of STO based on such layered structures—magnetic nano-contact and magnetic nano-pillar—are shown in Fig. 1 (a) and (b), correspondingly. In the case of a nano-contact [Fig. 1(a)] the "free" magnetic layer is not bounded in the layer plane, and, therefore, the magnetic oscillations excited by spin-polarized current, flowing through a nano-sized cross section of the contact (typically of the radius of 10–80 nm) formed by a dielectric layer on top of the "free" magnetic layer, can excite propagating spin waves in the magnetic material of the "free" layer. In the case of a magnetic nano-pillar [Fig. 1(b)] the "free" magnetic layer has finite lateral sizes and reflecting boundaries in the plane of the layer and represents a thin magnetic resonator. The spin wave eigenmodes of this resonator, that can be excited by the traversing spin-polarized current, might be spatially nonuniform and have discrete frequencies determined by the finite in-plane sizes of the pillar.

The analytical approach to the problem of spin wave generation by spin-polarized current developed in this paper is applicable to both nano-contact and nano-pillar geometries of STO as long as *only one* spin wave mode (standing or propagating) is excited in a magnetic nano-structure. Although, in the main part of this paper we assume for simplicity that the excited spin wave mode is spatially uniform (i.e., we use a simple macrospin approximation), it is shown in [5] that the account of the spatial structure of the excited spin wave leads only to the renormalization of parameters of the universal auto-oscillator model (5) used in our theory, but does not change neither the form of the model (5), nor the qualitative results obtained from it. The nature of spatially nonuniform spin wave modes that can be excited in magnetic nano-contacts and nano-pillars by spin-polarized current is briefly discussed in Appendix II.

It is important to stress, that in the both above described geometries the cross-sections of the bias electric current must be nano-sized (i.e., very small) to create very large current densities of the order of $10^7 - 10^8\ \mathrm{A/cm^2}$. These large current densities are necessary for the clear manifestation of the spin-transfer torque effect, which must compensate the natural magnetic damping in the material of the "free" layer to start the microwave oscillations of magnetization of this layer. More details about the spin-transfer torque effect in magnetic nano-structures and possible geometries of such nano-structures are given in the excellent review [6].

Before we start presenting the results of our approximate analytical theory, we would like to give a short review of the current literature on the problem of microwave generation in magnetic nano-structures caused by spin-polarized current. This review is rather brief and, certainly, is not complete, but we hope that it will give a reader who is starting to work in this rapidly developing field of research an idea about the most important experimental results and theoretical approaches.

The possibility of both the microwave spin wave excitation and magnetization reversal (switching) by spin-polarized current in magnetic multilayers was theoretically predicted by Slonczewski [1] and Berger [2]. Some authors, later, proposed alternative mechanisms of spin transfer (see, e.g., [7]) in GMR spin valves and investigated mechanisms of spin transfer in magnetic tunnel junctions [8].

The first indirect observations of microwave excitations caused by the spin-transfer torque effect were performed in a multilayered magnetic structure in a nano-contact geometry [3], [9], [10], where contact was made by a thin copper wire. A similar indirect evidence of spin-torque-induced microwave

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spin wave generation was observed in the magnetic nano-pillars [11], [12] which were formed by nano-lithography.

The first *direct* measurements of current-induced microwave magnetization precession were made in nano-pillar geometry in [13]–[15] and in nano-contact geometry in [16]–[19]. In these papers, the variations of the generated spectrum of magnetic excitations were studied as functions of the magnitude and direction of the bias magnetic field and the magnitude and direction of the bias direct current. An experiment reporting the telegraph switching between different precessional states excited by spin-polarized current was published in [20], while a systematic experimental study of current-induced microwave oscillations in the time domain was performed in [21] in the nano-pillar geometry.

Almost simultaneously the investigations of the current-induced spin wave excitations in nano-structures with a *single* magnetic layer were started. The experiments performed in a nano-contact geometry with a single magnetic layer were reported in [22], while nano-pillar geometry was studied in [23]. Obviously, the physics of spin-transfer torque effect in a single magnetic layer differs from that in a tri-layer magnetic structure Fig. 1, and the theoretical explanation of this effect was given in [24], [25]. Experimental study of current-induced dynamics in symmetric nano-pillars, where the thicknesses of the "free" and "fixed" layers are comparable, were done in [26].

The effect of switching of the magnetization direction in a "free" magnetic layer under the influence of the spin-polarized current in a relatively low external magnetic field was studied in magnetic nano-pillars in [11], [27]–[29] and in magnetic nanowires in [30]. The theoretical description of current-induced switching effect in the macrospin approximation based on the direct analysis of the Landau-Lifshitz equation was done in [31], while the analysis of the same effect based on the solution of the Fokker-Planck equation was given in [32][33] [34] [35].

At the same time, several authors were developing novel methods of experimental investigation of magnetic nano-structures in a sub-critical regime, when the bias current is not sufficient to overcome the magnetic damping in the "free" layer, i.e., is below the threshold of spin wave generation. A method of current-induced ferromagnetic resonance (FMR) was proposed and realized in the nano-pillar geometry by Sankey *et al.* [36]. This method allows one to measure the frequencies of linear spin wave eigenmodes and to determine from these measurements the geometrical or/and magnetic parameters of the *individual* magnetic nano-pillars. Another important experimental method to study magnetic nano-structures is the method of spatially resolved Brillouin light scattering which was used in the nano-contact geometry in [37].

Magnetic tunnel junctions (MTJ) with a thin dielectric spacer between the magnetic layers have a much larger tunneling magnetoresistance (TMR) than fully metallic GMR spin valves. Therefore, MTJ are very interesting for the development of practical STO. The generation of microwave oscillations in MTJ, caused by traversing direct current, was observed in [38], while the opposite phenomenon of the appearance of a constant voltage across an MTJ subjected to the radio-frequency alternating current (so-called "diode" effect) was experimentally observed in [4]. Theoretical explanation of the "diode" effect in MTJ was given in [39]. The measurements of thermal noise spectra in MTJ in a sub-critical regime (for bias currents that are lower than the microwave generation threshold) were done in [40]. Recently, a high-power microwave oscillations in MTJ devices have been observed in [41], [42].

Several authors studied different geometries of STO, where it was possible either to improve the STO characteristics, or to gain some new and useful properties. In particular, it was demonstrated analytically [43], numerically [44], and, finally, in experiment [45] that using a planar free layer and perpendicular polarizer it is possible to substantially increase the output microwave power generated in an STO based on a metallic GMR spin valve. It was, also, shown at first theoretically [46], [47], and then experimentally [48], [49], that in an asymmetric GMR spin valve the microwave generation can be obtained in a zero bias magnetic field.

In the cases when the free layer of an STO is not magnetized by the external magnetic field and when the anisotropies (shape and crystallographic) of this layer are insufficient to create a spatially uniform ground state the static magnetization of the "free" layer could be in a vortex ground state. Theory [50], [51] shows, that in such a case spin-polarized current can excite gyrotropic oscillations of the vortex core, and, it seems, that the recent experiment [52] performed in a nano-pillar geometry confirms this conclusion. Another interesting situation can occur at low bias magnetic fields in a nano-contact geometry, where the Oersted magnetic field of the bias current can create a vortex-like distribution of magnetization, and the spin-torque effect can excite gyrotropic oscillations of such a vortex (see [53] and [54]).

The forced dynamics of the GMR STO under the influence of an external periodic signal of a relatively low (compared to the microwave frequency generated by STO) frequency was studied experimentally in [55], and the frequency modulation effect was observed in these experiments. The phase-locking of a GMR STO to an external microwave signal, having frequency that is close to the STO generation frequency, was studied experimentally in [56], while the mutual phase-locking in a pair of STO was observed in [57]–[60].

The theoretical description of the process of microwave excitation by spin-polarized current was developed using many different techniques. Several authors developed theories based on the direct analysis of the Landau-Lifshitz equation [31], [60] [61]–[64], while others were using general methods of the nonlinear theory of dynamical system and bifurcation analysis [65]–[70]. The approach based on the theory of spin wave modes in magnetic films was formulated in [71]–[74], while the analysis based on the solution of the Fokker-Planck equation was given [75]–[78]. A theory of chaotic dynamics in STO in a macrospin approximation was presented in [79].

A lot of efforts were directed at the numerical analysis of current-induced generation of microwave oscillations. Numerical (macrospin and micromagnetic) simulations of the current-induced microwave dynamics in nano-pillars were done in [80]–[87], while the nano-contact geometry was studied in [87]–[89].

The theory of current-induced excitation of spatially nonuniform spin wave modes in different geometries, also, attracted a lot of attention. In particular, excitation of a linear prop-

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agating spin wave mode in a perpendicularly magnetized nano-contact was considered in [90], nonlinear corrections to the Slonczewski's theory [90] was given in [91], and an analytic theory of a self-localized nonlinear "bullet" mode in an in-plane magnetized nano-contact was given in [92]. Later, the analytical study of the self-localized and propagating spin wave modes (and hopping between these modes) in an obliquely magnetized nano-contact was given in [93]. The numerical confirmation of the existence of a self-localized "bullet" mode excited by spin-polarized current in a nano-contact geometry was provided in [94], [95]. The simultaneous excitation of two microwave modes in an STO in the framework of a spin wave theoretical approach was considered in [96], while the results of the experimental and theoretical study of mode hopping in a nano-pillar geometry were presented in [97].

For the practical applications of STO it is important to understand how these auto-oscillators will behave under the influence of external periodic signals and whether it is possible to organize a phase-locking (or synchronization) in a large array of STO. A theory of phase-locking of an STO to an external signal based on the spin wave approach was developed in [98], direct solution of the Landau-Lifshitz equation was used to solve the same problem in [99], [100], while methods of the dynamical systems theory were used in the phase-locking problem in [101]. The theory of mutual phase-locking in a pair of STO based on spin wave approach was presented in [102], [103] and a similar theory based on the direct solution of the Landau-Lifshitz equation was given in [104], [105]. A numerical study of a mutual phase-locking of two STO in a nano-contact geometry has been recently done in [106].

Since the microwave power generated by STO is comparable to the noise power in the magnetic material of the "free" layer at a room temperature, it is very important to study the influence of thermal noise on the performance of STO. Thermal noise also determines, for the most part, the generation linewidth in a current-driven STO. The first experimental and theoretical studies of the temperature-induced effects (such as generation linewidth) in STO based on magnetic nano-pillars were performed in [107], [108], while similar studies done in the nano-contact geometry were performed in [109]. Numerical study of thermal effects in STO made in the macrospin approximation were done in [110], [111]. The analytic theory of the STO linewidth and the power generated by STO in the presence of thermal noise developed in the framework of a spin wave approach was given in [112]–[114]. A different model of noisy dynamics in a current-driven STO was proposed in [115].

Finally, we can recommend several excellent reviews covering the field of spintronics in general [116], [117] and the current-induced microwave generation and switching in particular [6], [118], [119]. A good overview of the most recent advances in the field of spin-torque-induced magnetization dynamics is given in three tutorial papers published in [120]–[122].

III. Landau-Lifshitz-Gilbert-Slonczewski Equation of Magnetization Dynamics

Dynamics of the magnetization $\boldsymbol{M} = \boldsymbol{M}(t, \boldsymbol{r})$ of the "free" ferromagnetic layer (see Fig. 1) in the presence of spin-polarized current is described by the Landau-Lifshitz-Gilbert equation with an additional Slonczewski-Berger spin-transfer term [1], [2], [90]

$$\frac{\partial \boldsymbol{M}}{\partial t} = \gamma[\boldsymbol{H}_{\rm eff} \times \boldsymbol{M}] + \boldsymbol{T}_{\rm G} + \boldsymbol{T}_{\rm S}. \tag{1}$$

The first term in the right-hand side of this equation describes conservative precession of the magnetization vector $\boldsymbol{M}$ about the direction of the effective magnetic field $\boldsymbol{H}_{\rm eff}$ ($\gamma = g\mu_{\rm B}/\hbar \approx 2.8\ {\rm MHz/Oe}$ is the modulus of the gyromagnetic ratio, g is the spectroscopic Lande factor, $\mu_{\rm B}$ is the Bohr magneton, and $\hbar = h/2\pi$ is the reduced Planck constant), the second term $\boldsymbol{T}_{\rm G}$ is the dissipative torque describing energy dissipation, and the last term $\boldsymbol{T}_{\rm S}$ is the Slonczewski-Berger spin-transfer torque describing interaction of the magnetization $\boldsymbol{M}$ with the spin-polarized current traversing the "free" ferromagnetic layer.

The effective magnetic field $\boldsymbol{H}_{\rm eff}$ can be calculated as a variational derivative of the free energy $\mathcal{W}$ of the "free" ferromagnetic layer with respect to magnetization vector [123], [124]

$$\boldsymbol{H}_{\rm eff} = -\frac{\delta \mathcal{W}}{\delta \boldsymbol{M}}.$$

The main contributions into the free energy $\mathcal{W}$ are the Zeeman energy of interaction with the external magnetic field $\boldsymbol{H}_0$, magneto-dipolar energy (described by the magneto-dipolar field $\boldsymbol{H}_{\rm dip}$), energy of crystallographic anisotropy (described by the anisotropy field $\boldsymbol{H}_A$), energy of inhomogeneous intra-layer exchange (exchange field $\boldsymbol{H}_{\rm ex}$), and energy of interaction with the Oersted magnetic field $\boldsymbol{H}_I$ created by the current I. Consequently, the effective field $\boldsymbol{H}_{\rm eff}$ can be written as a sum of five terms

$$\boldsymbol{H}_{\rm eff} = \boldsymbol{H}_0 + \boldsymbol{H}_{\rm dip} + \boldsymbol{H}_A + \boldsymbol{H}_{\rm ex} + \boldsymbol{H}_I.$$

The other types of interactions often can be accounted for by the appropriate renormalization of various contributions to the effective field $\boldsymbol{H}_{\rm eff}$. For example, the static magneto-dipolar coupling between the "free" and "fixed" layers, as well as the Ruderman-Kittel-Kasuya-Yosida (RKKY) interaction between them (see, e.g., [125]) can be approximately described by the renormalization of the external magnetic field $\boldsymbol{H}_0$. In the case of noncircular nano-pillars, the in-plane anisotropy, induced by the magnetostatic interaction, can be accounted for by introducing of an effective shape-anisotropy field $\boldsymbol{H}_A$.

The dissipative processes in a magnetic system can be phenomenologically accounted for by an additional dissipative torque $\boldsymbol{T}_{\rm G}$ in the equation of motion for the magnetization (1) [124]. In the case of ferromagnetic metals an adequate description of the dissipation is given by the Gilbert-like damping torque [74], [126], [127]

$$\boldsymbol{T}_{\rm G} = \frac{\alpha(\xi)}{M_0}\left[\boldsymbol{M} \times \frac{\partial \boldsymbol{M}}{\partial t}\right]. \tag{2a}$$

Here, M_0 is the saturation magnetization of the "free" magnetic layer and $\alpha(\xi)$ is a positive-definite dimensionless function of the dimensionless parameter

$$\xi = \frac{|\partial \boldsymbol{M}/\partial t|^2}{\omega_M^2 M_0^2} \tag{2b}$$

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where $\omega_M = 4\pi\gamma M_0$. The parameter ξ is the dimensionless measure of the power of the magnetization precession, $\xi \sim |\boldsymbol{m}_\sim|^2/M_0^2$, where $\boldsymbol{m}_\sim$ is the variable part of the magnetization. The torque (2) is the most general phenomenological form of the dissipative torque under the assumptions that the dissipation is isotropic and local, has a dynamic origin (i.e., is determined only by the local and instant values of $\boldsymbol{M}$ and $\partial\boldsymbol{M}/\partial t$), and that the length of the magnetization vector is conserved, $|\boldsymbol{M}| = M_0 = \mathrm{const}$ (see [74] for details). In the framework of the phenomenological approach one can not reconstruct the function $\alpha(\xi)$. Nevertheless, for finite, but not very large angles of magnetization precession one can consider that $\xi \ll 1$ and represent $\alpha(\xi)$ as a Taylor series expansion

$$\alpha(\xi) \simeq \alpha_{\mathrm{G}}(1 + q_1\xi + q_2\xi^2 + \ldots)$$

where $\alpha_{\mathrm{G}} \ll 1$ is the linear Gilbert damping parameter and $q_i \sim 1$ are the phenomenological parameters characterizing nonlinear properties of the damping processes.

It should be noted, that for the qualitative (and even quantitative) description of "traditional" magnetic systems it is usually enough to take into account only the zero-order term in the expansion of $\alpha(\xi)$, i.e., to use linear Gilbert model $\alpha(\xi) = \alpha_{\mathrm{G}} = \mathrm{const}$. In the considered case of current-driven magnetic nano-structures, however, there are two circumstances that force us to use a more general nonlinear damping model with at least one nonvanishing (first-order) term $q_1 \neq 0$. First, the linear dissipation is fully compensated by the effective current-induced negative damping and, therefore, can not provide a mechanism for limiting amplitudes of the excited spin waves. Second, due to the fact that the spin wave spectrum in nano-sized elements becomes discrete, the nonlinear magnon-magnon interactions in such structures are effectively prohibited and, therefore, also can not limit the spin wave amplitudes. Therefore, one has to account for higher-order terms in the expansion of $\alpha(\xi)$, because without them the description of the experimental results will be even qualitatively incorrect.

It should be noted, that the Gilbert damping constant α_{G}, which is a dimensionless measure of the influence of the dissipative torque $\boldsymbol{T}_{\mathrm{G}}$ on the magnetization precession, is rather small ($\alpha_{\mathrm{G}} \sim 0.01$ in typical magnetic materials used for spin-torque experiments [permalloy (Py) or cobalt (Co)]. Therefore, the damping torque itself is small compared to the conservative precessional torque $\gamma[\boldsymbol{H}_{\mathrm{eff}} \times \boldsymbol{M}]$. The spin-transfer torque $\boldsymbol{T}_{\mathrm{S}}$ in (1) is, usually, comparable in magnitude with the damping torque $\boldsymbol{T}_{\mathrm{G}}$. Thus, both nonconservative torques in (1) are much smaller than the conservative one. This allows one to substitute in (2) the exact time derivative $\partial\boldsymbol{M}/\partial t$ by its "conservative" value $\gamma[\boldsymbol{H}_{\mathrm{eff}} \times \boldsymbol{M}]$

$$\boldsymbol{T}_{\mathrm{G}} = \frac{\gamma\alpha(\xi)}{M_0}\left[\boldsymbol{M} \times [\boldsymbol{H}_{\mathrm{eff}} \times \boldsymbol{M}]\right] \tag{3a}$$

$$\xi = \frac{\gamma^2\left|[\boldsymbol{H}_{\mathrm{eff}} \times \boldsymbol{M}]\right|^2}{\omega_M^2 M_0^2}. \tag{3b}$$

One may note that (3) describes dissipation processes using Landau-Lifshitz form of the damping torque [124]. In principle, the transformation between the Gilbert (2) and Landau-Lifshitz (3) forms of the dissipation torque can be made rigorously. The approximate substitution $\partial\boldsymbol{M}/\partial t \to \gamma[\boldsymbol{H}_{\mathrm{eff}} \times \boldsymbol{M}]$ used above has the accuracy of the order of $\alpha_{\mathrm{G}}^2 \sim 10^{-4}$, which is sufficient for all practical purposes, if we are dealing with weakly dissipative magnetic materials. At the same time, the Landau-Lifshitz form (3) of the damping torque has a clear advantage compared to the Gilbert form (2), as the equation of magnetization dynamics (1) with the Landau-Lifshitz form of the damping torque becomes an explicit differential equation for $\boldsymbol{M}$.

The last term in (1) in the Slonczewski-Berger spin-transfer torque $\boldsymbol{T}_{\mathrm{S}}$ describing transfer of spin angular momentum (or spin magnetic moment) from the spin-polarized current to the magnetization of the "free" magnetic layer. This term can be written as [1], [90]

$$\boldsymbol{T}_{\mathrm{S}} = \frac{\sigma_0 I f(\boldsymbol{r})}{M_0}\left[\boldsymbol{M} \times [\boldsymbol{M} \times \widehat{\mathbf{e}}_p]\right]. \tag{4a}$$

Here, I is the *electric* current traversing the "free" magnetic layer, dimensionless function $f(\boldsymbol{r})$ describes the spatial distribution of the current across the area of the "free" layer (e.g., in the case of nano-contact with constant current density $f(\boldsymbol{r}) = 1$ for $\boldsymbol{r}$ within the nano-contact area and $f(\boldsymbol{r}) = 0$ otherwise), $\widehat{\mathbf{e}}_p$ is the unit vector in the direction of the spin polarization of the current ($\widehat{\mathbf{e}}_p$ coincides with the direction of magnetization of the "fixed" magnetic layer), and coefficient σ_0 is given by

$$\sigma_0 = \frac{\varepsilon g \mu_{\mathrm{B}}}{2eM_0 LS} \tag{4b}$$

where ε is the dimensionless spin-polarization efficiency ($|\varepsilon| \leq 1$), describing the degree of spin polarization of the current, g is the spectroscopic Lande factor, μ_{B} is the Bohr magneton, e is the modulus of the electron charge, L is the thickness of the "free" magnetic layer, and S is the area of the current-carrying region.

Positive values of the current I correspond to the current flowing from the "fixed" magnetic layer to the "free" layer (note, that, since electron bears *negative* charge, the direction of motion of conducting electrons is opposite to the direction of the current). As it will be shown below, the bias current, having this positive direction, creates the instability of parallel orientations of the "free" and "fixed" magnetic layers.

Spin-polarization efficiency ε, in principle, depends on the mutual orientation of the magnetization vectors in the "fixed" and "free" magnetic layers, i.e., $\varepsilon = \varepsilon(\widehat{\mathbf{e}}_p \cdot \boldsymbol{M}/M_0)$. In most cases, however, this dependence does not lead to any qualitative or significant quantitative effects. The only difference is the case when the equilibrium orientations of magnetization vectors of the "free" and "fixed" layers are either exactly parallel or exactly orthogonal to each other. Even in this case, however, all the effects of the angular dependence of the polarization efficiency ε can be modeled by introducing a formal "misalignment" angle between the magnetization vectors of the layers. Therefore, one can assume that the polarization efficiency is constant $\varepsilon = \mathrm{const}$.

The product $\sigma_0 I$ has the dimension of frequency, and, as it will be shown below, represents the characteristic growth rate of the current-induced magnetization instability.

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IV. Universal Model of an Auto-Oscillator With Negative Damping

Auto-oscillatory systems are very common in nature and technology. Despite of a wide variety of different physical realizations, all auto-oscillatory systems have the same three key elements: i) "resonant element" that determines frequency of auto-oscillations, ii) "dissipative element", which is inevitable in any real device or mechanism, and iii) "active element" that compensates the energy losses and makes auto-oscillations possible. In many cases the effect of the active element can be described as a "negative damping," i.e., the presence of an active element in the system is reflected in the dynamical equations of motion by the presence of a term that has a form, similar to the term, which describes the natural damping, but with opposite (negative) sign.

The majority of auto-oscillators of this type, regardless of a particular physical realization, can be described by the same nonlinear oscillator model

$$\frac{dc}{dt} + i\omega\left(|c|^2\right)c + \Gamma_+\left(|c|^2\right)c - \Gamma_-\left(|c|^2\right)c = f(t) \qquad (5)$$

for the complex amplitude $c(t)$ of auto-oscillations, which measures both the power $p = |c|^2$ and phase $\phi = \arg(c)$ of the oscillations. In (5), $\omega(p)$ is the resonance frequency of the "resonant element," $\Gamma_+(p)$ is the damping rate for the natural energy dissipation (natural, or positive, damping), negative damping rate $\Gamma_-(p)$ describes the effective action of the external or internal energy source ("active element"), and the "driving" term $f(t)$ describes the interaction of the auto-oscillator with "the rest of the world", e.g., influence of external signals or/and thermal fluctuations. The resonance frequency $\omega(p)$ and both damping rates $\Gamma_\pm(p)$ are, in general, functions of the auto-oscillation power $p = |c|^2$, which makes the auto-oscillator a *nonlinear* dynamical system.

In this section we demonstrate explicitly how the nonlinear oscillator model (5) can be derived for the classical electrical auto-oscillatory circuit (van der Pol oscillator) and for the spin-torque oscillator in the simplest possible geometry (normally magnetized magnetic nano-pillar). We also discuss applicability and limitations of the model (5) for the description of a generic spin-torque oscillator (STO) and other dynamical auto-oscillatory systems.

A. Van der Pol "Negative-Resistance" Oscillator

In the 1920s, Dutch electrical engineer Balthasar van der Pol studied operation of electrical oscillators with nonlinear vacuum tube triodes or tetrodes. He was the first to realize that the action of the nonlinear active element (vacuum tube) can be described by an effective *negative resistance* (which describes negative damping for the case of electrical circuits), which brings instability into the system and leads to the excitation of self-sustained auto-oscillations. Today, the term "negative resistance" is commonly used in electrical engineering, and the van der Pol model is one of fundamental models for study of nonlinear auto-oscillations [128].

The simplified electrical circuit for the van der Pol oscillator is shown in Fig. 2(a). In contrast with the original van der Pol's scheme, we use a tunnel diode as an active element, which makes electrical circuit simpler and easier to analyze. The van der Pol oscillator contains the usual resonant LC circuit which determines the frequency of oscillations. The dissipation processes in the oscillator are modeled by a single resistance R, connected in series with the inductance L. The tunnel diode is biased by the constant voltage E_0, which acts as an external source of energy [see Fig. 2(a)].

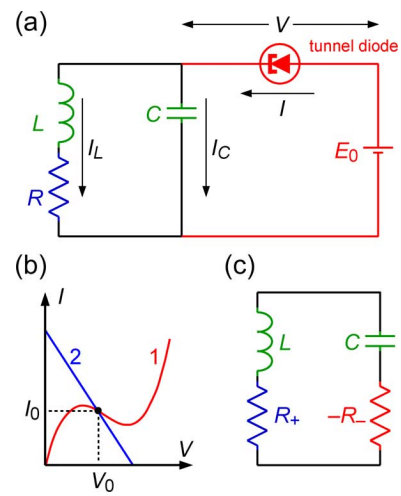

Fig. 2. (a) Electrical circuit of a van der Pol oscillator with a tunnel diode. (b) $I - V$ characteristic $I = F(V)$ of the tunnel diode (curve 1) and load line $I = (E_0 - V)/R$ for a constant current (line 2). These two lines intersect at the operation point $V = V_0$, $I = I_0$. (c) Equivalent circuit of the tunnel-diode oscillator for alternating current.

The current-voltage characteristic $I = F(V)$ (I is the current through the diode and V is the voltage drop at it) of the tunnel diode is shown in Fig. 2(b), curve 1. This characteristic is nonlinear and, what is more important, it has a region where the current I *decreases* with the *increase* of voltage V. In this region the differential resistance dV/dI of the tunnel diode is *negative* and, for small variations of the voltage and current, the diode can be approximated by a resistor with a *negative* resistance $R_{\text{eff}} = -|dV/dI|$. This resistance is added to the positive resistance R and can compensate it, leading to the situation when the circuit has a zero total damping.

The above qualitative consideration can be put in a more rigorous form by employing Kirchhoff's laws for the circuit shown in Fig. 2(a)

$$\begin{aligned} I &= I_L + I_C \\ E_0 &= V + V_L + V_R \\ E_0 &= V + V_C. \end{aligned}$$

Here, I and V are the current and voltage drop across the tunnel diode, I_L and $V_L = -LdI_L/dt$ are the current and voltage drop across the inductance L, $V_R = RI_L$ is the voltage drop at the resistance R, and $I_C = dQ/dt$ and $V_C = Q/C$ are the current and voltage drop across the capacitance C, while Q is the charge at this capacitance. Using the Kirchhoff's laws one can write the charge Q as

$$Q = C(E_0 - V)$$

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and obtain the following expression for the current I_L

$$I_L = I + C\frac{dV}{dt}.$$

Substituting these expressions for Q and I_L in the second Kirchhoff's law and using the current-voltage characteristic $I = F(V)$ of the tunnel diode, one obtains a closed-form equation for the voltage V:

$$E_0 = V + L\left[-S(V)\frac{dV}{dt} + C\frac{d^2V}{dt^2}\right] + R\left[F(V) + C\frac{dV}{dt}\right]. \tag{6}$$

Here, $S(V) = -dF(V)/dV$ is the slope of the tunnel diode characteristic, which is defined to be positive in the region of negative differential resistance of the diode.

For the constant voltage $(dV/dt = 0)$ (6) reduces to

$$F(V) = \frac{E_0 - V}{R}.$$

This equation determines the operation point $V = V_0$, $I = I_0$ (see Fig. 2(b)) of the oscillator as the intersection of the current-voltage characteristic $I = F(V)$ of the tunnel diode and the load line $I = (E_0 - V)/R$.

For the variable component of the voltage $v = V - V_0$ (6) gives

$$\frac{d^2v}{dt^2} + \left(\frac{R}{L} - \frac{S}{C}\right)\frac{dv}{dt} + \frac{1}{LC}v = 0 \tag{7}$$

where in the last term of (6) we have substituted $RF(V_0)$ for $RF(V)$ (which can be done for sufficiently small resistance R), and the slope S is calculated at the voltage $V = V_0 + v$.

One can see from (7), that the influence of the tunnel diode for the variable voltage $v(t)$ is exactly the same as the influence of an additional nonlinear resistor with resistance $-R_- = -LS/C$, which is negative in the region where $S > 0$. The equivalent electrical circuit corresponding to (7) is shown in Fig. 2(c), and it coincides with the scheme of a usual resonant circuit having inductance L, capacitance C, and two resistors—one with positive $R_+ = R$ and one with negative $-R_- = -LS/C$ resistances.

Now we will show how the real "electrical" model (7) can be rewritten in the universal complex form (5). This can be done by introducing complex "normal mode" variables that are commonly used in the theory of oscillations (see, e.g., [129, Sec. 1.4]). For our case we introduce this complex "normal mode" oscillation amplitude c as

$$c = v + i\left(\frac{1}{\omega_0}\frac{dv}{dt}\right) \tag{8}$$

where $\omega_0 = 1/\sqrt{LC}$ is the resonance frequency of the conservative electrical circuit (7). It is clear, that the complex amplitude c contains the information about both the power and the phase of the voltage oscillations. For simplicity, we consider the case of sufficiently small magnitudes $|v|$ of the variable voltage v and develop the slope $S(V_0 + v)$ in a Taylor series

$$S(V_0 + v) \approx S_0(1 - \alpha_1 v - \alpha_2 v^2). \tag{9}$$

Equations (7) and (9) allow us to obtain the exact first-order differential equation for the complex amplitude c

$$\frac{dc}{dt} + i\omega_0 c + \frac{R}{2L}(c - c^*) - \frac{S_0}{2C}\left[1 - \frac{\alpha_1}{2}(c + c^*) - \frac{\alpha_2}{4}(c + c^*)^2\right](c - c^*) = 0 \tag{10}$$

where c^* is the complex-conjugate of the amplitude (8).

In the conservative limit $R \to 0$, $S_0 \to 0$ (10) takes the simple form

$$\frac{dc}{dt} + i\omega_0 c = 0$$

with the general solution

$$c(t) = c_0 e^{-i\omega_0 t} \tag{11}$$

where c_0 is the constant amplitude of the oscillation.

If the resistance R and the slope S_0 of the $I-V$ characteristic are sufficiently small, the solution of (10) can, also, be written in the form (11), where c_0 is a slow function of time, i.e.,

$$c(t) = c_0(t)e^{-i\omega_0 t}, \qquad |dc_0(t)/dt| \ll \omega_0 |c_0(t)|. \tag{12}$$

Using the ansatz (12) in (10) one obtains an equation with terms having a different dependence on the "fast" variable $\omega_0 t$. For instance, the resistance term $(R/2L)(c - c^*)$ takes the form

$$\frac{R}{2L}(c - c^*) = \frac{R}{2L}ce^{-i\omega_0 t} - \frac{R}{2L}c^* e^{i\omega_0 t}.$$

The first term in the right-hand-side part of this equation has the same dependence on the fast variable $\omega_0 t$, as the conservative terms dc/dt and $i\omega_0 c$, i.e., it is *resonant* and, therefore, can lead to a substantial modification of the solution (12) on long time scales. In contrast, the second term $\propto e^{i\omega_0 t}$ is *nonresonant* and its influence, averaged over the period of "fast" motion $2\pi/\omega_0$, is zero. Therefore, this nonresonant term, in the first order on the resistance R, can be safely neglected. It is clear, that the resonant (in this sense) terms in (11) should have the form $c^n(c^*)^m$ with $n = m + 1$ and, therefore, any combination of them can be written as $f(|c|^2)c$, where $f(|c|^2)$ is some function that depends *only* on the oscillation power $p = |c|^2$.

Keeping only the resonant terms in (10) gives the equation

$$\frac{dc}{dt} + i\omega_0 c + \frac{R}{2L}c - \frac{S_0}{2C}\left(1 - \frac{\alpha_2}{4}|c|^2\right)c = 0$$

which can be rewritten in the universal form (5) (without the external "driving" term $f(t)$)

$$\frac{dc}{dt} + i\omega\left(|c|^2\right)c + \Gamma_+\left(|c|^2\right)c - \Gamma_-\left(|c|^2\right)c = 0 \tag{13a}$$

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where the frequency $\omega(p)$ and the damping rates $\Gamma_{\pm}(p)$ have the form

$$\omega(p) = \omega_0 = \text{const} \tag{13b}$$

$$\Gamma_{+}(p) = \frac{R}{2L} = \text{const} \tag{13c}$$

$$\Gamma_{-}(p) = \frac{S_0}{2C}\left(1 - \frac{\alpha_2}{4}p\right). \tag{13d}$$

Thus, we have shown that the dynamics of the classical van der Pol oscillator with a tunnel-diode (see Fig. 2) can be described by the universal oscillator model (13). In our derivation we used, essentially, only two assumptions. First, we approximated the slope of the tunnel diode I-V characteristic $S(V_0+v)$ by the Taylor expansion (9). It is clear, that this approximation does not limit the applicability of the general model (13a). Indeed, as it was explained above, any combination of resonant terms in the equation of motion for the complex amplitude c can be written as $f(|c|^2)c$ and, therefore, account of higher-order terms in (9) will simply change the dependence (13d) of the negative damping $\Gamma_{-}(p)$ on the oscillation power p, but will leave the functional form of (13a) unchanged.

The second approximation used in the derivation of (13) is the assumption of smallness of the nonconservative terms (proportional to the resistance R and the slope S_0) compared to the conservative one (proportional to the frequency ω_0). This approximation excludes from our analysis strongly "nonharmonic" oscillators (e.g., relaxation oscillators), for which the oscillations can not be described by the simple harmonic form (12). To estimate the accuracy of this assumption, we note, that the frequency of oscillations in (13) always equals to the "conservative" frequency ω_0 and is independent of the damping rates $\Gamma_{\pm}$. It is known, however, that the presence of dissipation changes the oscillation frequency in the second order on the dissipation rate. Therefore, the relative accuracy of our approach has the order of $(\Gamma_{\pm}/\omega)^2$, which is sufficient for many practical purposes.

B. Spin-Torque Oscillator Based on a Normally Magnetized Nano-Pillar

In this subsection,we consider the simplest possible geometry of spin-torque oscillator (STO)—the oscillator based on a normally magnetized isotropic magnetic nano-pillar shown schematically in Fig. 3. We will assume that the spin-polarized current is uniform across the area of the "free" layer and that only the spatially-uniform magnetization precession is excited. This simple case allows one to perform reduction of (1) to the universal oscillator model (5) exactly, and without complicated mathematics [71]. We would like to stress, that the similar reduction is possible to perform *approximately* for many other geometries of spin-torque oscillator with arbitrary direction of the bias magnetic field [5], [72].

The effective magnetic field $\boldsymbol{H}_{\text{eff}}$ in the considered case has the simple form

$$\boldsymbol{H}_{\text{eff}} = (H_0 - 4\pi M_z)\widehat{\mathbf{e}}_z$$

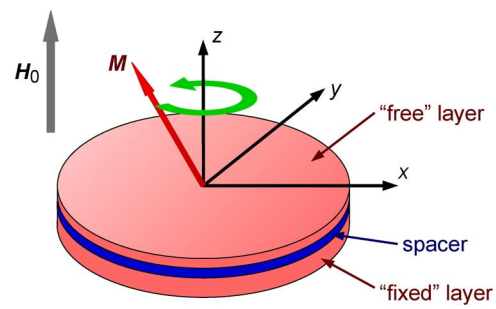

Fig. 3. Normally magnetized by the bias magnetic field $\boldsymbol{H}_0 = H_0\widehat{\mathbf{e}}_z$ magnetic nano-pillar. The magnetization $\boldsymbol{M}$ of the "free" magnetic layer precesses around the equilibrium direction z.

where $\widehat{\mathbf{e}}_z$ is the unit ort in the direction normal to the plane of the "free" layer (see Fig. 3), H_0 is the external bias field, and the term $-4\pi M_z$ describes the influence of the demagnetization (magneto-dipolar) field.

In all the practically interesting cases the influence of the first (conservative) torque $\gamma[\boldsymbol{H}_{\text{eff}} \times \boldsymbol{M}]$ in (1) is much larger than the influence of the nonconservative damping $\boldsymbol{T}_{\text{G}}$ and spin-transfer $\boldsymbol{T}_{\text{S}}$ torques. Thus, it is convenient to rewrite (1) using the "normal coordinates" of the conservative system, in which the conservative term has the simplest possible form. To get the insight about the possible form of the normal coordinates for the Landau-Lifshitz equation (1), one can consider conservative Landau-Lifshitz equation, linearized about the equilibrium orientation of magnetization $\boldsymbol{M} = M_0\widehat{\mathbf{e}}_z$

$$\left(\frac{dM_x}{dt}\right)_{\text{cons}} = -\omega_0 M_y, \qquad \left(\frac{dM_y}{dt}\right)_{\text{cons}} = +\omega_0 M_x$$

where

$$\omega_0 = \gamma(H_0 - 4\pi M_0)$$

is the ferromagnetic resonance (FMR) frequency of the normally magnetized magnetic film. Note, that the third component of the magnetization vector is constant in the linear approximation ($M_z = M_0$) due to the conservation of the length of the magnetization vector $|\boldsymbol{M}| = M_0$. Two *real* equations for the magnetization projections M_x and M_y can be rewritten as one *complex* equation for the dimensionless complex variable $\widetilde{c} = (M_x - iM_y)/2M_0$:

$$\left(\frac{d\widetilde{c}}{dt}\right)_{\text{cons}} = -i\omega_0\widetilde{c}.$$

Thus, the complex variable $\widetilde{c} = (M_x - iM_y)/2M_0$ provides the simplest possible description of the linear magnetization precession. The same simple description in the *nonlinear* regime is achieved by introducing the *complex dimensionless spin wave amplitude*

$$c = \frac{M_x - iM_y}{\sqrt{2M_0(M_0 + M_z)}}. \tag{14a}$$

It is clear, that in the linear regime ($M_z = M_0$)c and $\widetilde{c}$ coincide with each other.

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The magnetization vector $\boldsymbol{M}$ can be expressed through the complex amplitude c as

$$\boldsymbol{M} = M_0\left(1-2|c|^2\right)\widehat{\mathbf{e}}_z + M_0\sqrt{1-|c|^2}\left[(\widehat{\mathbf{e}}_x + i\widehat{\mathbf{e}}_y)c + (\widehat{\mathbf{e}}_x - i\widehat{\mathbf{e}}_y)c^*\right]. \quad (14b)$$

One can directly verify that (14b) automatically provides the conservation of the magnetization vector length $|\boldsymbol{M}| = M_0$.

Using (14) in the conservative Landau-Lifshitz equation (1), one can derive an equation for the complex amplitude c, valid for arbitrary large precession angles

$$\left(\frac{dc}{dt}\right)_{\text{cons}} = -i\omega\left(|c|^2\right)c \quad (15a)$$

where $\omega(|c|^2)$ is the *nonlinear* precession frequency

$$\omega\left(|c|^2\right) = \omega_0 + N|c|^2. \quad (15b)$$

Here N is the coefficient of the nonlinear frequency shift for the normally magnetized film

$$N = 2\omega_M$$

where $\omega_M = 4\pi\gamma M_0$.

One can see that (15a), which describes the *nonlinear* magnetization precession, has a very simple form and coincides with the conservative part of the universal model (5). The general solution of (15a) can be easily found in the form

$$c(t) = \sqrt{p}\exp\left[-i\omega(p)t + i\phi_0\right]$$

where p and ϕ_0 are, respectively, the dimensionless power and the initial phase of the precession. Such simple description of the nonlinear precession regime is made possible solely by the use of the proper "normal coordinates" of the system—complex spin wave amplitude c (14).

It should be noted, that the transformation (14) coincides, up to the normalization coefficient, with the well-known classical Holstein-Primakoff transformation [123]. In case of an arbitrary conservative magnetic system, this transformation leads to the following equation of motion:

$$\left(\frac{dc}{dt}\right)_{\text{cons}} = -i\left(\frac{\gamma}{2M_0}\right)\frac{\delta\mathcal{W}}{\delta c^*}$$

where $\mathcal{W}$ is the free energy of magnetic system. This equation is a Hamiltonian equation with the Hamiltonian $\mathcal{H} = \gamma\mathcal{W}/2M_0$. Thus, in a general case (14) describes the transformation from the components of the magnetization vector to the complex canonical variables of a magnetic system.

Now, we will consider corrections to the equation of motion (15) caused by the damping $\boldsymbol{T}_{\text{G}}$ and spin-transfer $\boldsymbol{T}_{\text{S}}$ torques. Using the transformation (14), one can obtain the following correction caused by the Gilbert $(\alpha(\xi) = \alpha_{\text{G}} = \text{const})$ damping torque (3)

$$\left(\frac{dc}{dt}\right)_{\text{damp}} = -\Gamma_+\left(|c|^2\right)c \quad (16a)$$

where

$$\Gamma_+\left(|c|^2\right) = \Gamma_{\text{G}}\left(1 + Q|c|^2 + Q'|c|^4\right) \quad (16b)$$

and

$$\Gamma_{\text{G}} = \alpha_{\text{G}}\omega_0, \quad Q = 2\omega_M/\omega_0 - 1, \quad Q' = -2\omega_M/\omega_0.$$

In all physically-relevant range of complex amplitudes c the damping function $\Gamma_+(|c|^2)$ is positive and (16a) leads to the reduction of spin wave amplitude c with time. For sufficiently small spin wave amplitudes, when one can neglect nonlinear terms in (16b), this reduction is exponential with the linear rate Γ_{G}, which represents the half-linewidth of the linear ferromagnetic resonance.

It is important to note, that the damping function $\Gamma_+(|c|^2)$ is *nonlinear*, even if one uses the usual "linear" form of the Gilbert damping torque $(\alpha(\xi) = \alpha_{\text{G}} = \text{const})$. Thus, the damping rate of spin wave excitations depends on the spin wave power $|c|^2$ and, generally, increases with $|c|^2$. The use of the nonlinear form of the damping torque $(\alpha(\xi) \neq \text{const})$ will lead to the same functional form of the damping correction (16), but with different values of nonlinear damping coefficients Q and Q'. This correction is qualitatively important in the case of an in-plane magnetized "free" layer (see [74] for details).

In a similar way one can find corrections to the dynamical equation for the spin wave amplitude c caused by the spin-transfer torque $\boldsymbol{T}_{\text{S}}$ (4). In the case when the bias current is spin-polarized along $\widehat{\mathbf{e}}_z$ direction, this correction has the form

$$\left(\frac{dc}{dt}\right)_{\text{spin}} = +\Gamma_-\left(|c|^2\right)c. \quad (17a)$$

Here, $\Gamma_-(|c|^2)$ is given by

$$\Gamma_-\left(|c|^2\right) = \sigma_0 I\left(1 - |c|^2\right). \quad (17b)$$

For $I > 0$ (current flows from the "fixed" to "free" magnetic layer) $\Gamma_-(|c|^2)$ is positive in physically-relevant range of complex spin wave amplitudes c $(|c|^2 < 1)$, and (17a) leads to the increase of the amplitude c with time, i.e., spin-transfer torque creates *effective negative damping* for the magnetization precession in the "free" magnetic layer.

Collecting together contributions from the all three torques (given by (15), (16), and (17)) we get

$$\frac{dc}{dt} = \left(\frac{dc}{dt}\right)_{\text{cons}} + \left(\frac{dc}{dt}\right)_{\text{damp}} + \left(\frac{dc}{dt}\right)_{\text{spin}}$$

and the dynamical equation for the spin wave amplitude c takes the form

$$\frac{dc}{dt} + i\omega\left(|c|^2\right)c + \Gamma_+\left(|c|^2\right)c - \Gamma_-\left(|c|^2\right)c = 0 \quad (18)$$

which exactly coincides with the universal oscillator model (5) without the external "driving" force $f(t)$.

Thus, the dynamics of the considered here simplest spin-torque oscillator can be *exactly* described by the universal model (18) with properly defined nonlinear frequency

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$\omega(p)$ [see (15b)], natural positive damping $\Gamma_+(p)$ [see (16b)], and effective negative current-induced damping $\Gamma_-(p)$ [see (17b)].

C. Universal Nonlinear Oscillator Model: General Overview

In the previous subsections we showed how the same nonlinear oscillator model (5) can be derived for two absolutely different physical systems. Then, the natural question is: under which conditions operation of an arbitrary auto-oscillator can be described by the model (5)? It turns out that these conditions can be formulated in a general case as follows.

1) Only one mode or oscillation type is excited in the auto-oscillatory system.
2) The auto-oscillatory system is weakly nonconservative, i.e., possible range of variations (increase or decrease) of the amplitude $|c|$ during one oscillation period $2\pi/\omega$ is relatively small.
3) The excitation of oscillations is *critical* or "soft," i.e., self-sustained oscillations start with zero amplitude $|c| \to 0$, which smoothly increases with the increase of some control parameter (bias current I in the case of spin-torque oscillator or bias voltage E_0 in the case of electrical tunnel-diode oscillator).

The first condition is necessary to allow the possibility of description of the auto-oscillator's dynamics using only one complex amplitude c—amplitude of the mode or oscillation type that is actually excited in the system. At a first glance, this condition seems to be very restrictive, especially for intrinsically multimode systems, like spin-torque oscillators based on magnetic nano-contacts or nano-pillars, that can support many different modes of magnetization oscillations. It turns out, however, that, in most cases, even such multimode auto-oscillatory systems can not support simultaneous self-excitation of more than one mode. Consider, for instance, the case of a usual laser generator. The resonator of the laser can support many different types of optical modes. The amplification bandwidth of the active medium of the laser is usually much larger than the frequency separation between individual resonator modes. Thus, one can assume that many modes will be excited, and a laser will generate simultaneously at many different frequencies. In reality, however, only one mode is excited, and the linewidth of the generated light is much smaller than the amplification bandwidth of the active medium. The reason for this behavior is that all the optical modes try to compensate their natural energy losses from the same energy source—the active laser medium. Thus, the larger is the amplitude of a particular optical mode, the smaller part of the available energy will go to all the other modes. In other words, all the modes compete for the same energy source, and this *mode concurrence* is the reason why, usually, only one mode can survive in an auto-oscillatory system.

The above presented statement about the concurrence process in a nonlinear system is applicable to the case of a spin-torque oscillator: regardless of the multimode nature of the "free" ferromagnetic layer, usually, only one spin wave mode is excited by the spin-polarized current. This single-mode character of the spin wave generation is confirmed by the laboratory and numerical experiments, where the excitation of more than one mode (manifesting itself as the appearance of several incommensurable frequencies in the output signal) was observed very rarely, and in a very limited range of parameters. Analytical study of two-mode excitation in a spin-torque oscillator [96], also, confirmed that only one spin wave mode survives in the steady-state regime of spin wave generation.

We would like to stress one more time, that the validity of the nonlinear oscillator model (5) for the description of a spin-torque oscillator is not limited by the case of a uniform magnetization precession, as it was assumed for simplicity in Section IV-B. The spin wave mode excited in an STO can have an *arbitrary spatial distribution*, and the model (5) can still be used, provided that this mode is the *only* mode excited in the STO [5]. The account for the spatial distribution of the excited spin wave mode will lead only to the appropriate renormalization of the coefficient in the universal auto-oscillator model (5), but will not change the qualitative properties of the auto-oscillation process (see [5] for details).

The second condition of applicability of the model (5), namely, the condition of the weakness of damping, was considered in details in Section IV-A. This condition allows one to neglect the nonresonant terms (e.g., terms, proportional to c^* or c^2) in the equation of motion for the complex amplitude c, which makes the description of the oscillator's dynamics much simpler.

In the case of a spin-torque oscillator the damping rates can be estimated as $\Gamma_+ \sim \Gamma_- \sim \alpha_{\rm G}\omega$, where $\alpha_{\rm G}$ is the Gilbert damping constant. Since for typical ferromagnetic metals $\alpha_{\rm G} \sim 0.01$, the accuracy of this assumption for spin-torque oscillator is rather good. Thus, the relative error introduced by this approximation is of the order of $(\Gamma_\pm/\omega)^2 \sim \alpha_{\rm G}^2 \sim 10^{-4}$ and can be safely tolerated.

The last condition (critical character of the mode excitation) is not really restrictive, and is needed, mostly, to make the physical interpretation of the results obtained in the framework of the model (5) simpler. Indeed, as it was shown in [92], the model (5) can be used even for an oscillator demonstrating *sub-critical* or "hard" excitation of a nonlinear ("bullet") spin wave mode (i.e., the mode that has a *finite* amplitude at the generation threshold), but in that case the quantity $p = |c|^2$ can no longer be interpreted as a power of the excited nonlinear spin wave mode.

Depending on the geometrical parameters, orientation, and magnitude of the bias magnetic field, spin-torque oscillators can demonstrate both critical and sub-critical instability scenarios (see Appendix II for a brief overview of different spin wave modes that can be excited by spin-polarized current). As it was mentioned above, a good example of sub-critically-unstable mode, having finite amplitude at the threshold of excitation, is the self-localized nonlinear "bullet" mode (see [92], [94], and [95] for details) which is excited in in-plane magnetized magnetic nano-contacts.

Below, we will consider only the case of a *critical* instability of the excited spin wave mode, when, similar to the simple geometry of Section IV-B, the modulus of the spin wave amplitude $|c|$ is proportional to the magnetization precession angle, and the dimensionless spin wave power $p = |c|^2$ is proportional to the experimentally measured microwave power of the STO.

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Thus, in the majority of cases the operation of a spin-torque oscillator can be described by the simple nonlinear auto-oscillator model (5). It should be noted, however, that the actual calculation of the dependences of the frequency $\omega(p)$ and damping rates $\Gamma_{\pm}(p)$ on the dimensionless spin wave power p for real geometries represent a complicated mathematical task. In most cases it is not possible to obtain these dependences for arbitrary values of p analytically, and, in the framework of an analytical approach, one can obtain only the weakly-nonlinear $(p \ll 1)$ Taylor expansions of these functions

$$\omega(p) \approx \omega_0 + Np \tag{19a}$$

$$\Gamma_+(p) \approx \Gamma_{\mathrm{G}}(1 + Qp) \tag{19b}$$

$$\Gamma_-(p) \approx \sigma I(1 - p). \tag{19c}$$

The explicit form of all the coefficients entering expansion (19) for an obliquely magnetized spin-torque oscillator is presented in Appendix I. Below, we will use the weakly-nonlinear expansion (19) for all the calculations and for the comparison of all the results obtained in the framework of the nonlinear oscillator model (5) with experimental data.

It should be noted, that in the case, when the expansion (19) does not provide a sufficient accuracy, the functional dependences of the frequency $\omega(p)$ and damping rates $\Gamma_{\pm}(p)$ on the power p of a spin-torque oscillator can be calculated numerically by using micromagnetic simulations. Then, one can use these dependences, obtained from the numerical simulations of a *single* spin-torque oscillator, to study more complicated tasks, e.g., phase-locking in a large array of oscillators (see Section VI), or the influence of thermal fluctuations (see Section VII) on the behavior of an STO array—the problems that can not be studied numerically using the existing computational capabilities.

As it was shown in Section IV-A, the frequency $\omega(p)$ of the van der Pol oscillator is independent of the oscillation power p [see (13b)]. This is the usual case for almost all types of conventional oscillators. Below, we will refer to such oscillators $(\omega(p) = \omega_0 = \mathrm{const})$ as "linear" auto-oscillators. In contrast, the frequency of a spin-torque oscillator (15b) strongly depends on the power p. Such oscillators will be called "nonlinear" auto-oscillators. Note, that this classification of oscillators as linear and nonlinear ones does not coincide with the pure mathematical classification. In a purely mathematical sense, a dynamical system (5) is nonlinear, if either the frequency $\omega(p)$ or the damping rates $\Gamma_{\pm}(p)$ depend on the power p. In this sense, *any* auto-oscillator is a nonlinear system, since the damping rates $\Gamma_{\pm}(p)$ in the auto-oscillator must depend on the generated power p to provide a mechanism for limiting the oscillation amplitude at a certain constant level and create a stable limit cycle in the phase space of the auto-oscillating system.

V. Autonomous Dynamics of Auto-Oscillators

In this section we will consider autonomous, or free, dynamics of an auto-oscillator, when the external "driving" force $f(t)$ in the right-hand side of (5) is absent

$$\frac{dc}{dt} + i\omega\left(|c|^2\right)c + \Gamma_+\left(|c|^2\right)c - \Gamma_-\left(|c|^2\right)c = 0. \tag{20}$$

We will analyze stationary solution of (20) and discuss its stability, compare theoretical results with experimental data, and, also, consider the case of an oscillator with slowly modulated parameters.

A. Stationary Regime of Generation

To find the stationary solution of the free auto-oscillator equation (20), we rewrite it as a system of two real equations for the power $p = |c|^2$ and phase $\phi = \arg(c)$ of the oscillations

$$\frac{dp}{dt} = -2\left[\Gamma_+(p) - \Gamma_-(p)\right]p \tag{21a}$$

$$\frac{d\phi}{dt} = -\omega(p). \tag{21b}$$

Here, the first equation determines the dynamics of the oscillation power p in an autonomous auto-oscillator, whereas the second one describes the dynamics of the oscillation phase ϕ or defines the generated frequency $-d\phi/dt$. Since (21a) is independent of the phase ϕ, it is convenient, first, to consider only the dynamics of the oscillation power p.

1) Stationary Power: According to (21a), there are two possible types of stationary $(dp/dt = 0)$ solutions. The first stationary solution $p = 0$ is trivial and corresponds to the absence of any oscillations. Linearizing (21a) near this zero-power solution $(p \to 0)$, one obtains the equation

$$\frac{dp}{dt} = -2\left[\Gamma_+(0) - \Gamma_-(0)\right]p.$$

It is clear from the linearized equation, that the zero-power solution is stable and, therefore, physically realizable, when $\Gamma_+(0) > \Gamma_-(0)$. Thus, the condition $\Gamma_+(0) = \Gamma_-(0)$ determines the threshold of appearance of an oscillation having a nonzero power, or, in other words, the threshold of generation in an auto-oscillator.

Using (19) defining the parameters of the universal auto-oscillator model (20) for an STO, one obtains the expression for the threshold current I_{th} of microwave generation in the case of a spin-torque auto-oscillator

$$I_{\mathrm{th}} = \frac{\Gamma_{\mathrm{G}}}{\sigma}. \tag{22}$$

This equation implicitly determines the dependence of the threshold current on the external bias field, geometrical sizes, and other parameters of the STO.

The second type of stationary solutions of (21a) is a nonzero solution $p = p_0$, where p_0 is determined from the condition

$$\Gamma_+(p_0) = \Gamma_-(p_0). \tag{23}$$

This equation has a clear physical meaning: in the stationary regime $(p = p_0)$, energy losses due to the natural energy dissipation $(\Gamma_+(p_0))$ should be *exactly* compensated by the energy supply from the external energy source $(\Gamma_-(p_0))$, and the total effective damping $\Gamma_+(p_0) - \Gamma_-(p_0)$ should be *exactly* equal to zero.

In general, (23) can have several solutions for the stationary power p_0, i.e., (20) can describe multistable regime of generation. For most physical systems, however, the natural positive damping $\Gamma_+(p)$ *increases* with the oscillation power p

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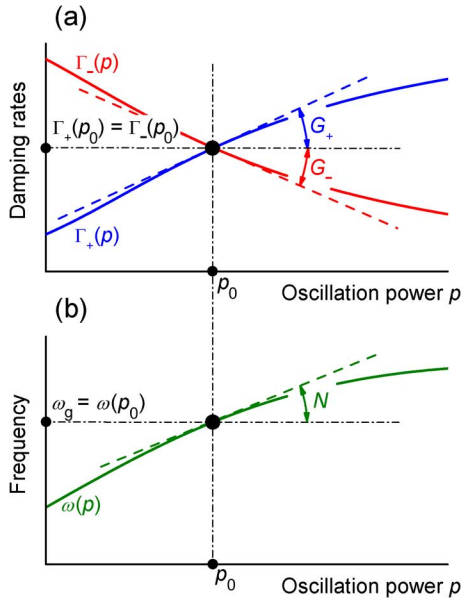

Fig. 4. Dependences of the damping rates $\Gamma_{\pm}(p)$ [panel (a)] and oscillation frequency $\omega(p)$ [panel (b)] on the oscillation power p. Intersection of the curves $\Gamma_{+}(p)$ and $\Gamma_{-}(p)$ determines the free-running operation point of the oscillator with the power p_0 and free-running generation frequency $\omega_{\rm g} = \omega(p_0)$. The slopes of the damping rates $\Gamma_{\pm}(p)$ and the frequency $\omega(p)$ at the free-running operation point $p = p_0$ define the nonlinear damping coefficients $G_{\pm} = d\Gamma_{\pm}(p)/dp$ and nonlinear frequency shift coefficient $N = d\omega(p)/dp$.

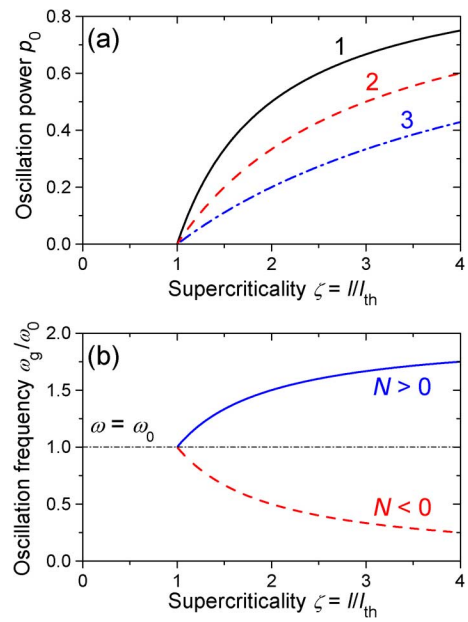

Fig. 5. (a) Dependence (24) of the stationary power p_0 of a spin-torque auto-oscillator on the supercriticality $\zeta = I/I_{\rm th}$ for several values of the nonlinear damping parameter Q: 1—$Q = 0$, 2—$Q = 1$, 3—$Q = 3$. (b) Dependence (29) of the stationary frequency $\omega_{\rm g} = \omega(p_0)$ of a spin-torque auto-oscillator on the supercriticality ζ for $Q = 0$. Solid line—positive nonlinear frequency shift coefficient $N = \omega_0 > 0$, dashed line—negative nonlinear frequency shift coefficient $N = -\omega_0 < 0$. Thin dash-dotted line shows the linear oscillation frequency ω_0.

$(d\Gamma_{+}(p)/dp > 0)$, whereas effective negative damping $\Gamma_{-}(p)$ *decreases* with p $(d\Gamma_{-}(p)/dp < 0)$. In this case (23) has a solution only above the linear generation threshold (i.e., for $\Gamma_{+}(0) > \Gamma_{-}(0)$), and this solution is unique. Fig. 4(a) shows schematically the dependences of the damping rates $\Gamma_{\pm}(p)$ on the power p and the operation point $p = p_0$, corresponding to the stable solution with nonzero power.

Using approximate (19), the stationary power p_0 for the spin-torque auto-oscillator can be written as

$$p_0 = \frac{\zeta - 1}{\zeta + Q} \tag{24}$$

where we have introduced the dimensionless *supercriticality parameter*

$$\zeta = \frac{I}{I_{\rm th}}. \tag{25}$$

The stationary power (24) equals to zero at the threshold of generation $\zeta = 1$ (i.e., $I = I_{\rm th}$) and gradually increases with the increase of the supercriticality ζ (or the bias current I) up to a limiting value $p_{\infty} = 1$ for $\zeta \to \infty$. For large supercriticalities $\zeta \gg 1$, however, (19) usually can not be used anymore, and this limit is interesting only academically. The dependence (24) of the stationary power p_0 on the supercriticality ζ for several different nonlinear damping coefficients Q is shown on Fig. 5(a).

To study the stability of the stationary solution $p = p_0$ we linearize (21a) for small power deviations $\delta p = p - p_0$ to get

$$\frac{d\delta p}{dt} = -2[G_{+} - G_{-}]p_0\delta p = -2\Gamma_{\rm p}\delta p. \tag{26}$$

Here, the parameters G_{+} and G_{-} are defined by

$$G_{+} = \frac{d\Gamma_{+}(p)}{dp}, \qquad G_{-} = \frac{d\Gamma_{-}(p)}{dp} \tag{27a}$$

where the derivatives are calculated at the stationary point $p = p_0$, and $\Gamma_{\rm p}$ is the damping rate for small power deviations δp

$$\Gamma_{\rm p} = (G_{+} - G_{-})p_0. \tag{27b}$$

The nonlinear damping coefficients G_{+} and G_{-} are determined by the slopes of the damping functions $\Gamma_{+}(p)$ and $\Gamma_{-}(p)$ at the operation point $p = p_0$ [see Fig. 4(a)].

It is clear, that the stationary solution $p = p_0$ is stable, if

$$\Gamma_{\rm p} > 0$$

or, equivalently, if

$$G_{+} > G_{-}.$$

For the majority of auto-oscillatory systems the natural positive damping $\Gamma_{+}(p)$ is an increasing function of the oscillation power, i.e., $G_{+} > 0$, while the effective negative damping $\Gamma_{-}(p)$ decreases with p, i.e., $G_{-} < 0$ [see Fig. 4(a)]. In that case the stationary solution (24) is always stable and physically realizable.

For the case of an STO (19) one obtains

$$G_{+} = Q\Gamma_{\rm G}, \qquad G_{-} = -\sigma I = -\zeta\Gamma_{\rm G}$$

and the effective damping rate $\Gamma_{\rm p}$ of small power deviations δp from the stationary solution $p = p_0$ is equal to

$$\Gamma_{\rm p} = (\zeta + Q)\Gamma_{\rm G}p_0 = (\zeta - 1)\Gamma_{\rm G}.$$

We would like to stress, that the value of the positive $\Gamma_{+}(p_0)$ or negative $\Gamma_{-}(p_0) = \Gamma_{+}(p_0)$ damping rate of the oscillator at

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the operation point $p = p_0$ *does not* determine the damping rate of the small power fluctuations δp and, in general, does not play any significant role in the free dynamics of an auto-oscillator.

2) Generated Frequency: The frequency of generation is one of the most important parameters of an auto-oscillator. Equation (21b) states that the instantaneous oscillator frequency $-d\phi/dt$ is, at each moment of time, equal to the nonlinear frequency $\omega(p(t))$ determined by the current value of the oscillation power $p(t)$. Thus, the frequency $\omega_{\rm g}$, generated in the stationary oscillation regime ($p = p_0 = \text{const}$), is given by the equation

$$\omega_{\rm g} = \omega(p_0) \tag{28}$$

which corresponds to the linear time dependence of the oscillation phase

$$\phi(t) = -\omega_{\rm g} t + \phi_0 = -\omega(p_0)t + \phi_0$$

where ϕ_0 is an arbitrary initial phase of the oscillations. Then, the full stationary solution $c_0(t)$ for the complex amplitude c of the auto-oscillation can be written as

$$c_0(t) = \sqrt{p_0}\exp[-i\omega_{\rm g} t + i\phi_0].$$

This solution can be used as a starting point in the study of the dynamics of a weakly perturbed oscillator (see, e.g., Section VI).

It should be noted, that, in contrast with the stationary power p_0, the value of the initial phase ϕ_0 is not fixed by the dynamical equation (20). In other words, the stationary solution is *absolutely* stable with respect to *power* fluctuations, but only *marginally* stable with respect to the *phase* fluctuations. This means, that even weak perturbations can lead to a substantial change of the phase dynamics at long times. This phase-invariance of the free oscillations is an inherent feature of all the *auto*-oscillators, and is directly connected to the time-invariance of the auto-oscillator dynamical equation of motion (20). This fundamental fact has far-reaching consequences. In particular, it makes possible the phase-locking of auto-oscillations to a weak external signal (see Section VI) and determines the main mechanism of broadening of the auto-oscillator linewidth under the action of thermal fluctuations (see Section VII-C).

Using (19) and (24), one can obtain an explicit expression for the frequency, generated by a spin-torque oscillator

$$\omega_{\rm g} = \omega_0 + N\frac{\zeta - 1}{\zeta + Q}. \tag{29}$$

This dependence is shown in Fig. 5(b) for positive ("blue" frequency shift) and negative ("red" frequency shift) values of N.

One can, also, find the *frequency tunability* $d\omega_{\rm g}/d\mu$ with respect to the variation of some control parameter μ of the oscillator

$$\frac{d\omega_{\rm g}}{d\mu} = \frac{d\omega}{d\mu} + N\frac{dp_0}{d\mu} \tag{30}$$

where

$$N = \frac{d\omega(p)}{dp} \tag{31}$$

is the *nonlinear frequency shift coefficient*, and all the derivatives are taken at the stationary operation point $p = p_0$. It is clear, that for a spin-torque oscillator the definitions (19a) and (31) of the nonlinear frequency shift coefficient N coincide.

The first term in (30) accounts for the direct dependence of the oscillator frequency $\omega(p)$ on the control parameter μ, while the second term describes the indirect changes in the generation frequency $\omega(p_0)$ due to the dependence of the stationary power p_0 on the parameter μ.

For instance, the eigen-frequency $\omega(p)$ of a spin-torque oscillator based on a metallic GMR spin-valve does not depend *explicitly* on the bias current I (as long as one can neglect the current-induced Oersted magnetic field), but the generated frequency $\omega(p_0)$ *does* depend on I through the dependence of the stationary power p_0 on the bias current I

$$\frac{d\omega_{\rm g}}{dI} = N\frac{dp_0}{dI} = \frac{N}{I_{\rm th}}\frac{1+Q}{(\zeta+Q)^2}. \tag{32}$$

A different situation can take place in oscillators based on MTJ with the MgO spacer between the magnetic layers. As it has been recently demonstrated in [130] the spin-transfer torque in MTJ can have a significant perpendicular (field-like) component which will create an explicit dependence of the generated frequency on the bias current (see, e.g., [41, Eq. (3)]). Thus, if the bias current I is chosen as a control parameter μ, in the case of an MTJ-based spin-torque auto-oscillator the first derivative in (30) could be nonzero and could play a significant role in the frequency tunability.

As it was emphasized above, the nonlinear frequency shift coefficient N for spin-torque oscillators, in contrast with the majority of conventional auto-oscillators, is very large. One can introduce the normalized dimensionless nonlinear frequency shift

$$\nu = \frac{N}{G_+ - G_-} \tag{33}$$

which measures the relative influence of the nonlinear frequency shift N in comparison with the nonlinear damping coefficients $G_\pm$. For most conventional oscillators $|\nu| \ll 1$, and one can totally neglect the dependence of the auto-oscillator frequency $\omega(p)$ on the power p. In contrast, for typical spin-torque oscillators $|\nu| \gg 1$, and the dependence of the oscillator frequency $\omega(p)$ on the oscillation power p determines many aspects of behavior of spin-torque oscillators.

It should also be noted, that the nonlinear frequency shift coefficient N for spin-torque oscillators strongly depends on the direction of the bias magnetic field and can be both positive and negative. Fig. 6(a) shows the nonlinear coefficient N for an *isotropic* magnetic film as a function of the *out-of-plane* angle θ_0 of the external bias magnetic field for several values of the magnitude H_0 of this field. The chosen values of the magnetic field are rather large ($H_0 > 4\pi M_0$), in order to saturate the magnetic film even in the case of normal magnetization $\theta_0 = 90°$. One can see, that N is a monotonically increasing function of θ_0, and is negative for an in-plane magnetized film ($\theta_0 = 0$) and positive for a normally magnetized film ($\theta_0 = 90°$). Apart from the small region near the "linear" angle $\theta_{0,\rm lin} \sim 50° - 80°$, at which N vanishes $N(\theta_{0,\rm lin}) = 0$, the nonlinear frequency shift is much larger than the nonlinear damping coefficients $G_\pm$, and

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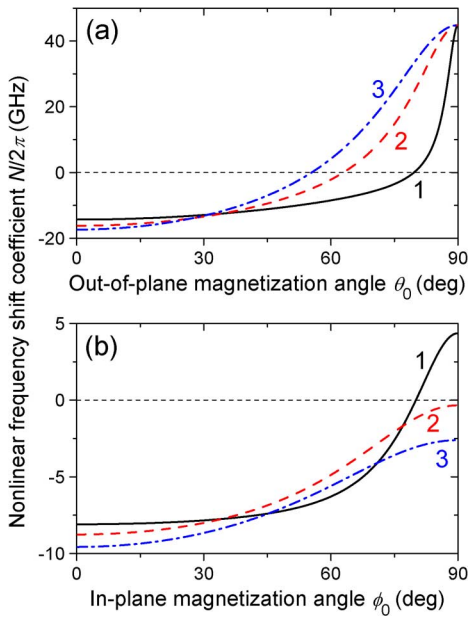

Fig. 6. (a) Nonlinear frequency shift coefficient N for an *isotropic* magnetic film (saturation magnetization $4\pi M_0 = 8$ kG) as a function of the *out-of-plane* magnetization angle θ_0 for several magnitudes of the external bias magnetic field H_0: 1—$H_0 = 10$ kOe, 2–$H_0 = 15$ kOe, 3—$H_0 = 20$ kOe. (b) Nonlinear frequency shift coefficient N for an *anisotropic* magnetic film (saturation magnetization $4\pi M_0 = 8$ kG, anisotropy field $H_A = 0.3$ kOe) as a function of the angle ϕ_0 between the *in-plane* bias magnetic field and the easy axis of the film for several magnitudes of the external magnetic field H_0: 1—$H_0 = 0.6$ kOe, 2—$H_0 = 1.2$ kOe, 3—$H_0 = 1.8$ kOe.

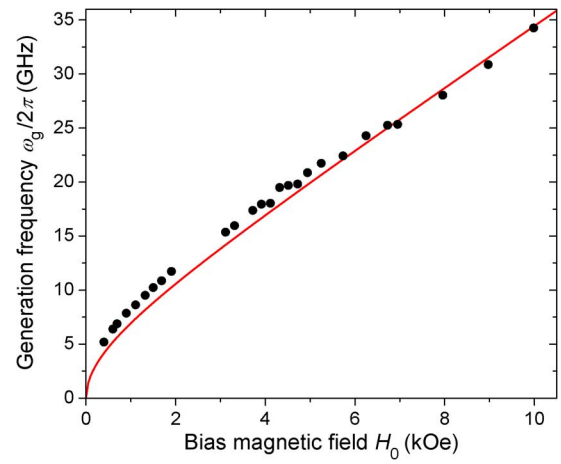

Fig. 7. Frequency of current-induced microwave oscillations ω_g in the in-plane magnetized nano-contact as a function of the bias magnetic field H_0. Solid circles—experimental data from [17], solid line—theoretical dependence from [72]. The following parameters have been used in theoretical calculations: saturation magnetization $4\pi M_0 = 8$ kG, nonlinear damping parameter $Q = 0$, supercriticality $\zeta = 1.56$.

the normalized dimensionless nonlinear frequency shift ν (33) is much larger than unity (typically $|\nu| \sim 10-50$). In the vicinity of the "linear" angle $\theta_0 \sim \theta_{0,\mathrm{lin}}$, from the other hand, $|\nu| \sim 0$ and a spin-torque oscillator behaves similar to a conventional oscillator with a constant generation frequency.

Fig. 6(b) shows the nonlinear coefficient N for an *anisotropic in-plane* magnetized "free" layer (with the anisotropy axis lying in the film plane) as a function of the angle ϕ_0 between the in-plane bias magnetic field and the easy axis of the magnetic film. In this case the magnitudes of the bias magnetic field H_0 are chosen to be much smaller, since the anisotropic in-plane magnetized film saturates at much smaller fields $H_0 > H_A$. In Fig. 6(b), one can see the same trend as in Fig. 6(a): the nonlinear frequency shift coefficient N is negative for easy-axis magnetization [$\phi_0 = 0$ at Fig. 6(b)] and monotonically increases when the direction of the bias magnetic field rotates toward the hard magnetization axis [$\phi_0 = 90°$ at Fig. 6(b)].

Thus, the nonlinear frequency shift coefficient N for a spin-torque oscillator can be varied in a wide range by changing the orientation and magnitude of the external magnetic field. This unique tunability of the parameters of STOs make them very interesting from the fundamental point of view, allowing investigations of fundamental aspects of the physics of strongly nonlinear auto-oscillatory systems.

3) Comparison With Experiments: The above described theory of free, or autonomous, dynamics of a spin-torque oscillator was derived, although on less general grounds, in [71]–[73]. This theory gave natural explanation of a number of qualitative features of microwave generation in spin-torque oscillators. First of all, it explained the dependences of the generation frequency ω_g and frequency-current slope $d\omega_g/dI$ on the direction and magnitude of the external bias field, allowed one to estimate the threshold current I_{th} of microwave generation, etc. Here we present several examples of experimental data that can be quantitatively explained by the simple nonlinear oscillator model (20). A more detailed comparison of theoretical and experimental results can be found in [72], [73].

Fig. 7 shows the dependence of the generation frequency ω_g on the bias magnetic field H_0 for an in-plane magnetized magnetic nano-contact [17]. One can see, that the nonlinear oscillator model of current-induced oscillations can give satisfactory quantitative explanation of the experimental data in all the range of bias magnetic fields H_0. While the shape of the dependence of the generation frequency ω_g on H_0 is qualitatively similar to the dependence of the FMR frequency ω_0 on the bias field H_0, the quantitative values of ω_g are significantly below ω_0. This is explained by the fact that the experiments [17] were performed significantly above the threshold of generation (where the supercriticality parameter was equal to $\zeta = 1.56$ [72]), and, since the nonlinear frequency shift coefficient N is negative for an in-plane magnetized film, the generation frequency $\omega_g = \omega(p_0)$ was shifted down from the FMR frequency ω_0; see (29).

Fig. 8 shows the dependence of the generation frequency ω_g on the out-of-plane magnetization angle θ_0 for two values of the bias magnetic field H_0. The points represent experimental data from [18], while the lines show theoretical calculation in the framework of nonlinear oscillator model [72]. Here, again, the simple theory gives a good quantitative description of the experimental data.

Fig. 9 shows the dependence of the frequency-current slope $d\omega_g/dI$ as a function of the out-of-plane magnetization angle θ_0. Solid and open circles represent experimental data from [18] and [17], respectively. The solid line shows theoretical dependence calculated using (32) with the nonlinear damping parameter $Q = 3$. It is clear that the simple analytical model

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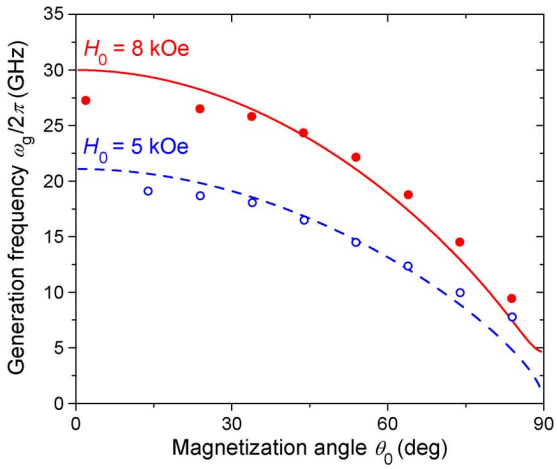

Fig. 8. Frequency of current-induced microwave oscillations ω_g as a function of the out-of-plane magnetization angle θ_0 for $H_0 = 8$ kOe (solid circles, solid line) and $H_0 = 5$ kOe (open circles, dashed line). The solid and open circles represent the experimental data from [18]; solid and dashed lines—theoretical calculation [72]. The following parameters have been used in theoretical calculations: saturation magnetization $4\pi M_0 = 8$ kG, nonlinear damping parameter $Q = 0$, supercriticality $\zeta = 1.26$.

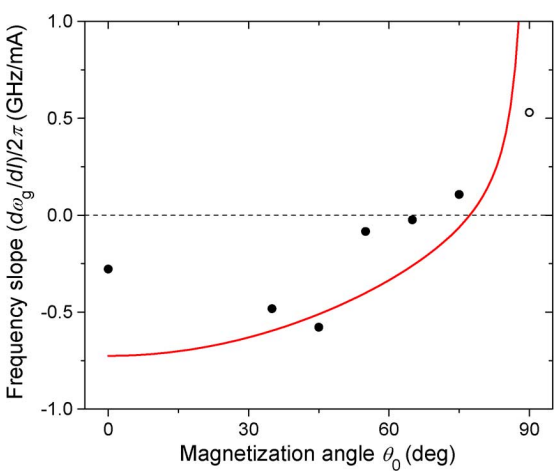

Fig. 9. Slope of the frequency ω_g versus current I as a function of the out-of-plane magnetization angle θ_0. Solid circles—experimental data from [18], open circle—experiment [17], solid line—theoretical dependence (32) with nonlinear damping parameter $Q = 3$, supercriticality $\zeta = 1.26$, saturation magnetization $4\pi M_0 = 8$ kG, and bias field $H_0 = 8$ kOe.

gives a good qualitative and even a reasonably good quantitative description for both experiments. The frequency-current slope $d\omega_g/dI$ has the same angular dependence as the nonlinear frequency shift N [see Fig. 6(a)], i.e., it is negative ("red" frequency shift) for in-plane magnetized free layer ($\theta_0 = 0$), monotonically increases with the increase of θ_0, and for normally magnetized film ($\theta_0 = 90°$) is always positive ("blue" frequency shift).

It is interesting to note, that the theoretical description of the frequency-current slope, performed in [72], gave the same *qualitative* dependence as shown in Fig. 9 (including the value of the "linear" angle $\theta_{0,\mathrm{lin}} \simeq 75°$, at which slope changes sign), but the absolute *quantitative* values of the slope $d\omega_g/dI$ were several times larger than in the experiments [17], [18]. This was caused by the fact that in the theory [72] the natural positive damping was assumed to be linear, i.e., independent of the oscillation power p, $\Gamma_+(p) = \Gamma_G = \mathrm{const}$, which corresponds to zero nonlinear damping parameter $Q = 0$ in our current approach. It is clear from (32), that such an approximation gives qualitatively correct angular dependence of the slope $d\omega_g/dI$ (which is determined by the angular dependence of the nonlinear frequency coefficient N, see Fig. 6(a)), but overestimates absolute value of the slope $|d\omega_g/dI|$ by $\sim (1+Q)$ times (for $\zeta \sim 1$).

Thus, the simple nonlinear oscillator model (20) provides reasonably good quantitative description of microwave generation in spin-torque oscillators.

B. Frequency Modulation of Nonlinear Oscillators

To demonstrate the power of the above described general theoretical approach, based on the nonlinear oscillator model (5), we consider the problem of the auto-oscillator dynamics under the slow frequency modulation of its parameters. In particular, we will consider the case when the effective negative damping $\Gamma_-(p)$ is modulated with time as

$$\Gamma_-(p) \to \Gamma_-(t,p) = [1 + \epsilon\cos(\omega_m t)]\,\Gamma_-(p). \tag{34}$$

Here, ϵ is the modulation depth and ω_m is the modulation frequency. In the case of a spin-torque oscillator, such modulation can be easily achieved by modulating the bias current $I \to [1 + \epsilon\cos(\omega_m t)]I$.

In general, the frequency-modulated dynamics of an oscillator can be described using the simple model (20) with time-dependent coefficients only for sufficiently small modulation frequencies $\omega_m \ll \omega_g$. Otherwise, one should take into account terms that were neglected in the derivation of (20). Note, also, that, since the coefficients of (20) now depend on time, the frequency-modulated dynamics of an auto-oscillator is not, strictly speaking, a dynamics of an exactly autonomous system.

Assuming that the modulation depth ϵ is small $\epsilon \ll 1$, we can represent the oscillation power $p(t)$ as $p(t) = p_0 + \delta p(t)$, where p_0 is the stationary oscillation power, corresponding to an unmodulated oscillator ($\epsilon = 0$), and $\delta p(t)$ is a small ($\delta p(t) \sim \epsilon$) modulation of the output power. Keeping in (20) only the terms that are linear in ϵ, one obtains a system of equations for the power modulation δp and the phase $\phi = \arg(c)$ of the modulated auto-oscillator

$$\frac{d\delta p}{dt} = -2\Gamma_p\delta p + 2\epsilon\Gamma_-(p_0)p_0\cos(\omega_m t) \tag{35a}$$

$$\frac{d\phi}{dt} = -\omega_g - N\delta p. \tag{35b}$$

In comparison with (26) for power fluctuations δp of free auto-oscillations, (35a) has an additional nonhomogeneous term proportional to the modulation depth ϵ.

A stationary solution of the system (35) has the form

$$\delta p(t) = \epsilon\frac{2\Gamma_-(p_0)p_0}{\sqrt{\omega_m^2 + 4\Gamma_p^2}}\cos(\omega_m t + \psi) \tag{36a}$$

$$\phi(t) = -\omega_g t - \Delta\phi\sin(\omega_m t + \psi) + \phi_0 \tag{36b}$$

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where $\psi = -\arctan(\omega_{\rm m}/2\Gamma_{\rm p})$, ϕ_0 is an arbitrary initial oscillation phase, and

$$\Delta\phi = \epsilon \frac{2N\Gamma_-(p_0)p_0}{\omega_{\rm m}\sqrt{\omega_{\rm m}^2 + 4\Gamma_{\rm p}^2}} \tag{37}$$

is the amplitude of the induced phase modulation of the generated signal.

The power modulation $\delta p(t)$ is always of the order of the modulation depth ϵ, regardless of the modulation frequency $\omega_{\rm m}$. In contrast, the modulation of the phase [$\Delta\phi \sim \epsilon/\omega_{\rm m}$, see (37)] diverges as $\omega_{\rm m} \to 0$ and, therefore, can be large even for a very weak modulation depth ϵ. This difference between the power and phase deviations is connected with the discussed above different stability types of oscillator's power and phase, i.e., absolute stability for the power and marginal stability for the phase fluctuations.

Taking into account that the power modulation δp is always weak (for small ϵ), one can neglect it and write the solution for the complex amplitude $c(t) = \sqrt{p(t)}e^{i\phi(t)}$ approximately as

$$c(t) \approx \sqrt{p_0}\exp\left[-i\omega_{\rm g}t - i\Delta\phi\sin(\omega_{\rm m}t + \psi) + i\phi_0\right].$$

Developing this solution in a Fourier series, one obtains the spectrum of the frequency-modulated oscillations

$$c(t) = \sqrt{p_0}e^{-i\omega_{\rm g}t + i\phi_0}\sum_{n=-\infty}^{\infty} J_n(\Delta\phi)e^{in(\omega_{\rm m}t+\psi)} \tag{38}$$

where $J_n(x)$ is the Bessel function of the nth order. Besides the central peak at the frequency $\omega_{\rm g} = \omega(p_0)$, the spectrum (38) consists of an infinite number of sidebands at the frequencies $\omega_{\rm g} + n\omega_{\rm m}$ having relative amplitudes $J_n(\Delta\phi)$.

Equations (37) and (38) predict a nontrivial dependence of the intensity of the spectrum sidebands on the modulation frequency $\omega_{\rm m}$ and, in particular, suggest a direct way of experimental determination of one of the most important oscillator parameters—effective damping rate $\Gamma_{\rm p}$.

An experimental study of the frequency-modulated regime of operation of a spin-torque oscillator was undertaken in [55]. The experimental results qualitatively coincide with the predictions of the nonlinear oscillator model. In the experiments [55], however, the level of current modulation was relatively large ($\epsilon \sim 0.1$) and, to perform a quantitative comparison with this experiment, one needs to take into account higher-order terms in the expansion of coefficients of (20). Also, the data presented in [55] were taken at a single fixed modulation frequency $\omega_{\rm m}/2\pi = 40$ MHz, which makes the verification of the frequency dependence (37) and determination of the effective damping rate $\Gamma_{\rm p}$ from these experimental data impossible. Further experiments performed at several different modulation frequencies are needed for the experimental determinations of these important parameters of an STO.

The theoretical methods developed in this section can, also, be used to study other interesting regimes of operation of spin-torque auto-oscillators (or auto-oscillators of any other nature), such as the transient dynamics after the abrupt changes in the bias current or/and magnetic field, dynamics under the weak modulation of the direction or/and magnitude of the bias magnetic field, etc.

VI. Forced Dynamics of Oscillators

In this section, we consider dynamics of an auto-oscillator (5) under the action of an external *deterministic* harmonic force $f(t) \sim e^{-i\omega_{\rm e}t}$. It is well-known that, under the action of a harmonic external signal, *passive* oscillators ($\Gamma_- = 0$) demonstrates the phenomenon of *resonance*–strong increase of the oscillation amplitude when the external frequency $\omega_{\rm e}$ approaches the natural oscillator frequency ω. An active auto-oscillatory system demonstrates a qualitatively different behavior under the action of a weak external signal. Namely, when the difference $\omega_{\rm g} - \omega_{\rm e}$ between the free-running frequency of the auto-oscillator $\omega_{\rm g}$ and the frequency of an external signal $\omega_{\rm e}$ is sufficiently small, the auto-oscillator starts to generate at *exactly* the external frequency $\omega_{\rm e}$ with the phase, controlled by the phase of the external signal. This effect, known as *phase-locking* (other names for the same effect are *injection locking* and *frequency entrainment*), allows one to control a strong output of an auto-oscillator by a weak external signal and is common to auto-oscillators of any nature. This effect, also, can be used to reduce the generation linewidth of an oscillator and to increase the output power of oscillations. Examples of phase-locking behavior can be found not only in physical systems, but also in the auto-oscillatory systems studied in chemistry, biology, economics, etc.

In this section, we consider several important cases of the phase-locking. In Section VI-A we consider locking of an auto-oscillator to an *external* harmonic signal. In such a case the amplitude, phase, and frequency of the external "driving" signal $f(t)$ are independent of the auto-oscillator dynamics. Section VI-B is devoted to the case of the *mutual* phase-locking of two weakly-coupled auto-oscillators, when each of the interacting oscillators provides the locking signal for the other one. Finally, in Section VI-C we present a general theoretical framework for the investigation of phase-locking phenomena in a system of *many* interacting auto-oscillators.

A. Phase-Locking to an External Signal

The phase-locking of an auto-oscillator to an external signal is described by the model (5) where the "driving" term $f(t) = f_{\rm e}e^{-i\omega_{\rm e}t}$ represents an external harmonic signal

$$\frac{dc}{dt} + i\omega\left(|c|^2\right)c + \Gamma_+\left(|c|^2\right)c - \Gamma_-\left(|c|^2\right)c = f_{\rm e}e^{-i\omega_{\rm e}t}. \tag{39}$$

Here, $f_{\rm e}$ and $\omega_{\rm e}$ are, respectively, the (complex) amplitude and frequency of the external signal.

Depending on the amplitude $|f_{\rm e}|$ and frequency $\omega_{\rm e}$ of the external signal, (39) can describe a wide range of physical phenomena, including bi-stability and chaotic behavior of an auto-oscillator [99]. Below, we will consider only the case of a *weak* external signal $|f_{\rm e}| \to 0$ which is the most interesting from the practical point of view, as it is important to be able to control a relatively large auto-oscillation amplitude by a relatively weak external "driving" force. This case allows one to perform a simple and fully analytical study of the forced auto-oscillator dynamics. Other regimes of forced auto-oscillator dynamics can, also, be investigated using the model (39) but, in general, would involve numerical methods.

In the case of a spin-torque oscillator the external signal $f(t)$ can be created in several different ways. The simplest way is the

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microwave modulation of the bias current $I \to I + \Delta I \cos(\omega_e t)$. In such a case the amplitude f_e of the locking signal can be written approximately as [98]

$$f_e = -\sigma \Delta I \tan(\gamma_0)/(2\sqrt{2}) \tag{40a}$$

where γ_0 is the angle between the equilibrium orientations of the magnetization vectors of the "free" and "fixed" magnetic layers.

Another simple way of creating an external locking signal is to add a microwave component to the bias magnetic field $\boldsymbol{H}_0 \to \boldsymbol{H}_0 + \boldsymbol{h}\cos(\omega_e t)$. In this case, the amplitude f_e may be represented, approximately, as [98]

$$f_e = \gamma(h_y + ih_x)/(2\sqrt{2}). \tag{40b}$$

Here, γ is the gyromagnetic ratio and the axis z is assumed to be directed along the equilibrium direction of the magnetization in the "free" magnetic layer.

It should be noted, that, strictly speaking, interaction of a spin-torque oscillator with a microwave current or a microwave magnetic field depends on the power of the excited spin wave mode $|c|^2$, i.e., the effective amplitude f_e of the locking signal in (39) is a function of $|c|^2$. For simplicity, we neglect this dependence, assuming in (40) the limit $|c|^2 \to 0$. The theory of phase-locking developed in this section under this assumption can be easily generalized to the case of an arbitrary dependence of the "driving" force on the auto-oscillation power $f_e(|c|^2)$—one simply has to replace in all equations the constant amplitude f_e by the value $f_e(p_0)$, where p_0 is the free-running power of the auto-oscillator.

It is convenient to rewrite one complex equation (39) as two real equations for the power $p = |c|^2$ and phase $\phi = \arg(c)$ of the auto-oscillations

$$\frac{dp}{dt} + 2\left[\Gamma_+(p) - \Gamma_-(p)\right]p = 2F_e\sqrt{p}\cos(\omega_e t + \phi - \psi_e) \tag{41a}$$

$$\frac{d\phi}{dt} + \omega(p) = -\frac{F_e}{\sqrt{p}}\sin(\omega_e t + \phi - \psi_e) \tag{41b}$$

where $F_e = |f_e|$ is the (real) amplitude and $\psi_e = \arg(f_e)$ is the initial phase of the external force.

As it was mentioned earlier, one of the characteristic features of spin-torque oscillators is the strong dependence of the oscillator frequency $\omega(p)$ on the oscillation power p. To better understand the influence of this fact on the process of phase-locking, it would be interesting to consider first the phase-locking of a "linear" auto-oscillator, where the oscillation frequency is constant $\omega(p) = \omega_0 = \text{const}$, and then to analyze the phase-locking for a "nonlinear" oscillator $\omega(p) \neq \text{const}$.

1) "Linear" Auto-Oscillator: For a "linear" auto-oscillator ($\omega(p) = \omega_0 = \text{const}$) (41b) takes the form

$$\frac{d\phi}{dt} + \omega_g = -\frac{F_e}{\sqrt{p}}\sin(\omega_e t + \phi - \psi_e). \tag{42}$$

Here, we wrote $\omega_g = \omega(p_0)$ instead of a more convenient ω_0 (clearly, they coincide in the case of a "linear" auto-oscillator) to keep notations the same as in the case of a nonlinear auto-oscillator, see Section VI-A2.

For sufficiently small amplitudes F_e of the external force one can neglect the difference between the instantaneous oscillation power p and its free-running value p_0 in the right-hand side part of (42). In this case (42) becomes a closed-form equation describing the dynamics of the auto-oscillator phase ϕ:

$$\frac{d\phi}{dt} + \omega_g = -\frac{F_e}{\sqrt{p_0}}\sin(\omega_e t + \phi - \psi_e). \tag{43}$$

Introducing a slow auto-oscillator phase (i.e., phase of the auto-oscillator relative to the phase of the external force $\psi_e - \omega_e t$)

$$\Phi = \phi + \omega_e t - \psi_e \tag{44}$$

we can rewrite (43) in a very simple form

$$\frac{d\Phi}{dt} = \delta - \frac{F_e}{\sqrt{p_0}}\sin(\Phi) \tag{45}$$

where

$$\delta = \omega_e - \omega_g$$

is the mismatch between the frequency of the external signal ω_e and the free-running auto-oscillator frequency ω_g.

Equation (45) was first proposed by R. Adler [131] in 1940s for the description of the phase-locking effect in an electrical oscillator based on a standard LC-circuit (similar to the oscillator considered in Section IV-A), and since then it was extensively used for many other auto-oscillatory systems.

The phase-locked regime of the auto-oscillator operation corresponds to a constant relative phase $\Phi = \Phi_0 = \text{const}$ of the generated auto-oscillation (it is clear, that in this case the auto-oscillator frequency $-d\phi/dt$ exactly coincides with the external frequency ω_e). The relative phase difference Φ_0 can be found from (45) as

$$\Phi_0 = \arcsin\left(\frac{\delta}{\Delta_0}\right) \tag{46}$$

where

$$\Delta_0 = \frac{F_e}{\sqrt{p_0}}. \tag{47}$$

Using (45), one can easily check that the solution (46), when it exists, is stable and, therefore, can be observed experimentally. It is, also, clear that the solution (46) exists only for $|\delta| < \Delta_0$ and, consequently, Δ_0 represents the *phase-locking bandwidth* of a linear auto-oscillator.

For the frequency mismatch δ outside the phase-locking interval $|\delta| > \Delta_0$ (45) has a periodic (with the period $\tau = 2\pi/\xi$) solution of the following form:

$$\Phi(t) = 2\arctan\left\{\frac{\Delta_0 + \xi\tan\left[\xi(t+t_0)/2\right]}{\delta}\right\}$$

where

$$\xi = \sqrt{\delta^2 - \Delta_0^2}$$

and t_0 is an arbitrary integration constant. The phase $\Phi(t)$ depends on the time t in a nonlinear way, which means that the oscillation amplitude $c(t)$ becomes phase-modulated, and its spectrum contains an infinite number of sidebands $\omega_e + n\xi$, where n is an integer number. The spectral power of sidebands rapidly decreases with the increase of $|n|$, and one can consider this

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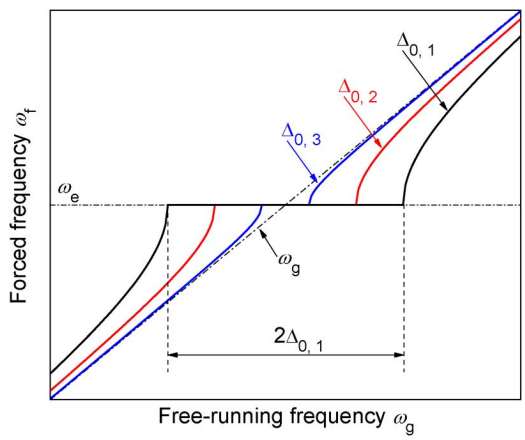

Fig. 10. Schematic dependence of the "forced" auto-oscillator frequency ω_f (48) on the free-running frequency ω_g of the "linear" auto-oscillator for several values of the phase-locking bandwidth Δ_0: $\Delta_{0,1} > \Delta_{0,2} > \Delta_{0,3}$. Dash-dotted lines show the free-running frequency ω_g and the frequency of the external signal ω_e.

regime of oscillations as a quasi-harmonic one with the frequency given by $\omega_e \pm \xi$, where the sign depends on the sign of the frequency mismatch $\delta = \omega_e - \omega_g$.

Thus, the "forced" frequency ω_f of the auto-oscillation is equal to

$$\omega_f = \omega_e \tag{48a}$$

within the phase-locking interval $|\omega_e - \omega_g| < \Delta_0$, and is determined by the expression

$$\begin{aligned}\omega_f &= \omega_e + \mathrm{sign}(\delta)\xi \\ &= \omega_e + \mathrm{sign}(\omega_g - \omega_e)\sqrt{(\omega_g - \omega_e)^2 - \Delta_0^2}\end{aligned} \tag{48b}$$

outside the phase-locking band $|\omega_e - \omega_g| > \Delta_0$. The dependence (48) for several magnitudes of the phase-locking bandwidths Δ_0 is shown schematically in Fig. 10.

One can see from Fig. 10, that outside the phase-locking range $|\omega_e - \omega_g| > \Delta_0$ the forced frequency ω_f gradually deviates from the external frequency ω_e and approaches the free-running frequency ω_g only for relatively large frequency mismatches. This effect, known as *frequency pulling*, shows that the weak external signal can significantly influence the oscillator dynamics even outside the phase-locking region.

2) "Nonlinear" Oscillator: For a "nonlinear" auto-oscillator $\omega(p) \neq \mathrm{const}$ one has to consider simultaneously two equations (41) for the power p and phase ϕ of the auto-oscillation. In the limit of a weak external signal $F_e \to 0$ one can expand (41) in a Taylor series using the "driving" force amplitude F_e and the power deviation $\delta p = p - p_0 \sim F_e$ as small parameters. In the first order on these parameters one obtains

$$\frac{d\delta p}{dt} + 2\Gamma_p \delta p = 2F_e\sqrt{p_0}\cos(\omega_e t + \phi - \psi_e) \tag{49a}$$

$$\frac{d\phi}{dt} + \omega_g = -\frac{F_e}{\sqrt{p_0}}\sin(\omega_e t + \phi - \psi_e) - N\delta p \tag{49b}$$

where Γ_p and N are defined by (27b) and (31), respectively. In comparison with the case of a "linear" auto-oscillator, (49b) for the phase of a "nonlinear" auto-oscillator has an additional "forcing" term $-N\delta p$, which, for a strong nonlinearity $|N|$, can be substantially more important than the first "linear" term proportional to the "driving" force amplitude F_e.

One can simplify (49b) by introducing an effective phase $\widetilde{\phi}(t)$ of a nonlinear auto-oscillator

$$\widetilde{\phi} = \phi - \nu\frac{\delta p}{2p_0} \tag{50}$$

where ν is the normalized nonlinear frequency shift coefficient defined by (33). We note, that the effective "nonlinear" auto-oscillator phase $\widetilde{\phi}(t)$ defined by the renormalizing transformation (50) is a natural variable for the description of the phase dynamics of nonlinear auto-oscillators and, therefore, it plays a very important role in the theoretical description of the deterministic and stochastic dynamics of weakly coupled auto-oscillator arrays discussed below.

It is clear, that, since for a weak external force the power deviation δp always remain small, the effective phase $\widetilde{\phi}$ only slightly deviates from the "linear" oscillator phase ϕ. Using (49), one can derive dynamical equation for the renormalized "nonlinear" effective phase $\widetilde{\phi}$ in the following form:

$$\frac{d\widetilde{\phi}}{dt} + \omega_g = -\frac{\widetilde{F}_e}{\sqrt{p_0}}\sin\left(\omega_e t + \widetilde{\phi} - \widetilde{\psi}_e + \nu\frac{\delta p}{2p_0}\right) \tag{51}$$

where

$$\widetilde{F}_e = \sqrt{1+\nu^2}F_e, \qquad \widetilde{\psi}_e = \psi_e - \arctan(\nu). \tag{52}$$

Since the nonlinear phase term $\nu\delta p/2p_0$ is small (of the order of the small external signal amplitude F_e), one can neglect it in the right-hand-side part of (51). In this approximation (51) *exactly* coincides with the phase equation (43) for a "linear" auto-oscillator, but with the amplitude F_e and phase ψ_e of the external "driving" force renormalized according to (52)

$$\frac{d\widetilde{\phi}}{dt} + \omega_g = -\frac{\widetilde{F}_e}{\sqrt{p_0}}\sin(\omega_e t + \widetilde{\phi} - \widetilde{\psi}_e).$$

This fact allows us to easily generalize the previously derived results to the case of a "nonlinear" auto-oscillator.

In particular, the phase-locking bandwidth Δ of a "nonlinear" oscillator is given by

$$\Delta = \sqrt{1+\nu^2}\Delta_0 = \sqrt{1 + \left(\frac{N}{G_+ - G_-}\right)^2}\frac{F_e}{\sqrt{p_0}}. \tag{53}$$

One can see, that the nonlinearity of the oscillator frequency $\omega(p)$, characterized by the normalized nonlinear frequency shift coefficient ν (33), leads to the increase of the phase-locking bandwidth by the factor of $\sqrt{1+\nu^2}$. For conventional quasi-linear auto-oscillators, where $|\nu| \ll 1$, this increase is negligible and the "linear" Adler's theory of phase-locking [131] works very well. In contrast, spin-torque auto-oscillators are characterized by a substantial frequency nonlinearity $|\nu| \gg 1$, and the effective renormalization (52) of the external "driving" force should always be taken into account, as it qualitatively affects the phase-locking behavior of a nonlinear auto-oscillator.

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One may argue, that in a strongly nonlinear auto-oscillator $|\nu| \gg 1$ the phase-locking is achieved by a qualitatively different mechanism. Namely, in this case even small changes in the auto-oscillation power δp can result in matching of the nonlinear eigenfrequency $\omega(p) \approx \omega_{\rm g} + N\delta p$ of the auto-oscillator to the frequency of the external signal $\omega_{\rm e}$. Thus, the primary phase-locking mechanism of a strongly nonlinear auto-oscillator is a *nonlinear frequency matching* with the external signal, and this nonlinear mechanism is very different from the classical phase-locking mechanism.

The expression (53) for the phase-locking bandwidth of a nonlinear auto-oscillator was first derived in [98] for a particular case of a spin-torque oscillator. The experimental studies of the phase-locking in spin-torque oscillators were undertaken in [56], [132]. In both cases, the microwave modulation of the bias current was used to create the locking signal. It should be noted, that although this method of creation of the external locking signal is the simplest for the experimental realization, it is not the best one. In this case the external locking signal (40a) is proportional to the tangent of the angle γ_0 between the equilibrium orientations of the magnetization vectors in the "free" and "fixed" layers, which is usually small (of the order of several degrees) and difficult to estimate theoretically. Thus, the locking force (40a) has a small prefactor and, to achieve a reasonable amplitudes of the locking signal [and, consequently, of the phase-locking bandwidth (53)], one has to use relatively large magnitudes of microwave current ΔI (current modulation $\Delta I/I \sim 0.1 - 0.2$; see [132]). Note that in the experiments [56], [132] the microwave current, introduced as a phase-locking signal, was either comparable or larger than the microwave current generated by the experimental STO in the free-running regime. Thus, to study the fundamental properties of the phase-locking effect in a nonlinear STO it would have been much better to create the microwave modulation of the bias magnetic field, since the small prefactor of the external signal amplitude in not present in that case, and relatively small external signals could lead to the observable phase-locking effect.

The experiment [132] was performed on a normally-magnetized nano-pillar, for which the nonlinear frequency shift coefficient N is positive and much larger than the nonlinear damping coefficient $(G_+ - G_-)$. In this case of a strongly-nonlinear auto-oscillator the expression for the phase-locking bandwidth (53) can be simplified to

$$\Delta \approx |\nu| \frac{F_{\rm e}}{\sqrt{p_0}} = \frac{|N|}{G_+ - G_-} \frac{F_{\rm e}}{\sqrt{p_0}} \tag{54}$$

i.e., the locking bandwidth Δ is linearly proportional to the absolute value of the nonlinear frequency shift $|N|$. Experimental study [132] directly confirmed this prediction.

As it was pointed out above, the nonlinear frequency shift coefficient N for spin-torque oscillator strongly depends on the orientation of the bias magnetic field (see Fig. 6). Then, one can expect nontrivial dependence of the phase-locking bandwidth Δ on the magnetization angle of the "free" layer. In Fig. 11 we show the phase-locking bandwidth (53) of an STO as a function of the magnetization angle θ_0 (angle that the external bias magnetic field $\boldsymbol{H}_0$ makes with the plane of the "free" magnetic layer)

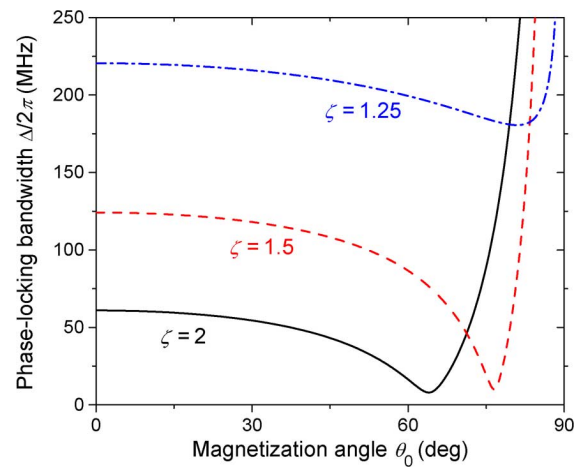

Fig. 11. Dependence (53) of the phase-locking bandwidth Δ of a spin-torque auto-oscillator on the out-of-plane magnetization angle θ_0 for different values of the supercriticality ζ (from [98]). The parameters of the spin-torque auto-oscillator are: saturation magnetization $4\pi M_0 = 9$ kG, bias magnetic field $H_0 = 7$ kOe, Gilbert damping parameter $\alpha_{\rm G} = 0.01$, nonlinear damping parameter $Q = 0$. The phase-locking bandwidth corresponds to the amplitude of microwave current $\Delta I = 1$ mA for misalignment angle $\gamma_0 = 5^\circ$ [see (40a)], or to the amplitude of the external microwave magnetic field $|\boldsymbol{h}| \approx 5.7$ Oe [see (40b)].

for typical parameters of a spin-torque auto-oscillator [98]. The phase-locking bandwidth Δ has a strong minimum for the magnetization angle around $\theta_0 \approx 80^\circ$. This minimum occurs at the "linear" magnetization angle, for which the nonlinear frequency shift coefficient vanishes $N = 0$. At this point STO behaves as a conventional "linear" oscillator, and its phase-locking bandwidth can be calculated using the classical equation (47). Far from the "linear" angle $\theta_0 \approx 80^\circ$, the phase-locking bandwidth Δ of an STO drastically increases due to the large values of the normalized nonlinear frequency shift coefficient ν.

The renormalization (52), also, allows one to easily generalize for the "nonlinear" case other results obtained in the forced dynamics of a "linear" auto-oscillator. For instance, outside the locking bandwidth $|\omega_{\rm e} - \omega_{\rm g}| > \Delta$ one observes the phenomenon of frequency pulling with the averaged oscillation frequency given by (48b), where the "linear" locking bandwidth Δ_0 [see (47)] should be replaced by the "nonlinear" one Δ [see (53)]

$$\omega_{\rm f} = \omega_{\rm e} + {\rm sign}(\omega_{\rm g} - \omega_{\rm e})\sqrt{(\omega_{\rm g} - \omega_{\rm e})^2 - \Delta^2}. \tag{55}$$

It is interesting, also, to analyze the behavior of the relative phase of the auto-oscillator $\Phi = \phi + \omega_{\rm e} t - \psi_{\rm e}$ (difference between the auto-oscillator phase ϕ and the phase $-\omega_{\rm e} t + \psi_{\rm e}$ of the external signal, see (44)). Within the locking bandwidth $|\delta| < \Delta$ this relative phase is independent of time $\Phi = \Phi_0 = {\rm const}$. Using the solution (46) for a linear auto-oscillator, renormalization of the external force (52), and neglecting the small nonlinear term $\nu\delta p/2p_0$ in (50), one obtains the relative phase Φ_0 for a nonlinear auto-oscillator

$$\begin{aligned}\Phi_0 &= \arcsin\left(\frac{\delta}{\Delta}\right) - \arctan(\nu) \\ &= \arcsin\left(\frac{\omega_{\rm e} - \omega_{\rm g}}{\Delta}\right) - \arctan\left(\frac{N}{G_+ - G_-}\right).\end{aligned} \tag{56}$$

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Thus, "nonlinear" auto-oscillator has an additional phase shift $-\arctan(\nu)$ which is not present in the "linear" case. For a strongly "nonlinear" oscillator ($|\nu| \gg 1$) this phase shift is equal to $\pm\pi/2$, depending on the sign of the nonlinear frequency shift coefficient N. For a spin-torque oscillator this additional phase shift has been discovered numerically in [100], [133] for the case of the exact frequency matching between the oscillator frequency with the frequency of an external signal ($\omega_{\rm e} = \omega_{\rm g}$). It is clear from the above presented analysis, that this additional nonlinear phase shift is not a specific feature of an STO, but is a generic property of a strongly "nonlinear" auto-oscillator of any nature.

B. Mutual Phase-Locking of Two Auto-Oscillators

One of the most important cases of phase-locking between oscillators is the *mutual phase-locking* of auto-oscillators. When two or more auto-oscillators weakly interact with each other, oscillations of each of them provide the external phase-locking signals for other oscillators. If the interaction between the oscillators is strong enough to overcome the differences in free-running frequencies of the individual oscillators, the whole system of oscillators starts to generate *coherently*, i.e., at exactly the same frequency and phase. The output power of coherent oscillations can be added coherently, i.e., this power increases as a square of the number of generating auto-oscillators (instead of a linear dependence on the oscillator's number that is obtained when noncoherent oscillations are combined). Besides the possibility of a significant increase of the output power, an array of mutually phase-locked auto-oscillators has a much smaller generation linewidth than a single oscillator in the same array, which is also a very important advantage from the practical point of view.

The increase of the output power and reduction of the generation linewidth, that can be achieved in a phase-locked array, are especially important for the nano-sized spin-torque auto-oscillators. The output power of a single spin-torque auto-oscillator is very small (of the order of about 0.1 nW for fully metallic spin valve structures and about 10 nW for dielectric MTJ structures) and, because of that, such oscillators are more vulnerable to the action of thermal fluctuations, than a conventional macro-sized auto-oscillator. Thus, the use of large (with number of oscillators of about 10^4 for metallic spin valve structures and about 10–100 for MTJ structures) phase-locked arrays of STO seems to be a natural way for the development of practical nano-scale active microwave devices.

The practical development of large phase-locked arrays of spin-torque oscillators requires a deep understanding of the basic physical mechanisms responsible for the phase-locking effect. Below we will consider in detail process of the mutual phase-locking in a pair of auto-oscillators—the simplest and the most physically transparent problem of the theory of mutual phase-locking, that can be solved analytically in the framework of the nonlinear oscillator model (5).

1) Generic Model of Mutual Phase-Locking: The dynamics of two weakly-coupled "nonlinear" oscillators can be described by a system of two coupled nonlinear equations for the complex amplitudes $c_j(t)$, $j = 1, 2$, of the oscillations in the jth oscillator:

$$\frac{dc_1}{dt} + i\omega_1\left(|c_1|^2\right)c_1 + \Gamma_{+,1}\left(|c_1|^2\right)c_1 - \Gamma_{-,1}\left(|c_1|^2\right)c_1 = \Omega_{1,2}e^{i\beta_{1,2}}c_2 \tag{57a}$$

$$\frac{dc_2}{dt} + i\omega_2\left(|c_2|^2\right)c_2 + \Gamma_{+,2}\left(|c_2|^2\right)c_2 - \Gamma_{-,2}\left(|c_2|^2\right)c_2 = \Omega_{2,1}e^{i\beta_{2,1}}c_1. \tag{57b}$$

Here, $\omega_j(p_j)$ and $\Gamma_{\pm,j}(p_j)$ are the frequencies and damping rates of the jth oscillator ($p_j = |c_j|^2$ is the power of oscillations in the jth oscillator), and $\Omega_{j,j'}$ and $\beta_{j,j'}$ are the (real) coupling frequencies and phases describing the interaction between the jth and j'th oscillators. For simplicity, we assumed that the coupling frequencies $\Omega_{j,j'}$ and phases $\beta_{j,j'}$ are independent of the oscillations amplitudes c_j, i.e., we take into account only the linear coupling between the oscillators.

The coupling frequencies $\Omega_{j,j'}$ describe the strength of the interaction between the oscillators. The coupling phases $\beta_{j,j'}$ depend on the physical nature of the coupling mechanism. For instance, if the coupling between the oscillators is conservative, then the interaction terms in the right-hand-side parts of (57) can be calculated as variational derivatives of a certain interaction Hamiltonian $\mathcal{H}_{\rm int}$, i.e.,

$$\left(\frac{dc_1}{dt}\right)_{\rm int} = -i\frac{\partial\mathcal{H}_{\rm int}}{\partial c_1^*}, \quad \left(\frac{dc_2}{dt}\right)_{\rm int} = -i\frac{\partial\mathcal{H}_{\rm int}}{\partial c_2^*}.$$

In such a case, the phases $\beta_{1,2}$ and $\beta_{2,1}$ are connected by the relation

$$\beta_{1,2} + \beta_{2,1} = -\pi$$

and, by a proper normalization of the complex amplitudes c_1 and c_2, the phases can be chosen to be equal to

$$\beta_{1,2} = \beta_{2,1} = -\pi/2.$$

In a more general case when the coupling mechanism is dissipative or mixed the phases $\beta_{j,j'}$ can be arbitrary. Such a situation takes place when there are several different coupling mechanisms that act simultaneously, and the complex coupling frequency $\Omega_{j,j'}e^{i\beta_{j,j'}}$ in (57) is a sum of partial complex frequencies, corresponding to each type of the coupling.

The coupling phases $\beta_{j,j'}$ can, also, account for the time delay of the coupling signal. Let us consider the situation when two oscillators are separated by a distance $a_{1,2}$ and coupling signals, carrying the information about the oscillation amplitude of a particular oscillator, can travel with the velocity $v_{1,2}$. Obviously, in such a situation each oscillator gets the coupling signal from the other one with the time delay $\tau_{1,2} = a_{1,2}/v_{1,2}$. Thus, to describe this delayed coupling one should write a *time-delayed* complex amplitude of, e.g., the second oscillator $c_2(t - \tau_{1,2})$ in the right-hand-side part of (57a) for the first oscillator. If the time delay is sufficiently small (namely, if $\Omega_{2,1}\tau_{1,2} \ll 1$), one can neglect the influence of coupling during the time interval

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$\tau_{1,2}$ and approximate the delayed amplitude of the second oscillator $c_2(t-\tau_{1,2})$ as

$$c_2(t-\tau_{1,2}) \approx e^{i\omega_{g,2}\tau_{1,2}} c_2(t)$$

where $\omega_{g,2}$ is the free-running generation frequency of the second oscillator. Thus, the delay of the coupling signal can be approximately taken into account by the renormalization of the coupling phases $\beta_{j,j'}$

$$\beta_{j,j'} \to \beta_{j,j'} + \omega_{g,j'}\tau_{j,j'}.$$

The system of coupled equations (57) for the particular case of spin-torque oscillators was first derived in [102] without account of the time delay ($\beta_{j,j'} = 0$), and later a general derivation of this system was given in [134]. We would like to stress, that the system of coupled oscillator equations (57) is absolutely generic, and can be used for the description of mutual phase-locking of oscillators of any nature. The specific features of STO are hidden in the coupling amplitudes $\Omega_{j,j'}$ and phases $\beta_{j,j'}$, and in the dependence of these quantities on such system parameters as the distance between the oscillators, orientation and magnitude of the bias magnetic field, etc.

The coupling mechanisms specific for the interaction of spin-torque oscillators will be considered in the following subsection.

2) Mechanisms of Coupling of Spin-Torque Oscillators: The coupling between the spin-torque oscillators can be caused by several different physical mechanisms. The simplest one is the *coupling through the common bias current*. When two (or more) spin-torque oscillators are connected, for instance, in parallel and feeded by a source of constant current, the magnetization precession in each oscillator will cause changes of the resistance of this oscillator due to giant magnetoresistance (GMR) effect. This change, in its turn, will alter the current flowing through all the other oscillators. In a different topology, when spin-torque oscillators are connected in series and feeded by a voltage source, the precession-induced variations in oscillators' electrical resistance will change the common current flowing through all the oscillators. Such a coupling mechanism between spin-torque oscillators has been recently considered theoretically in [104]. In both cases, however, the coupling signal is proportional to the variations of the bias current ΔI; see (40a); and, therefore, is proportional to the small GMR ratio of metallic spin-valve structures. Due to the smallness of the coupling signal created by this mechanism and its dependence on the electrical scheme (e.g., parallel or serial connection of oscillators) we will not consider this coupling mechanism explicitly. Nevertheless, we would like to note, that the coupling through the common current might be much more efficient in spin-torque oscillators based on magnetic tunnel junctions (MTJ), because of a much larger value of the magnetoresistance ratio created by TMR effect in MTJ structures.

Another physical mechanism of mutual coupling of spin-torque oscillators is the *coupling through the common magneto-dipolar field*, created by the precessing magnetization. This mechanism of coupling can be important for STO using both nano-pillar and nano-contact geometries (see Fig. 1). The coupling frequency $\Omega_{\rm MD}$ due to the magneto-dipolar coupling has the form [102]

$$\Omega_{\rm MD} = \frac{V_{\rm eff}}{a^3}\frac{\omega_M^2}{\omega_{\rm g}} f(\boldsymbol{H}_0). \tag{58}$$

Here, both interacting spin-torque oscillators were assumed to be identical, and $V_{\rm eff}$ is the effective volume of each of the oscillators, a is the distance between the oscillators, $\omega_M = 4\pi\gamma M_0$, $\omega_{\rm g}$ is the generated frequency, and $f(\boldsymbol{H}_0)$ is the dimensionless function of the order of unity that depends on the amplitude and orientation of the bias magnetic field $\boldsymbol{H}_0$ (see [102] for details). The coupling frequency $\Omega_{\rm MD}$ decreases with the increase of the oscillator separation a as $1/a^3$, i.e., in the same way as the magnetic field of a point magnetic dipole.

The effective volume $V_{\rm eff}$ in (58) is equal to the nominal volume of a "free" magnetic layer in the case of a spin-torque oscillator based on a magnetic nano-pillar [see Fig. 1(b)]. In the case of an oscillator based on a nano-contact (see Fig. 1(a)) of the radius $R_{\rm c}$ made in the "free" magnetic layer of the thickness L the effective volume $V_{\rm eff}$ involved in oscillation is larger than the volume $V = \pi R_{\rm c}^2 L$ of the magnetic material in the contact area, because the excited spin wave mode is not strictly localized within the geometric sizes of the nano-contact. The effective volume $V_{\rm eff}$ depends on the spatial profile of the excited spin wave mode and, in the case of the excitation of a propagating Slonczewski's mode [90] in a perpendicularly magnetized film, it is given by the following expression derived in [102]:

$$V_{\rm eff} \approx \left(1 + 1.86\frac{\omega_M}{\Gamma_{\rm G}}\lambda_{\rm ex}^2 R_{\rm c}^2\right) V \tag{59}$$

where $\Gamma_{\rm G}$ is the ferromagnetic resonance (FMR) half-linewidth and $\lambda_{\rm ex}$ is the exchange length. The first term in the bracket of (59) describes the volume of magnetic material within the nano-contact area, while the second term describes the volume occupied by the excited spin wave, which propagates away from the current-driven nano-contact.

The magneto-dipolar coupling mechanism is, clearly, conservative. Therefore, the coupling phases $\beta_{\rm MD}$ are equal to

$$\beta_{\rm MD} = -\frac{\pi}{2}.$$

In the case of spin-torque oscillators based on magnetic nano-contacts [see Fig. 1(a)] there is another important type of oscillators' coupling, namely, the *coupling through the radiation of propagating spin waves*. Indeed, the spin waves, excited by one of the nano-contacts, propagate towards the other nano-contact. This leads to an effective coupling between the two nano-contact oscillators. The coupling frequency $\Omega_{\rm SW}$, corresponding to this type of interaction, was calculated in [102]

$$\Omega_{\rm SW} \approx 0.65\Gamma_{\rm G}\sqrt{\frac{R_{\rm c}}{a}}\exp(-\Gamma_{\rm G} a/v_{\rm gr}). \tag{60}$$

Here, the excited spin wave modes are assumed to be the propagating Slonczewski's modes [90], and the radii of the two interacting nano-contacts are assumed to be equal to $R_{\rm c}$. In (60), $v_{\rm gr} \approx 1.7\omega_M\lambda_{\rm ex}^2/R_{\rm c}$ is the group velocity of the excited propagating spin wave mode. The factor $1/\sqrt{a}$ in (60)

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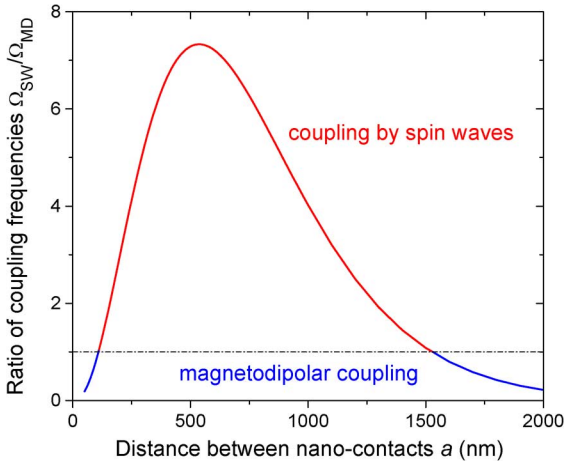

Fig. 12. Dependence of the ratio of the coupling frequencies $\Omega_{\rm SW}$ (60), caused by propagating spin waves, and $\Omega_{\rm MD}$ (58), caused by the direct magneto-dipolar interaction, on the distance a between the spin-torque nano-contact oscillators (from [102]).

describes geometrical $\sim 1/\sqrt{r}$ decrease of the circular propagating spin wave in a two-dimensional geometry, while the factor $\exp(-\Gamma_{\rm G} a/v_{\rm gr})$ describes exponential decay of the spin wave amplitude due to the energy dissipation in the magnetic material of the "free" layer.

Since the velocity of the propagating spin waves is finite, this interaction mechanism involves a certain time delay of the coupling signal. As it was explained above, this time delay can be accounted for by the coupling phase $\beta_{\rm SW}$ [134]

$$\beta_{\rm SW} = k_{\rm c} a \tag{61}$$

where $k_{\rm c} \approx 1.2/R_{\rm c}$ is the wave number of the excited propagating spin wave. Thus, in this coupling mechanism, the change in the distance a between the nano-contact oscillators leads not only to the change in the coupling frequency $\Omega_{\rm SW}$, but, also, to the change in the coupling phase $\beta_{\rm SW}$. As it will be shown below, this effect significantly modifies the process of mutual phase-locking of two spin-torque oscillators based on magnetic nano-contacts.

The ratio of the coupling frequencies caused by the spin wave (60) and magneto-dipole (58) coupling mechanisms calculated in [102] as a function of the contact separation a for typical parameters of nano-contact spin-torque oscillators is shown in Fig. 12. It is clear from this figure, that the coupling through the propagating spin waves $\Omega_{\rm SW}$ is dominant ($\Omega_{\rm SW}/\Omega_{\rm MD} > 1$) in all practically interesting range of distances between the nano-contacts 100 nm $\leq a \leq 1.5\ \mu$m. For larger nano-contact separations $a > 1.5\ \mu$m both coupling mechanisms are too weak to provide a reasonable coupling strength, while for smaller separations $a < 100$ nm the interacting nano-contacts are too close to each other to be considered as weakly interacting and independent of each other.

Thus, the coupling through the propagating spin waves is the dominant mechanism of coupling of spin-torque oscillators based on magnetic nano-contacts. This theoretical conclusion, obtained in [102], was later directly confirmed experimentally in [59]. The authors of [59] managed to cut the free magnetic layer between the two nano-contacts, thus preventing the spin waves from propagating between the nano-contacts. At the same time, this cut did not affect the direct magneto-dipolar coupling between the nano-contacts. In was clearly demonstrated in [59] that the cut of the "free" layer totally destroyed the phase-locking between the nano-contacts, which indicates that propagating spin waves rather than dipolar magnetic fields provide the main part of coupling between the nano-contact STOs.

Strictly speaking, any mechanism of interaction between the spin-torque oscillators is nonlinear, i.e., in general, the coupling frequencies $\Omega_{j,j'}$ and phases $\beta_{j,j'}$ should depend on the power generated by the interacting auto-oscillators. In a majority of cases, however, the nonlinear corrections are small and can be ignored. As it will be shown in Section VI-B5, the *linear* model (57) of coupling between the oscillators gives the values of the mutual phase-locking bandwidth for two nano-contact STOs that quantitatively coincide with the experimentally observed ones.

A different coupling model, that, in contrast with (57), takes into account *only the nonlinear* coupling between the spin-torque oscillators was proposed in [103]. In that model the strength of interaction between the oscillators was phenomenologically described by a dimensionless coupling constant λ, which has the meaning of the relative amplitude of the oscillations excited by the first nano-contact at the center of the second nano-contact. The nonlinear coupling model proposed in [103] predicts, that the phase-locked regime should start at the critical coupling value $\lambda \approx 0.66$ (see [103, Fig. 2]). It is easy to check that, if the mode similar to the propagating Slonczewski's mode [90] is excited in the system, the coupling constant $\lambda \approx 0.66$ is achieved when the distance a between the centers of the two interacting nano-contacts is smaller than the diameter of each of the nano-contacts (i.e., the nano-contacts are overlapping). Thus, the purely nonlinear mechanism of nano-contact interaction suggested in [103] can not explain experimentally observed mutual phase-locking of spin-torque oscillators [57], [58] separated by the distance of up to $a = 500$ nm (or $a > 20R_{\rm c}$), and the nonlinear corrections to the coupling are much weaker than the linear coupling terms taken into account in the model (57).

3) Phase Equations for Coupled Oscillators: To find the conditions of mutual phase-locking of two oscillators we can analyze system of (57) using the same methods as in the case of phase-locking to an external signal (see Section VI-A). Indeed, the coupling terms in the right-hand-side part of (57) for jth oscillator can be considered as weak external forces

$$f_j(t) = \Omega_{j,j'} e^{i\beta_{j,j'}} c_{j'}(t)$$

with the (real) amplitude

$$F_{{\rm e},j} = \Omega_{j,j'} \sqrt{p_{j'}}$$

and phase

$$\psi_{{\rm e},j} = \beta_{j,j'} + \phi_{j'}$$

where $p_{j'} = |c_{j'}|^2$ and $\phi_{j'} = \arg(c_{j'})$ are the power and phase of the oscillation in the j'th oscillator, respectively.

Assuming that the power deviations $\delta p_j = p_j - p_{0,j}$ (where $p_{0,j}$ is the free-running power of jth oscillator) are small and

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using the phase renormalization (50) for each oscillator, one obtains the equations for the renormalized oscillation phases $\widetilde{\phi}_j$, similar to (51). Using the explicit form of the amplitude $F_{\mathrm{e},j}$ and phase $\psi_{\mathrm{e},j}$ of the force acting on the jth oscillator, these equations can be written as

$$\frac{d\widetilde{\phi}_1}{dt} + \omega_{\mathrm{g},1} = \widetilde{\Omega}_{1,2}\sin(\widetilde{\phi}_2 - \widetilde{\phi}_1 + \widetilde{\beta}_{1,2}) \tag{62a}$$

$$\frac{d\widetilde{\phi}_2}{dt} + \omega_{\mathrm{g},2} = \widetilde{\Omega}_{2,1}\sin(\widetilde{\phi}_1 - \widetilde{\phi}_2 + \widetilde{\beta}_{2,1}) \tag{62b}$$

where $\omega_{\mathrm{g},j} = \omega_j(p_{0,j})$ is the free-running generation frequency of the jth oscillator and the renormalized coupling frequencies $\widetilde{\Omega}_{j,j'}$ and coupling phases $\widetilde{\beta}_{j,j'}$ have the form

$$\widetilde{\Omega}_{j,j'} = \sqrt{1+\nu_j^2}\sqrt{\frac{p_{0,j'}}{p_{0,j}}}\Omega_{j,j'} \tag{63a}$$

$$\widetilde{\beta}_{j,j'} = \beta_{j,j'} - \arctan(\nu_j). \tag{63b}$$

Here, ν_j is the normalized nonlinear frequency shift coefficient (33) for jth oscillator. Equations (63) defining the renormalized coupling frequencies take into account both the nonlinear properties of oscillators (through the terms with ν_j) and the difference in the operational regime of oscillators, i.e., the fact that the free-running powers $p_{0,j}$ of oscillators could be different (through the factor $\sqrt{p_{0,j'}/p_{0,j}}$ in (63a)).

Equations (62) for the oscillator phases, describing mutual phase-locking of two nonlinear oscillators, allow one to perform a complete analysis of the process of phase-locking in a general case. For instance, using these equations it is possible to find analytical conditions for the existence of a phase-locked regime, analytical expressions for the phase difference of two oscillators $(\widetilde{\phi}_1 - \widetilde{\phi}_2)$, the generated frequency $-d\widetilde{\phi}_1/dt = -d\widetilde{\phi}_2/dt$ in the phase-locked regime, and, also, to describe analytically the regime of mutual frequency pulling. For the sake of mathematical simplicity, here we will consider only the case of almost identical oscillators (see next subsection for exact definition). Some results about the mutual phase-locking of two nonidentical oscillators can be found in [102] (see, also, Fig. 15).

4) Phase-Locking of Almost Identical Oscillators: The main parameter that controls the phase-locking behavior of two oscillators is the difference δ between their free-running frequencies

$$\delta = \omega_{\mathrm{g},2} - \omega_{\mathrm{g},1} = \omega_2(p_{0,2}) - \omega_1(p_{0,1}). \tag{64}$$

In this subsection we will consider a simple, but still nontrivial, case of phase-locking of two "almost identical" oscillators, i.e., oscillators, that have all the parameters, except the free-running frequencies $\omega_{\mathrm{g},j}$, identical

$$p_{0,j} = p_0, \qquad \nu_j = \nu,$$
$$\Omega_{j,j'} = \Omega, \qquad \beta_{j,j'} = \beta.$$

Introducing the average oscillator phase

$$\Phi = \frac{\widetilde{\phi}_1 + \widetilde{\phi}_2}{2}$$

and the oscillators' phase difference

$$\Psi = \widetilde{\phi}_1 - \widetilde{\phi}_2$$

we can rewrite system (62) as

$$\frac{d\Phi}{dt} = -\overline{\omega} + \Omega\sqrt{1+\nu^2}\sin(\beta - \arctan\nu)\cos\Psi \tag{65a}$$

$$\frac{d\Psi}{dt} = \delta - 2\Omega\sqrt{1+\nu^2}\cos(\beta - \arctan\nu)\sin\Psi \tag{65b}$$

where

$$\overline{\omega} = \frac{\omega_{\mathrm{g},1} + \omega_{\mathrm{g},2}}{2}$$

is the average free-running frequency of two oscillators.

In the phase-locked regime the phase difference Ψ does not depend on time and is constant $\Psi(t) = \Psi_0 = \mathrm{const}$. Since $|\sin\Psi_0| \leq 1$, it is clear from (65b) that the mutual phase-locking is possible only for

$$|\delta| \leq \Delta_2$$

where Δ_2 is the mutual phase-locking bandwidth for two oscillators [134]

$$\Delta_2 = 2\Omega\sqrt{1+\nu^2}\,|\cos(\beta - \arctan\nu)|\,. \tag{66}$$

One can see that the nonlinearity of the oscillators ν and the coupling phase (or delay of the coupling signal) β significantly modify the phase-locking bandwidth. For a given magnitude of Ω and ν, the maximum bandwidth

$$\Delta_{2,\max} = 2\Omega\sqrt{1+\nu^2} \tag{67}$$

is achieved for the optimum coupling phases defined by

$$\beta_{\mathrm{opt}} = \arctan\nu + n\pi \tag{68a}$$

where n is an integer number. On the other hand, for the coupling phases given by

$$\beta_* = \arctan\nu + \left(n + \frac{1}{2}\right)\pi \tag{68b}$$

the locking bandwidth Δ_2 is zero, i.e., the mutual phase-locking is not possible at all. If the interacting oscillators are not identical (e.g., $\Omega_{1,2} \neq \Omega_{2,1}$), than, for the anti-optimum coupling phases (68b), the locking bandwidth will not be zero, but will have a local minimum.

For nano-contact spin-torque oscillators coupled by propagating spin waves, the coupling phase β linearly depends on the distance a between the contacts (see (61)). This means, that the dependence of the locking bandwidth Δ_2 on the distance a will not be a monotonously decreasing function (as one can expect from the analysis of the coupling frequency (60)), but will have local maxima and minima determined by the conditions (68a) and (68b), respectively. This fact is illustrated by Fig. 13 [134], where the mutual phase-locking bandwidth Δ_2 is shown as a function of the distance a between the oscillators (see solid line in Fig. 13) for the typical parameters of magnetic nano-contacts (see caption to Fig. 13). The dashed line in Fig. 13 shows the

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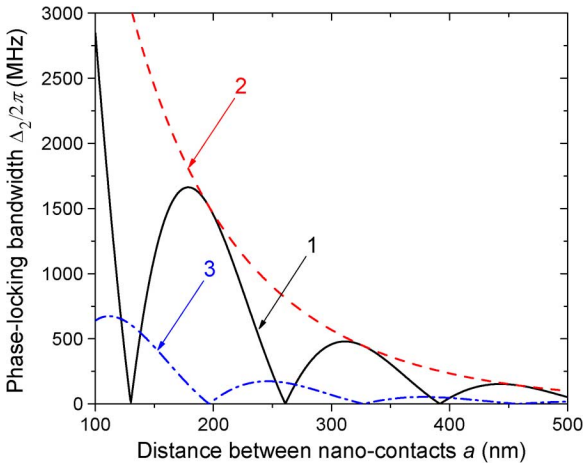

Fig. 13. Dependence of the phase-locking bandwidth Δ_2 of two spin-torque oscillators on the distance a between the oscillators: 1—bandwidth Δ_2 (66) with account of time delay β and frequency nonlinearity ν, 2—maximum possible bandwidth $\Delta_{2,\max}$ (67), 3—bandwidth $\Delta_{2,\mathrm{lin}}$ (69) for linear oscillators $\nu = 0$, multiplied by 10. Parameters of spin-torque oscillators: normally magnetized (magnetic field $H_0 = 15$ kOe) circular nano-contacts of the radius $R = 50$ nm, saturation magnetization $4\pi M_0 = 8$ kOe, Gilbert damping constant $\alpha_{\mathrm{G}} = 0.01$, nonlinear damping parameter $Q = 3$, exchange length $\lambda_{\mathrm{ex}} = 5$ nm, supercriticality parameter $\zeta = 1.5$.

maximum locking bandwidth $\Delta_{2,\max}$ (67), which would be expected for the optimum coupling phase β. Dash-dotted line in Fig. 13 shows the multiplied by 10 "linear" locking bandwidth

$$\Delta_{2,\mathrm{lin}} = 2\Omega|\cos\beta| \tag{69}$$

i.e., the locking bandwidth obtained without account of the nonlinear frequency shift ($\nu = 0$). One can see that the nonlinearity of spin-torque oscillators not only significantly increases the bandwidth of mutual phase-locking (in the same way as it increased the bandwidth for the locking to an external signal, see Section VI-A), but, also, shifts the positions of the optimum distances between the nano-contacts.

The importance of the time delay of a coupling signal for the problem of mutual phase-locking of two spin-torque oscillators was first pointed out in [105]. By numerically simulating (in a macrospin approximation) the phase-locking of a pair of oscillators coupled by a common bias current, the authors of [105] noted, that the introduction of a small artificial time delay drastically increases the range of possible difference in the oscillators' parameters, for which the phase-locking could take place. The optimum delay time was approximately equal to the quarter of the period of the magnetization precession in the oscillator, i.e., the optimum coupling phase was found to be equal to $\beta_{\mathrm{opt}} \approx \pi/2$. Taking into account that the spin-torque oscillators studied in [105] operated in a strongly nonlinear regime ($|\nu| \gg 1$, $|\arctan\nu| \approx \pi/2$), this numerical result directly follows from our analytical expression (68a).

Equations (65), also, allow one to obtain, besides the explicit expression for the phase-locking bandwidth (66), all the other characteristics of the phase-locking regime. For instance, the stationary phase difference Ψ_0 between the two phase-locked oscillators is given by

$$\Psi_0 = \begin{cases} \arcsin(\delta/\Delta_2), & \cos(\beta - \arctan\nu) > 0 \\ \pi - \arcsin(\delta/\Delta_2), & \cos(\beta - \arctan\nu) < 0 \end{cases}. \tag{70}$$

Although both solutions (70) for the stationary phase are possible for any value of β, only the combinations presented above are stable and, therefore, describe real experimentally observable phase-locking.

The stationary phase shift Ψ_0 should be taken into account when one calculates the total power generated by two oscillators

$$|c_1 + c_2|^2 = 4p_0\cos^2\left(\frac{\Psi_0}{2}\right).$$

It is clear, that this power varies between the "coherent" value $4p_0$, corresponding to the exact frequency match $\delta = 0$ ($\Psi_0 = 0$), to the "incoherent" value $2p_0$, which is achieved at the edge of the locking zone $\delta = \pm\Delta_2$ [$\Psi_0 = \pm\pi/2$; see (70)].

The expression for the frequency $\omega_1 = -d\Phi/dt$, generated by both oscillators in the phase-locked regime, directly follows from (65a) and (70):

$$\omega_1 = \overline{\omega} - \sqrt{\Delta_2^2 - \delta^2}\tan(\beta - \arctan\nu).$$

It is interesting to note, that the frequency generated in the phase-locked regime *does not* coincide with the average free-running frequency $\overline{\omega}$ even in the considered case of almost identical oscillators.

5) Comparison With Experiments: The detailed experimental study of the mutual phase-locking of two spin-torque oscillators have not been done yet, and many results of the above presented theory of phase-locking require experimental confirmation. In particular, it would be very interesting to check experimentally the dependence of the phase-locking bandwidth Δ_2 on the delay time of the coupling signal (e.g., by changing the distance between the nano-contact oscillators, or by introducing an artificial delay for electrically coupled oscillators) and on the level of frequency nonlinearity ν of the oscillators (e.g., by changing the orientation of the bias magnetic field).

Nevertheless, the first experimental studies of mutual phase-locking of a pair of spin-torque oscillators, performed in [57], [58], demonstrated the possibility of achieving the regime of phase-locked generation in a relatively large frequency bandwidth. Both experiments were performed on STOs based on magnetic nano-contacts magnetized almost perpendicularly to the plane of the "free" layer, when the normalized nonlinear frequency shift is positive and large $\nu \gg 1$. In such a case, according to (66), the phase-locking bandwidth is large, because it contains the large factor $\sqrt{1+\nu^2}$.

In [57] the individual characteristics of both nano-contact oscillators were approximately the same, and the theoretical approach developed in Section VI-B4 can be directly applied in this case. Using independent current biasing of two spin-torque oscillators, the authors of [57] were able to continuously change the free-running frequency of one oscillator, while keeping the generation frequency of the other one almost constant. The phase-locking bandwidth measured experimentally in this case

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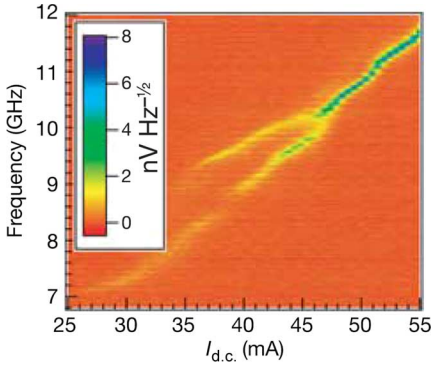

Fig. 14. Color map of the spectrum amplitude versus frequency and bias current for two nano-contact spin-torque oscillators [from [58], Fig. 4(a)]. Nominal radii of nano-contacts $R_{c,\mathrm{nom}} = 40$ nm, distance between nano-contacts $a = 150$ nm.

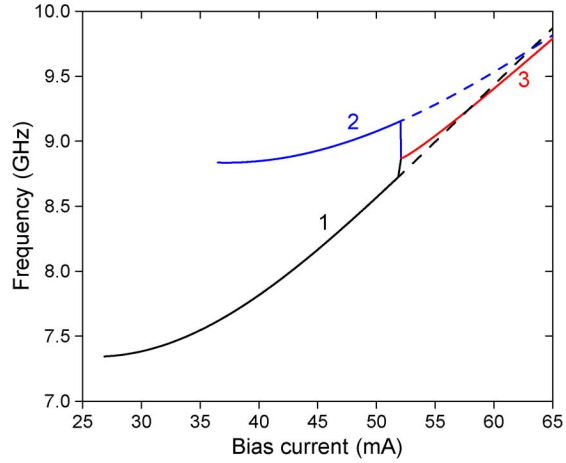

Fig. 15. Theoretical dependence of the frequencies, generated by two nano-contact oscillators, on the bias current (from [102], Fig. 3). Solid lines: 1—first nanocontact, 2—second nano-contact, 3—common generation frequency in the phase-locked regime. Dotted lines show the generated frequencies in the absence of coupling.

was around $2\Delta_2/2\pi \approx 200$ MHz (see Fig. 2a in [57]). The theoretical analysis [102] of mutual phase-locking, based on the nonlinear oscillator model (57) without account of time delay, gives, for the experimental parameters of [57], the magnitude of the phase-locking bandwidth of $2\Delta_2/2\pi \approx 160$ MHz (see Fig. 2 in [102]). We believe that, taking into account the uncertainty of the experimental parameters, this is a rather good estimation of the observed phase-locking bandwidth.

Fig. 14 shows an example of the mutual phase-locking of two spin-torque oscillators, observed in [58] (see Fig. 4a in [58]). In this experiment, STOs were connected in parallel and biased by a common current. One can see, that for small currents (<47 mA) two nano-contacts generate independently at rather different frequencies. Note, also, that in spite of the fact that the two oscillators studied in [58] were considered to be nominally identical, all the major parameters (threshold current, frequency, generated at the threshold, frequency-current slope, etc.) of them are quite different. With the increase of the bias current the difference between the two generated frequencies becomes smaller (due to the different frequency-current slopes for two oscillators). The transition to the phase-locked regime occurs at $I = 47$ mA, when the difference between the individual free-running frequencies was $\delta/2\pi = \Delta_2/2\pi \approx 400$ MHz.

The theoretical analysis of the experiment Fig. 14 was performed in [102] using the coupled nonlinear oscillator model (57) without account of the time delay. Since the experimentally observed characteristics of two nano-oscillators were substantially different, the analysis was performed using the full model (57) for nonidentical oscillators (see [102] for details). The result of the theoretical analysis is shown in Fig. 15. It is clear, that the simple oscillator theory gives a reasonably good qualitative description of the experiment (compare Fig. 14 and Fig. 15). Note, also, that the theoretical phase-locking bandwidth $\Delta_2/2\pi \approx 400$ MHz (see Fig. 15) *quantitatively* coincides with the experimentally observed one (see Fig. 14).

C. Mutual Phase-Locking in a Large Array of Auto-Oscillators

Since the output power of a single nano-scale spin-torque oscillator based on a GMR spin valve is very low (of the order of 0.1 nW), practical realization of a microwave oscillator based on the spin-torque effect requires coherent phase-locked operation of a large number (of the order of $10^2 - 10^4$) of individual nano-sized oscillators. Although output power of spin-torque oscillators based on dielectric MTJ is much larger (of the order of 10 nW), it is still unsufficient for connection to conventional electronics and phase-locking of about 10–100 MTJ spin-torque oscillators is required. It is impossible to model such a large magnetic system using full-scale micromagnetic simulations. At most, currently existing computational capabilities allow one to simulate phase-locking in a system of two oscillators [106].

The nonlinear oscillator model (5) provides a computationally effective way of simulating mutual phase-locking in a large array of strongly nonlinear oscillators. The generalization of the coupled equations (57) for the case of a large number of weakly coupled oscillators has the form

$$\frac{dc_j}{dt} + i\omega_j(p_j)c_j + \Gamma_{+,j}(p_j)c_j - \Gamma_{-,j}(p_j)c_j = \sum_{j'} \Omega_{j,j'} e^{i\beta_{j,j'}} c_{j'}. \quad (71)$$

Here, indices j and j' enumerate oscillators in a coupled array, c_j and $p_j = |c_j|^2$ are the complex amplitude and power of the oscillation in the jth oscillator, respectively, and all the other terms have the same meaning as in (57).

The coupling frequencies $\Omega_{j,j'}$ and phases $\beta_{j,j'}$ depend on the distance $a_{j,j'}$ between the jth and j'th oscillators, as well as on physical characteristics of these oscillators (e.g., effective volume of the oscillators), and on the type of the coupling mechanism. By choosing the proper form of $\Omega_{j,j'}$ and $\beta_{j,j'}$, an arbitrary geometry of an array of oscillators can be modeled using (71).

It is, also, possible to take into account the inevitable randomness in the distribution of physical parameters of different oscillators. For instance, in the case of spin-torque oscillators, the random distribution of anisotropy fields (caused, e.g., by a polycrystalline structure of the "free" magnetic layer) can be

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modeled by an additive random frequency shift $\delta\omega_j$ for each oscillator:

$$\omega_j(p_j) \to \omega_j(p_j) + \delta\omega_j.$$

In the same way, the distribution of radii of nano-contact STOs can be accounted for by a similar renormalization of the effective negative damping $\Gamma_{-,j}(p_j)$ and the coupling frequencies $\Omega_{j,j'}$.

In the case when the coupling frequencies $\Omega_{j,j'}$ are sufficiently small, one can further simplify system (71) by using the technique of the "effective oscillator phase," which was used before for the description of the phase-locking of a "nonlinear" oscillator to an external signal (see Section VI-A2) and mutual phase-locking of two oscillators (see Section VI-B3). Namely, introducing the effective oscillator phases

$$\widetilde{\phi}_j = \phi_j - \nu_j \frac{(p_j - p_{0,j})}{2p_{0,j}} \tag{72}$$

where ν_j and $p_{0,j}$ are the normalized nonlinear frequency shift (see (33)) and free-running power of the jth oscillator, respectively, one can reduce (71) to a closed-form system of equations for the effective phases $\widetilde{\phi}_j$

$$\frac{d\widetilde{\phi}_j}{dt} + \omega_{\mathrm{g},j} = \sum_{j'} \widetilde{\Omega}_{j,j'} \sin\left(\widetilde{\phi}_{j'} - \widetilde{\phi}_j + \widetilde{\beta}_{j,j'}\right) \tag{73}$$

where $\omega_{\mathrm{g},j} = \omega_j(p_{0,j})$ is the free-running frequency of the jth oscillator and the renormalized coupling frequencies $\widetilde{\Omega}_{j,j'}$ and phases $\widetilde{\beta}_{j,j'}$ are given by (63).

The phase equations (73) describe dynamics of each oscillator using only one real variable—effective phase $\widetilde{\phi}_j$, i.e., they give the simplest possible description of the dynamics of coupled oscillators.

One may note that (73) for $\widetilde{\beta}_{j,j'} = 0$ coincides with the classical Kuramoto model of oscillators' coupling [135]. The generalizations of the Kuramoto model for the case of nonzero coupling phases $\widetilde{\beta}_{j,j'} \neq 0$ have been, also, investigated previously (see, e.g., [136], [137]), and these classical results can be used for the description of the dynamics of coupled STO arrays in the phase model (73).

The dynamical models of weakly coupled arrays of oscillators (71) and (73) do not only provide an effective framework for numerical study of mutual phase-locking and similar phenomena in a large arrays of oscillators, but, also, they allow one to obtain a number of interesting and useful analytical results [135].

It should be noted that the dynamical behavior of a large number of weakly coupled oscillators is much richer than the dynamics of two coupled oscillators. Besides the *state of global synchronization*, when all the oscillators in an array generate coherently at the same frequency, and *incoherent state*, when all the oscillators generate at their own frequencies, there is a number of different intermediate states. For instance, a certain part of oscillators can generate coherently, while all the others can continue to generate at their own frequencies, i.e., a so-called *partially synchronized state* can be achieved. Another possibility is the state when several coherent clusters with different generation frequencies exist simultaneously in an array. In general, the nonzero coupling phases ($\widetilde{\beta}_{j,j'} \neq 0$) could introduce *frustration* into the system of oscillators [135]. In the case of frustration the reference oscillator configuration, which makes the coupling terms in (73) vanish, is incoherent, i.e., $\widetilde{\phi}_j \neq \widetilde{\phi}_{j'}$. Also, for certain values of the coupling phases $\widetilde{\beta}_{j,j'}$ the interaction can lead to the *repulsion of oscillators*, i.e., the globally synchronized state can not be reached even for infinitely large coupling frequencies $\widetilde{\Omega}_{j,j'}$.

VII. Stochastic Dynamics of Oscillators

Thermal fluctuations and other noise sources play an important role in the dynamics of conventional auto-oscillators. Thermal fluctuations (or noise) determine one of the most important, from the practical point of view, parameters of the auto-oscillator—generation linewidth $\Delta\omega$. The influence of thermal noise is even more important for the nano-scale spin-torque oscillators. Due to the extremely small sizes of STO, the characteristic operational energy of these oscillators can be comparable with the thermal energy $k_{\mathrm{B}}T$ (where k_{B} is the Boltzmann constant and T is the thermodynamic temperature). As a result, not only the generation linewidth, but all the other characteristics of spin-torque oscillators can be dependent on the temperature. In particular, thermal fluctuations lead to the blurring of the generation threshold in STO and to a finite level of power fluctuations in the above-threshold operational regime, in which the influence of thermal fluctuations is, usually, negligible for conventional macro-sized oscillators.

In this section, we describe how the influence of noise can be accounted for in the nonlinear oscillator model (Section VII-A) and consider several important noise-related issues, namely, the influence of noise on the generation power of an oscillator (Section VII-B) and the broadening of the oscillator linewidth under the action of thermal fluctuations (Section VII-C).

A. Stochastic Nonlinear Oscillator Model

The stochastic dynamics of an auto-oscillator can be described in the framework of the nonlinear oscillator model (5) with external force $f_{\mathrm{n}}(t)$ representing the action of thermal fluctuations

$$\frac{dc}{dt} + i\omega\left(|c|^2\right)c + \Gamma_+\left(|c|^2\right)c - \Gamma_-\left(|c|^2\right)c = f_{\mathrm{n}}(t). \tag{74}$$

For any particular oscillator scheme (74) can be derived from the stochastic "material" equations. For instance, to obtain (74) for the Van der Pol oscillator (see Section IV-A) one has to add to the electrical scheme shown in Fig. 2(a) an additional stochastic voltage source that describes thermal Johnson-Nyquist noise connected with the resistor R. For a spin-torque oscillator stochastic equation (74) can be derived from the Landau-Lifshitz-Gilbert-Slonczewski equation (1) with an additional random thermal magnetic field.

For an auto-oscillator of an arbitrary nature, stochastic force $f_{\mathrm{n}}(t)$ in (74) can be considered as a phenomenological term describing action of thermal fluctuations. In such a case, the statistical properties of $f_{\mathrm{n}}(t)$ should be determined from the condition of a proper thermodynamical behavior of the oscillator in the state of thermal equilibrium (i.e., when the negative

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damping is vanishing $\Gamma_-(|c|^2) = 0$). This is achieved by selecting $f_{\rm n}(t)$ to be a white Gaussian noise with the zero mean and second-order correlator given by [114]

$$\langle f_{\rm n}(t) f_{\rm n}(t')\rangle = 0, \qquad \langle f_{\rm n}(t) f_{\rm n}^*(t')\rangle = 2D_{\rm n}\delta(t-t'). \quad (75)$$

Here, $D_{\rm n}$ is the effective "diffusion coefficient" that characterizes the noise amplitude. To describe correctly the stochastic dynamics of a *nonlinear* oscillator with arbitrary dependences of the frequency $\omega(p)$ and natural damping $\Gamma_+(p)$ on the oscillation power p, one has to assume that the diffusion coefficient $D_{\rm n}$ also depends on p as [114]

$$D_{\rm n}(p) = \Gamma_+(p)\eta(p) = \Gamma_+(p)\frac{k_{\rm B}T}{\lambda\omega(p)} \quad (76)$$

where $\eta(p) = k_{\rm B}T/\lambda\omega(p)$ is the effective noise power in the nonlinear regime. As it will be shown below, this form of the diffusion coefficient provides correct equilibrium properties [see (80)] for the *arbitrary* power dependences $\omega(p)$ and $\Gamma_+(p)$.

In (76) factor λ is a scale factor that relates the oscillator energy $\mathcal{E}(p)$ to the dimensionless oscillator power $p = |c|^2$

$$\mathcal{E}(p) = \lambda \int_0^p \omega(p')dp'. \quad (77)$$

Clearly, λ depends on the normalization of the oscillator amplitude c. For our choice of normalization of the amplitude c of the spin-torque oscillator (see (15)) λ has the form

$$\lambda = V_{\rm eff} M_0/\gamma$$

where $V_{\rm eff}$ is the effective volume of the magnetic material of the "free" layer involved in the auto-oscillation, M_0 is the saturation magnetization of the "free" layer, and γ is the gyromagnetic ratio.

For many problems, instead of solving the nonlinear stochastic Langevin equation (74) for the complex oscillation amplitude $c(t)$, it is convenient to consider the dynamics of the probability density function (PDF) $\mathcal{P}(t,p,\phi)$ that gives the probability of the auto-oscillator to have the power p and phase ϕ at the time t (given some probability distribution at the initial moment of time). While the dynamics of the complex amplitude $c(t)$ is described by the *stochastic* and *nonlinear* equation (74), the evolution of the PDF $\mathcal{P}$ is governed by the *deterministic* and *linear* Fokker-Planck equation that can be obtained from (74) using a standard formalism [138]

$$\frac{\partial \mathcal{P}}{\partial t} - \frac{\partial}{\partial p}\left[2p(\Gamma_+ - \Gamma_-)\mathcal{P}\right] - \omega\frac{\partial \mathcal{P}}{\partial \phi} = \frac{\partial}{\partial p}\left(2pD_{\rm n}\frac{\partial \mathcal{P}}{\partial p}\right) + \frac{\partial}{\partial \phi}\left(\frac{D_{\rm n}}{2p}\frac{\partial \mathcal{P}}{\partial \phi}\right). \quad (78)$$

Terms in the left-hand-side part of (78) originate from the deterministic terms in (74), whereas the terms in the right-hand-side part of (78) describe the stochastic action of the thermal noise $f_{\rm n}(t)$. One can see, that the noise-related terms in (78) have the form of a usual diffusion operator (in the power-phase coordinates) with the power-dependent diffusion coefficient $D_{\rm n}$. Thus, thermal noise leads to the diffusion of the oscillator in the phase space.

Equation (74) or (78) can be applied for the analysis of a number of noise-related problems of oscillator dynamics. It is clear, that one can easily add to this stochastic scheme additional action of deterministic external signals [132] or mutual coupling between oscillators. It is, also, possible to generalize (74) to include effects of other sources of noise, e.g., in the case of spin-torque oscillators, flicker noise related to the bias current or mag-noise related to the fluctuations in the orientation of the "fixed" layer magnetization.

B. Power Generated by an Oscillator in the Presence of Noise

In this subsection we consider the problem of determination of the power, generated by an auto-oscillator in the presence of thermal fluctuations. In the framework of the nonlinear auto-oscillator model one can find a complete analytical solution of this problem, valid both below the generation threshold (when, like in a passive oscillator, the oscillations exist only due to the thermal noise) and in the above-threshold generation regime (when the thermal noise leads to the fluctuations of the generated power). The obtained general results, applied to the case of an STO, suggest a precise method for the determination of the threshold current $I_{\rm th}$ and quantitatively describe experimental data in the near-threshold region, when the influence of the thermal fluctuations can not be ignored.

1) Theoretical Results: To find the average power, generated by an oscillator in the presence of thermal noise, it is sufficient to find the stationary solution $\mathcal{P}_0$ of the Fokker-Planck equation (78). It should be noted, that, since (78) does not depend on the oscillator phase ϕ explicitly (which is connected with the phase-invariance of the Langevin equation (74) or, in more general terms, with an autonomous character of the auto-oscillator's dynamics), the stationary PDF $\mathcal{P}_0$ is independent of the phase ϕ. Thus, in the stationary state any value of the oscillator phase has equal probability, and stationary PDF $\mathcal{P}_0$ is a function of only the oscillator power $p : \mathcal{P}_0 = \mathcal{P}_0(p)$. In this case (78) becomes an ordinary second-order differential equation

$$\frac{d}{dp}\left[2p(\Gamma_+ - \Gamma_-)\mathcal{P}_0 + 2pD_{\rm n}\frac{d\mathcal{P}_0}{dp}\right] = 0$$

and a physically-consistent solution of this equation has the form [114]

$$\mathcal{P}_0(p) = \mathcal{N}_0 \exp\left[-\frac{\lambda}{k_{\rm B}T}\int_0^p \omega(p')\left(1 - \frac{\Gamma_-(p')}{\Gamma_+(p')}\right)dp'\right]. \quad (79)$$

Here, $\mathcal{N}_0$ is the normalization constant determined from the normalization condition

$$\int_0^\infty \mathcal{P}_0(p)dp = 1.$$

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One can see from (79), that at the thermal equilibrium ($\Gamma_-(p) = 0$) stationary PDF $\mathcal{P}_0(p)$ takes the form

$$\mathcal{P}_{\rm eq}(p) = \mathcal{N}_0 \exp\left[-\frac{\lambda}{k_{\rm B}T}\int_0^p \omega(p')dp'\right] = \mathcal{N}_0 \exp\left[-\frac{\mathcal{E}(p)}{k_{\rm B}T}\right] \quad (80)$$

which coincides with the standard Boltzmann distribution. We would like to stress, that our approach gives a correct equilibrium distribution (80) and, hence, satisfies the fluctuation-dissipation theorem [139] for an *arbitrary* dependence of the oscillator's frequency $\omega(p)$ and damping $\Gamma_+(p)$ on the power p, even if these functions significantly change on the thermal energy scale $k_{\rm B}T$.

Equation (79) allows one to determine any stationary characteristic of a nonlinear auto-oscillator in the presence of noise. For instance, the mean oscillation power $\overline{p}$ is found to be

$$\overline{p} = \int_0^\infty p\mathcal{P}_0(p)dp.$$

The power fluctuations Δp^2 can be calculated as

$$\Delta p^2 = \overline{p^2} - (\overline{p})^2$$

where

$$\overline{p^2} = \int_0^\infty p^2\mathcal{P}_0(p)dp.$$

When the temperature T is small, the argument of the exponential function in (79) is large and (79) can be simplified in two important cases. First, in the below-threshold regime (when $\Gamma_-(p) < \Gamma_+(p)$ for all p) the distribution $\mathcal{P}_0(p)$ has a maximum at $p = 0$. Expanding the integrand in (79) in a Taylor series near $p = 0$ and keeping only the first nonzero term, one obtains an approximate expression for the PDF $\mathcal{P}_0(p)$ in the below-threshold regime

$$\mathcal{P}_<(p) \approx \frac{1-\zeta}{\eta(0)}\exp\left[-\frac{1-\zeta}{\eta(0)}p\right]$$

where the supercriticality is defined as $\zeta = \Gamma_-(0)/\Gamma_+(0)$, which coincides with the previously used notation (25) for the case of a spin-torque oscillator. The average power $\overline{p}$ and power fluctuations Δp^2 in this regime are given approximately by the expressions

$$\overline{p} = \frac{\eta(0)}{1-\zeta} = \left(\frac{k_{\rm B}T}{\lambda\omega(0)}\right)\frac{\Gamma_+(0)}{\Gamma_+(0)-\Gamma_-(0)} \quad (81a)$$

$$\Delta p^2 = (\overline{p})^2 = \left(\frac{\eta(0)}{1-\zeta}\right)^2. \quad (81b)$$

The below-threshold PDF $\mathcal{P}_<(p)$ has approximately the same Boltzmann-like form $\mathcal{P}_<(p) \sim \exp(-p/\overline{p})$ as the equilibrium distribution $P_{\rm eq}(p)$ [note, that for small temperature T one can approximate $\mathcal{E}(p)/k_{\rm B}T$ as $\lambda\omega(0)p/k_{\rm B}T = p/\eta(0)$ in (80)], but with the increased average power level $\overline{p}$. One may interpret this effect as an effective increase of the temperature $T \to T/(1-\zeta)$ (see, e.g., [32] and [33]). We would like to note, however, that this interpretation is not completely correct, as the negative damping $\Gamma_-(p)$ in this regime does not only increase the power $\overline{p}$, but, also, reduces the linewidth of oscillations (see Section VII-C), which can not be described by any changes of the effective temperature.

In the above-threshold generation regime stationary PDF $\mathcal{P}_0(p)$ has a maximum at the stationary power $p = p_0$ [see (24)], for which $\Gamma_+(p_0) = \Gamma_-(p_0)$, and the integrand in (79) is zero. By expanding the integrand in (79) in a Taylor series near $p = p_0$, one can derive an approximate expression for the PDF $\mathcal{P}_0(p)$ in the above-threshold regime

$$\mathcal{P}_>(p) \approx \frac{1}{\sqrt{2\pi}\Delta p}\exp\left[-\frac{(p-p_0)^2}{2\Delta p^2}\right]$$

where

$$\Delta p^2 = \left(\frac{\Gamma_+(p_0)}{\Gamma_{\rm p}}\right)p_0\eta(p_0) \approx \left(\frac{\Gamma_+(p_0)}{\Gamma_{\rm p}}\right)\frac{k_{\rm B}T}{\mathcal{E}(p_0)}p_0^2 \quad (82)$$

is the level of power fluctuations of the oscillator in the above-threshold regime ($\Gamma_{\rm p}$ is defined by (27b)). The mean oscillation power $\overline{p}$ in this regime is approximately equal to the zero-temperature value p_0 (24).

One can see, that the probability distribution function $\mathcal{P}_>(p)$ in the above-threshold regime has a functional form that is substantially different from the form of the below-threshold PDF $\mathcal{P}_<(p)$. It is also interesting to note, that the power fluctuations in this regime increase with temperature as $\Delta p \sim \sqrt{T}$, while below the generation threshold $\Delta p \sim T$. Since for typical auto-oscillators $\Gamma_{\rm p} \sim \Gamma_+(p_0)$, the relative level of power fluctuations $\Delta p/p_0 \sim \sqrt{\eta(p_0)/p_0}$ decreases with p_0 (i.e., with the increase of the supercriticality) and, for sufficiently large p_0, power fluctuations become negligible.

Neglecting the dependence of the oscillator frequency $\omega(p)$ on the power p ($\eta(p) = \eta = \text{const}$), which has only a minor influence on the form of the PDF, and using (19), one can derive from (79) an explicit general expression for the stationary probability distribution function of a spin-torque oscillator valid in the *whole range of bias currents*

$$\mathcal{P}_0(p) = \sqrt{\frac{2\zeta}{\pi\eta}}\frac{\exp\left[-(1-\zeta+\zeta p)^2/2\zeta\eta\right]}{1+\operatorname{erf}\left[(\zeta-1)/\sqrt{2\zeta\eta}\right]} \quad (83a)$$

for $Q = 0$ and

$$\mathcal{P}_0(p) = \frac{Q\exp\left[-(\zeta+Q)(1+Qp)/Q^2\eta\right]}{(1+Qp)^\beta E_\beta\left((\zeta+Q)/Q^2\eta\right)} \quad (83b)$$

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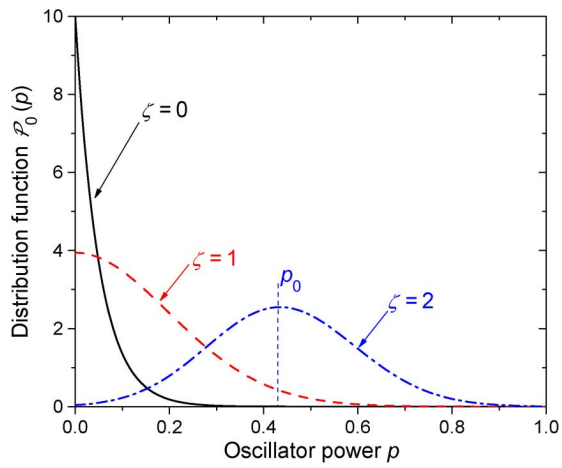

Fig. 16. Stationary probability distribution function $\mathcal{P}_0(p)$ (83) for several values of the supercriticality parameter ζ (from [114]). Nonlinear damping coefficient $Q = 0.3$, noise level $\eta = 0.05$. Data for $\zeta = 0$ (thermal equilibrium) have been divided by 2.

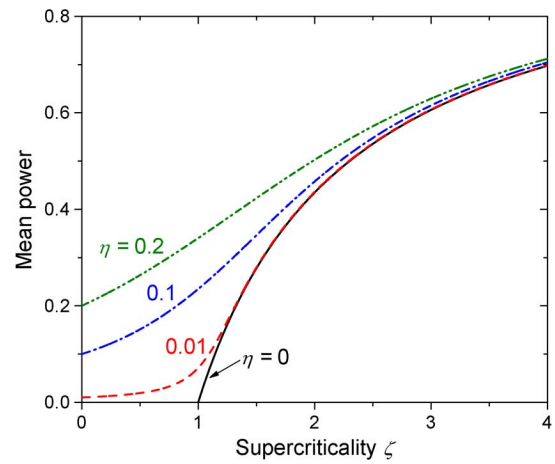

Fig. 17. Dependence of the mean oscillator power $\overline{p}$ (84) on the supercriticality ζ for several noise levels η (from [114]). Nonlinear damping coefficient $Q = 0.3$. The lower curve for $\eta = 0$ coincides with the deterministic value (24) of the oscillator power p_0.

for $Q \neq 0$. Here $\beta = -(1+Q)\zeta/Q^2\eta$, $\mathrm{erf}(x) = (2/\sqrt{\pi})\int_0^x e^{-t^2}dt$ is the error function, and $E_n(x) = \int_1^\infty e^{-xt}/t^n dt$ is the exponential integral function. Fig. 16 shows the profile of the probability distribution function (83) in the below-threshold ($\zeta = 0$), threshold ($\zeta = 1$), and above-threshold ($\zeta = 2$) regimes.

Using the analytical expression (83) for the probability distribution function, it is possible to obtain an analytical expression for the mean oscillation power $\overline{p}$ of a spin-torque oscillator:

$$\overline{p} = \sqrt{\frac{2\eta}{\pi\zeta}}\frac{\exp\left[-(\zeta-1)^2/2\zeta\eta\right]}{1+\mathrm{erf}\left[(\zeta-1)/\sqrt{2\zeta\eta}\right]} + \frac{\zeta-1}{\zeta} \tag{84a}$$

for $Q = 0$ and

$$\overline{p} = \frac{Q\eta}{Q+\zeta}\left[1+\frac{\exp(-(\zeta+Q)/Q^2\eta)}{E_\beta\left((\zeta+Q)/Q^2\eta\right)}\right] + \frac{\zeta-1}{\zeta+Q} \tag{84b}$$

for $Q \neq 0$. Fig. 17 shows the dependence (84) of the mean power $\overline{p}$ on the supercriticality ζ for several values of the noise power η.

2) Comparison With Experiments: To illustrate the application of the above described theory to a real experimental situation we compare below the theoretical expressions for the auto-oscillator power in the presence of noise with the recent measurements of the generated power in a GMR spin-valve auto-oscillator based on a metallic nano-pillar [108]. The experimental data from [108] are shown by black dots in the main frame of Fig. 18. To analyze these experimental data it is important, first of all, to determine the threshold of microwave generation, i.e., the magnitude of the bias current at which the spin-torque oscillator, instead of passively filtering its eigen-frequency from the thermal noise, starts to actively self-generate. We believe that this can be done by analyzing the experimental data for very small values of the bias current. In the inset of Fig. 18 we show by dots the experimental dependence of the *inverse* mean power $1/\overline{p}$ on the bias current I. These dots can be fitted to the theoretical expression (81a) which gives a linear

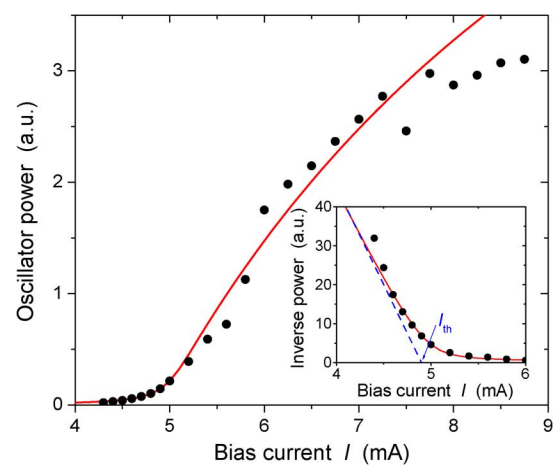

Fig. 18. Main panel: Dependence of the mean power $\overline{p}$ on the bias current I. Dots—experiment [108] for $T = 225$ K, solid line—theoretical dependence (84) for $I_{\mathrm{th}} = 4.9$ mA, $\eta = 4.2 \cdot 10^{-4}$, $Q = 0.3$ (from [114]). Inset shows the same data for *inverse* power $1/\overline{p}$ in near-threshold range of currents. Dashed line corresponds to the approximate expression (85) valid for small currents. Intersection of this line with x axis gives the value of the threshold current I_{th}.

dependence of the inverse power on the bias current for small values of current

$$\frac{1}{\overline{p}} \propto (I_{\mathrm{th}} - I). \tag{85}$$

This linear dependence, taking place in the below-threshold regime $I < I_{\mathrm{th}}$ and shown by a dashed line in the inset of Fig. 18, crosses the horizontal axis (axis of the bias current) at the point $I = I_{\mathrm{th}}$ and, therefore, allows one to determine the I_{th} with a reasonably high precision.

Thus, the precise measurement of the oscillator output power as a function of the bias current *for low values of current* in combination with the theoretical formula (81a) provides a simple method for the precise determination of the threshold current I_{th} of microwave generation in a spin-torque oscillator in situations when the influence of thermal fluctuations is strong, and

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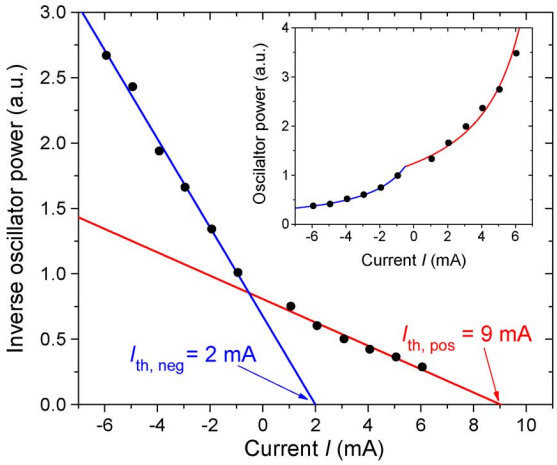

Fig. 19. Main panel: Dependence of the inverse power $1/\overline{p}$ on the bias current I. Dots—experiment [40] for a magnetic tunnel junction (the direction of the bias current was changed to coincide with our sign agreement), solid lines—theoretical dependences (85) demonstrating different values of the threshold current, determined from the data for positive $(I_{\mathrm{th,pos}})$ and negative $(I_{\mathrm{th,neg}})$ current directions. Inset shows the same set of data for the direct power $\overline{p}$.

the determination of I_{th} by other means is difficult. In particular, from the inset of Fig. 18 one can immediately determine the value of the threshold current of $I_{\mathrm{th}} = 4.9\ \mathrm{mA}$, which is substantially larger than the current $I_* \approx 4.2\ \mathrm{mA}$, at which the thermally-induced oscillations become observable in the experiment [108].

Using thus determined value of the threshold current $I_{\mathrm{th}} = 4.9\ \mathrm{mA}$ to calculate the supercriticality parameter $\zeta = I/I_{\mathrm{th}}$ in the theoretical expression (84), it is possible to describe theoretically the power $\overline{p}$, generated by an STO, in a wide range of bias currents (see main panel in Fig. 18). We would like to stress, that the correct determination of the threshold current I_{th} (and, therefore, the parameter ζ) is essential for the comparison of (84) with experiment, since the dependence of the oscillator power $\overline{p}$ on the other parameters of the model (like noise power η and nonlinear damping parameter Q) in the above-threshold region $(\zeta > 1)$ is relatively weak.

Fig. 19 demonstrates another example of the threshold current determination using the linear dependence (85) in the below-threshold region. In this case, the above described method of the threshold current determination is applied to a spin-torque device based on a magnetic tunnel junction (MTJ). Dots in the main panel of Fig. 19 show the experimental data for the inverse power $1/\overline{p}$ measured in the below-threshold regime in an MTJ nano-pillar (see[40]), in which the spacer between the "free" and "fixed" magnetic layers [see Fig. 1(b)] was made of a nonmagnetic *insulator* $\mathrm{Al_2O_3}$. Due to a much larger resistance of the MTJ nano-pillar (compared to fully-metallic GMR spin valve nano-pillars, where the spacer is made of a nonmagnetic *metal*), the authors of [40] were able to measure the power of the thermally-induced oscillations of the device even for the negative values of the bias currents $I < 0$, when the spin-transfer torque creates effective *positive* damping and, according to (85), *reduces* the measured oscillation power $\overline{p}$.

One can see from Fig. 19 that the experimental dependences of the inverse output power on the bias current for positive $I > 0$ and negative $I < 0$ current directions are both linear (85), but have substantially different slopes. Respectively, values of the threshold current I_{th}, estimated using (85) from the experimental data for positive $(I_{\mathrm{th,pos}} = 9\ \mathrm{mA})$ and negative $(I_{\mathrm{th,neg}} = 2\ \mathrm{mA})$ currents, are significantly different. Note, that both $I_{\mathrm{th,pos}}$ and $I_{\mathrm{th,neg}}$ correspond *to the same magnetization configuration*, but were calculated *for different current polarities*. Since the threshold current (22) is inversely proportional to the spin-polarization efficiency ε, this difference in I_{th} suggests that the spin-polarization efficiency ε for MTJ depends not only on the mutual orientation of the magnetizations of the "free" and "fixed" layers, but, also, on the direction of the bias current. In the case shown in Fig. 19, the ratio of spin-polarization efficiencies for positive and negative currents is $\varepsilon_{\mathrm{pos}}/\varepsilon_{\mathrm{neg}} = 2/9 \approx 0.22$.

Thus, the measurements of the current dependence of the noise-induced output power in MTJ-based STO can provide direct information about the spin-polarization efficiency, and can be used for verification of theories of spin-dependent tunneling in magnetic nano-structures.

We would like to mention, that the threshold current, estimated in [40] using the measurements of the linewidth and the data for the *direct* (rather than *inverse*) power $\overline{p}$, is $I_{\mathrm{th}} = 20 \pm 5\ \mathrm{mA}$, which is far above our estimates. We attribute this difference to the fact that other sources of noise can give a substantial contribution to the linewidth of a highly-resistive MTJ-based device, and, therefore, the estimation of the threshold current based on the linewidth data could be unreliable. On the other hand, the dependence of the direct power $\overline{p}$ on the bias current I (see inset in Fig. 19) is nonlinear, and it is much more difficult to make an estimate of the threshold current based on such a nonlinear dependence. Thus, the difference in the slopes for currents of different polarity, that is clearly seen in Fig. 19, which demonstrates the current dependence of the inverse power, is hardly noticeable in the inset of Fig. 19, where the current dependence of the direct power is presented.

C. Generation Linewidth of an Auto-Oscillator

Thermal fluctuations do not only blur the stationary distribution $\mathcal{P}_0(p)$ of the oscillation power, but, also, alter the dynamical behavior of an auto-oscillator. In the presence of noise the auto-oscillation amplitude $c(t)$ is not anymore a harmonic function of time $c(t) \sim e^{-i\omega_{\mathrm{g}}t}$, corresponding to a simple monochromatic oscillation, but is, instead, a rather complicated stochastic function, and its spectrum

$$c_\Omega = \int_{-\infty}^{+\infty} c(t) e^{i\Omega t} dt$$

has, strictly speaking, nonzero spectral components, corresponding to every frequency Ω.

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The *power spectrum* of the auto-oscillation $\mathcal{S}(\Omega)$, that can be defined as

$$\langle c_\Omega c^*_{\Omega'} \rangle = 2\pi\delta(\Omega - \Omega')\mathcal{S}(\Omega)$$

is a convenient quantity that gives a detailed description of the auto-oscillation spectrum in the presence of noise. Here the angular brackets $\langle \ldots \rangle$ stay for the averaging over the statistics of thermal fluctuations. For a coherent monochromatic oscillation with the frequency $\omega_{\rm g}$ the power spectrum in singular $\mathcal{S}(\Omega) \sim \delta(\Omega - \omega_{\rm g})$. In the presence of noise $\mathcal{S}(\Omega)$ has a finite width (called sometimes *generation linewidth*) $\Delta\omega$, which provides a simple way for a single-parameter characterization of a noisy auto-oscillation spectrum. The generation linewidth $\Delta\omega$ is one of the most important parameters of an auto-oscillator, especially with regards to its practical applications.

Alternatively, the coherence of oscillations can be characterized by an *autocorrelation function*

$$\mathcal{K}(\tau) = \langle c(t+\tau)c^*(t) \rangle .$$

The descriptions of noisy oscillations by a power spectrum $\mathcal{S}(\Omega)$ or an autocorrelation function $\mathcal{K}(\tau)$ are fully equivalent, since these two functions are connected by the Fourier transform

$$\mathcal{S}(\Omega) = \int \mathcal{K}(\tau) e^{i\Omega\tau} d\tau.$$

As it will be shown below, in many cases the power spectrum of an auto-oscillation can be approximately represented in a simple Lorentzian form

$$\mathcal{S}(\Omega) \approx \frac{2\Delta\omega\overline{p}}{(\Omega - \overline{\omega})^2 + \Delta\omega^2} \tag{86a}$$

where $\overline{p}$ is the average oscillation power, and $\overline{\omega}$ is the mean oscillation frequency. Note, that in (86a) and in the following text $\Delta\omega$ is defined as *half*-linewidth at a half-power level. The autocorrelation function, corresponding to (86a), has a simple exponential form

$$\mathcal{K}(\tau) = \overline{p} \exp\left(-i\overline{\omega}\tau - \Delta\omega|\tau|\right). \tag{86b}$$

It is important to stress, that in many cases the experimental data for the generation linewidth are fitted to the simple Lorentzian function (86), even if the actual lineshape $\mathcal{S}(\Omega)$ (or autocorrelation function $\mathcal{K}(\tau)$) has more complicated form. Thus, usually, the generation linewidth $\Delta\omega$ is understood in a sense of an experimentally determined quantity obtained as a best fitting parameter in the Lorentzian expression (86).

In the next section, we derive expressions for generation linewidth $\Delta\omega$ of an auto-oscillator is several important particular cases, using the stochastic Langevin equation (74).

1) Below-Threshold Regime: In the below-threshold regime $(\Gamma_-(p) < \Gamma_+(p))$, when the characteristic oscillation power $p \sim \eta$ is small, one can neglect all the nonlinearities in (74). In such a case (74) becomes an equation describing a usual linear oscillator in the presence of noise

$$\frac{dc}{dt} + i\omega(0)c + [\Gamma_+(0) - \Gamma_-(0)]\, c = f_{\rm n}(t).$$

One can easily find a solution of this linear equation in the form

$$c(t) = \int_{-\infty}^{t} f_{\rm n}(t') \exp\left\{-[i\omega(0) + \Gamma_+(0) - \Gamma_-(0)]\,(t-t')\right\} dt'.$$

Using the statistical properties of the thermal fluctuations (75), it is easy to show that the autocorrelation function $\mathcal{K}(\tau)$ in this linear case has a simple exponential form (86b) with the average power $\overline{p}$ given by (81a), mean frequency defined by $\overline{\omega} = \omega(0)$, and the full linewidth given by

$$2\Delta\omega = 2\Gamma_+(0) - 2\Gamma_-(0) = 2\Gamma_+(0)(1-\zeta). \tag{87}$$

It is clear, that the power spectrum $\mathcal{S}(\Omega)$ of the oscillation in this linear case has exactly the Lorentzian shape (86a).

For a spin-torque oscillator the expressions for the positive and negative damping are given by (19), and the linewidth $2\Delta\omega$ of such an oscillator decreases linearly with the bias current I as

$$2\Delta\omega = 2\Gamma_{\rm G} - 2\sigma I = 2\Gamma_{\rm G}\left(1 - \frac{I}{I_{\rm th}}\right). \tag{88}$$

The measurements of the current dependence of the linewidth $\Delta\omega$ in the below-threshold regime provide an alternative method of the experimental determination of the threshold current $I_{\rm th}$ for current-induced excitations. As it was discussed above, this method may not be very accurate for highly-resistive STO based on magnetic tunnel junctions (MTJ).

Recently, the linear dependence (88) of the linewidth $\Delta\omega$ on the bias current I has been observed both in fully metallic GMR spin valves [21] and in MTJ based on the TMR effect [40].

2) Above-Threshold Regime: "Linear" Oscillator: Far above the generation threshold the power fluctuations in an auto-oscillator are negligible compared to the mean generated power. In this regime the complex amplitude $c(t)$ becomes a pure phase-modulated process $c(t) \approx \sqrt{p_0} e^{i\phi(t)}$. Consequently, the generation linewidth $\Delta\omega$ and the overall shape of the auto-oscillation power spectrum $\mathcal{S}(\Omega)$ is determined only by the phase fluctuations. This property is general for auto-oscillators of any nature, and is connected with different types of stability of the oscillator dynamics with respect to the power and phase fluctuations (see Section V-A2).

Thus, to find a generation linewidth of an auto-oscillator in the above-threshold generation regime, it is sufficient to determine the statistical properties of the auto-oscillator phase $\phi(t)$. Here we will consider this problem for a "linear" oscillator, i.e., for an auto-oscillator with frequency independent of the oscillation power $\omega(p) = \omega_{\rm g} = \omega_0 = \text{const}$. As it was noted before, the majority of conventional oscillators belong to this class, whereas spin-torque oscillators, in general, are characterized by a strong dependence of the oscillation frequency $\omega(p)$ on the power p (this case will be considered in the following subsection).

Using (74), one can derive a stochastic equation for the phase $\phi = \arg(c)$ of an auto-oscillator

$$\frac{d\phi}{dt} + \omega_{\rm g} = \frac{1}{\sqrt{p}} \mathrm{Im}\left[\widetilde{f}_{\rm n}(t)\right] \tag{89}$$

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where $\widetilde{f}_{\rm n}(t) = f_{\rm n}(t)e^{-i\phi(t)}$. Note, that the statistical properties of the function $\widetilde{f}_{\rm n}(t)$ are the same as for the function $f_{\rm n}(t)$, i.e., $\widetilde{f}_{\rm n}(t)$ is a white Gaussian stochastic process with the second-order correlator given by (75).

Far above the generation threshold one can substitute the stationary power p_0 for p in the right-hand-side part of (89), thus reducing this equation to a closed-form equation for the phase ϕ, which has the following solution for the phase as a function of time:

$$\phi(t) = \phi(0) - \omega_{\rm g} t + \frac{1}{\sqrt{p_0}} \int_0^t {\rm Im}\left[\widetilde{f}_{\rm n}(t')\right] dt'$$

where it was assumed that the phase at the initial moment of time $t = 0$ has fixed value $\phi(0)$. This solution describes the "Brownian motion" of the phase $\phi(t)$ under the action of thermal fluctuations. It is clear, that $\phi(t)$ is a Gaussian process with the mean value

$$\langle \phi(t) \rangle = \phi(0) - \omega_{\rm g} t \tag{90a}$$

and the variance

$$\Delta\phi^2(t) = \left\langle \phi^2(t) \right\rangle - \left[\langle \phi(t) \rangle\right]^2 = \frac{\Gamma_+(p_0)\eta(p_0)}{p_0}|t|. \tag{90b}$$

The phase variance $\Delta\phi^2(t)$ increases linearly with the time interval $|t|$, exactly as the square of a particle displacement for a usual Brownian motion.

The autocorrelation function $\mathcal{K}(t)$, corresponding to the Gaussian random phase $\phi(t)$, can be written as

$$\begin{aligned}\mathcal{K}(t) &= p_0 \left\langle \exp\left\{i\left[\phi(t) - \phi(0)\right]\right\}\right\rangle \\ &= p_0 e^{i\langle \phi(t) - \phi(0)\rangle} \exp\left[-\Delta\phi(t)^2/2\right].\end{aligned}$$

For a "linear" auto-oscillator [$\Delta\phi^2(t) \sim |t|$; see (90b)] the autocorrelation function $\mathcal{K}(t)$ has the Lorentzian form (86b) with the average frequency $\overline{\omega} = \omega_{\rm g}$ and full linewidth

$$2\Delta\omega_0 = \frac{\Delta\phi^2(t)}{|t|} = \Gamma_+(p_0)\frac{\eta(p_0)}{p_0}.$$

The ratio of the noise power $\eta(p_0)$ to the stationary auto-oscillation power p_0 in the above derived linewidth expression can be rewritten as a ratio of corresponding energies $k_{\rm B}T/\mathcal{E}(p_0)$, yielding a physically transparent expression for the Lorentzian generation linewidth of a "linear" auto-oscillator

$$2\Delta\omega_0 = \Gamma_+(p_0)\frac{k_{\rm B}T}{\mathcal{E}(p_0)}. \tag{91}$$

To the best of our knowledge, the generation linewidth in a general form (91) was first written in [113].

The damping rate $\Gamma_+(p_0)$ in (91) determines the overall scale of the possible linewidth variations, and the ratio $k_{\rm B}T/\mathcal{E}(p_0)$ describes the linewidth reduction due to a smaller influence of noise (having the constant energy $k_{\rm B}T$) for above-threshold oscillations, the energy $\mathcal{E}(p_0)$ of which increases with the oscillation power p_0. Note, also, that if we formally use (91) for the evaluation of the linewidth in the state of thermodynamic equilibrium ($\mathcal{E}(p_0) \to k_{\rm B}T$, $\Gamma_+(p_0) \to \Gamma_+(0)$), we obtain the linewidth value that is 2 times smaller than the value given by (87) in the same limit ($\Gamma_-(0) \to 0$). This difference is explained by the fact that *only phase fluctuations* lead to the linewidth broadening of an auto-oscillator in the above-threshold generation regime, whereas, for a passive oscillator, both phase and amplitude fluctuations play the same role.

The expressions for the generation linewidth of many different types of conventional ("linear") auto-oscillators have been derived previously (see, e.g., a classical book [140]). It is important to note, that all these expressions can be rewritten in the general form (91). In particular, the full generation linewidth of an electrical auto-oscillator (similar to the one considered in Section IV-A) is given by the classical formula [140]:

$$2\Delta\omega_0 = \frac{k_{\rm B}TR\omega_0^2}{U_0^2}$$

where R is the resistance of the circuit, $\omega_0 = 1/\sqrt{LC}$ is the resonance frequency of the electrical circuit having inductance L and capacitance C, and U_0 is the voltage amplitude of self-sustained electrical oscillations on the capacitor C. This expression can be exactly rewritten in the form (91), taking into account that, for an electrical oscillator, the damping rate is $\Gamma_+ = R/(2L)$, and the oscillation energy is $\mathcal{E} = CU_0^2/2$.

The general expression (91) for the oscillator linewidth, sometimes, can be used even for the description of quantum systems. For example, the quantum limit for the linewidth of a single-mode laser (Schawlow-Townes limit) leads to the following expression [141], [142]

$$2\Delta\omega_0 = \frac{\hbar\omega_0}{2}\frac{\Gamma_0^2}{P_{\rm out}}.$$

Here, ω_0 is the generation frequency, Γ_0 is the loss rate, which is assumed to be caused only by the energy radiation out of the laser cavity (losses in the active medium were neglected), and $P_{\rm out}$ is the output power of the laser. This expression, also, can be rewritten in the form (91), if one uses the energy of quantum fluctuations $\hbar\omega_0/2$ instead of the thermal noise energy $k_{\rm B}T$, and takes into account, that the output power $P_{\rm out}$ of the laser is connected with the oscillation energy $\mathcal{E}$ by the expression $P_{\rm out} = \Gamma_0\mathcal{E}$.

An attempt to calculate the generation linewidth of a spin-torque oscillator has been undertaken in [112], and resulted in an expression (see equation (28) in [112]), that, also, can be rewritten in the form (91). However, the dependence of the generated frequency of the generation power was ignored in [112], and, therefore, the linewidth expression obtained in [112] is correct only qualitatively. Thus, it is necessary to consider the case of a substantially "nonlinear" auto-oscillator (i.e., an oscillator with a large nonlinear frequency shift) to quantitatively describe the generation linewidth of an STO.

3) Above-Threshold Regime: "Nonlinear" Oscillator: The expression for the auto-oscillator generation linewidth (91), which was verified for many physical realizations of auto-oscillators, underestimates the generation linewidth of an STO

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by one-two orders of magnitude. The reason for such a large difference is that in the derivation of (91) the oscillation frequency was assumed to be independent of the oscillation power $\omega(p) = \text{const}$. For spin-torque oscillators, as it was already emphasized, the influence of the nonlinear frequency shift is very important. To find the correct expression for the generation linewidth of an STO, as well as for any other "nonlinear" oscillator, one has to explicitly take into account the power dependence of the frequency $\omega(p)$ in the theory.

In the above-threshold generation regime the power fluctuations $\delta p(t) = p(t) - p_0$ are small even for a "nonlinear" oscillator $\delta p(t) \ll p_0$. However, to obtain a correct result for the generation linewidth one should not ignore them altogether, but, instead, should use this smallness to linearize (74) near the stationary generation power $p = p_0$ to derive an approximate system of equations for the power fluctuations $\delta p(t)$ and the phase $\phi(t)$ in a "nonlinear" auto-oscillator

$$\frac{d\delta p}{dt} + 2\Gamma_{\rm p}\delta p = 2\sqrt{p_0}\mathrm{Re}\left[\widetilde{f}_{\rm n}(t)\right] \tag{92a}$$

$$\frac{d\phi}{dt} + \omega_{\rm g} = \frac{1}{\sqrt{p_0}}\mathrm{Im}\left[\widetilde{f}_{\rm n}(t)\right] - N\delta p \tag{92b}$$

where, as in the previous subsection, $\widetilde{f}_{\rm n}(t) = f_{\rm n}(t)e^{-i\phi(t)}$ is a stochastic process with the same properties as $f_{\rm n}(t)$. The left-hand side of (92a) is the same as (26) for power fluctuations δp of free auto-oscillations.

One can see from (92b), that the frequency nonlinearity N creates an additional noise term $-N\delta p$ in the phase equation (92b). The physical mechanism behind this additional noise is clear: for a nonlinear auto-oscillator the power fluctuations $\delta p(t)$ lead to the frequency modulation $\omega(p(t)) = \omega(p_0 + \delta p(t)) \approx \omega_{\rm g} + N\delta p$. In some sense, this additional noise term can be considered as an inhomogeneous broadening of the oscillator linewidth: oscillators with different (due to thermal fluctuations) powers p have different oscillation frequencies $\omega(p)$.

Since the stochastic system (92) is a *linear* system of equations, and the noise $\widetilde{f}_{\rm n}(t)$ is a Gaussian process, both $\delta p(t)$ and $\phi(t)$ are the Gaussian processes also. One can easily obtain a complete set of statistical characteristics of these processes. Namely, the mean value of the power fluctuations following from (92) is zero $\langle\delta p(t)\rangle = 0$, whereas its correlation function has the form

$$\langle\delta p(t)\delta p(t')\rangle = \frac{\Gamma_+(p_0)}{\Gamma_{\rm p}}\eta(p_0)p_0 e^{-2\Gamma_{\rm p}|t-t'|}. \tag{93}$$

The mean value of the phase $\phi(t)$ (for a fixed value $\phi(0)$ of the phase at the initial moment of time $t = 0$) is given by

$$\langle\phi(t)\rangle = \phi(0) - \omega_{\rm g}t \tag{94a}$$

and the variance of the phase fluctuations is expressed as

$$\Delta\phi^2(t) = 2\Delta\omega_0\left[(1+\nu^2)|t| - \nu^2\frac{1-e^{-2\Gamma_{\rm p}|t|}}{2\Gamma_{\rm p}}\right] \tag{94b}$$

where ν is the dimensionless nonlinear frequency shift coefficient (33).

In contrast to the case of a "linear" oscillator (90b), the variance $\Delta\phi^2(t)$ (94b) has a nonlinear dependence on the time interval $|t|$. This is caused by a finite correlation time [see (93)] of the additional "nonlinear" noise term $-N\delta p(t)$ in (92b). Only for the time intervals $|t|$, that are much larger than the correlation time $1/\Gamma_{\rm p}$ of the power fluctuations, the phase variance $\Delta\phi^2(t)$ starts to grow linearly with $|t|$. Consequently, the spectrum of a "nonlinear" auto-oscillator is, in general, non-Lorentzian. There are, however, two limiting cases when a rather complicated expression (94b) for the phase variance of a nonlinear auto-oscillator can be substantially simplified.

If the temperature T and, consequently, the generation linewidth $\Delta\omega$ is sufficiently small $\Delta\omega \ll \Gamma_{\rm p}$, one can neglect the exponential factor in the second term of the right-hand-side part of (94b) at the characteristic decoherence time scale of $|t| \sim 1/\Delta\omega$. Then, the following expression for the phase variance is obtained in the low-temperature limit:

$$\Delta\phi^2(t) \approx 2\Delta\omega_0(1+\nu^2)|t| - \frac{\nu^2}{2\Gamma_{\rm p}}.$$

In this limit, the autocorrelation function $\mathcal{K}(t)$ becomes exponential (86b), and the power spectrum $\mathcal{S}(\Omega)$ has a Lorentzian shape (86a) with the full linewidth of generation given by the expression

$$2\Delta\omega = 2\Delta\omega_0(1+\nu^2) = (1+\nu^2)\Gamma_+(p_0)\frac{k_{\rm B}T}{\mathcal{E}(p_0)}. \tag{95}$$

One can see, that the nonlinearity of the auto-oscillator frequency $\omega(p)$ leads to the increase of the generation linewidth by $(1+\nu^2)$ times. The expression for the generation linewidth of a "nonlinear" auto-oscillator (95) was derived on the basis of a nonlinear oscillator model and published for the first time in [113].

The expression (95) can, also, be derived in a more straightforward way, noting the analogy between the system of (49), describing the phase-locking of a "nonlinear" auto-oscillator to an external periodic signal, and the linearized stochastic equations (92), describing the interaction of the auto-oscillator with thermal noise. Using the phase renormalization transformation given by (50), we obtain that the effective noise amplitude in the equation for the auto-oscillator phase ϕ is multiplied by $\sqrt{1+\nu^2}$ [see (52)], which corresponds to the increase of the effective temperature and, therefore, the linewidth by $(1+\nu^2)$ times. When the expression for the generation linewidth (95) is obtained using the above described approach, it can be interpreted as a phase-locking bandwidth of a "nonlinear" auto-oscillator to a stochastic external signal provided by thermal noise.

One can show, that the condition of applicability of the phase renormalization (50) exactly coincides with the condition $\Delta\omega \ll \Gamma_{\rm p}$, which can be rewritten as a condition for the temperature

$$k_{\rm B}T \ll \left(\frac{\Gamma_{\rm p}}{\Gamma_+(p_0)}\right)\frac{\mathcal{E}(p_0)}{1+\nu^2} \tag{96a}$$

or, using (82), as a condition for the relative power of thermal fluctuations

$$\frac{\Delta p}{p_0} \ll \frac{1}{\sqrt{1+\nu^2}}. \tag{96b}$$

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Thus, (95) represents a low-temperature limit of the generation linewidth of a "nonlinear" auto-oscillator. The estimations for the typical parameters of a spin-torque oscillator (permalloy circular nano-pillar of the radius $R_c = 50$ nm and thickness of the "free" layer $L = 5$ nm, generation frequency $\omega_g/2\pi \simeq 30$ GHz) show that the expression (95) is quantitatively correct for the temperatures $T \leq 10 - 100$ K, depending on the supercriticality parameter ζ.

Another limiting case, which allows an easy analysis of (94b), is the opposite limiting case of relatively long (compared to the inverse linewidth $1/\Delta\omega$) correlation times of the power fluctuations $1/\Gamma_p$, or the case of relatively large generation linewidths $\Delta\omega \gg \Gamma_p$. In this case the exponential function in (94b) can be developed in a Taylor series to give

$$\Delta\phi^2(t) \approx 2\Delta\omega_0 \left(|t| + \nu^2\Gamma_p t^2\right).$$

For sufficiently large frequency nonlinearities $\nu \gg 1$ one can drop the first (linear in $|t|$) term in the brackets, and retain only the second term that is one quadratic in time. Then, the phase variance $\Delta\phi^2(t) \sim t^2$, and the power spectrum takes the form

$$\mathcal{S}(\Omega) \sim \exp\left[-\frac{(\omega(p_0) - \Omega)^2}{2\Delta\omega_*^2}\right]$$

with the characteristic linewidth

$$\Delta\omega_* = |\nu|\sqrt{\Gamma_+(p_0)\Gamma_p}\sqrt{\frac{k_B T}{\mathcal{E}(p_0)}}. \tag{97}$$

In this limit, the auto-oscillator linewidth is large $\Delta\omega_* \gg \Gamma_p$ and proportional to the normalized nonlinear frequency shift coefficient ν in the first power. Also, in contrast with the low-temperature linewidth expression (95), giving the linear dependence on temperature, the expression (97) for the auto-oscillator linewidth is proportional to $\sqrt{T}$. It is interesting to note, that formula for the auto-oscillator linewidth, which is qualitatively similar to (97), was for the first time proposed in [107] for the case of a spin-torque auto-oscillator (see (2) in [107]). In the same paper the $\sqrt{T}$ temperature dependence of an STO linewidth was obtained as a result of direct numerical simulations.

The expression (97) for the generation linewidth is valid in the temperature range

$$\left(\frac{\Gamma_p}{\Gamma_+(p_0)}\right)\frac{\mathcal{E}(p_0)}{\nu^2} \ll k_B T \ll \left(\frac{\Gamma_p}{\Gamma_+(p_0)}\right)\mathcal{E}(p_0) \tag{98a}$$

or, in terms of power fluctuations,

$$\frac{1}{|\nu|} \ll \frac{\Delta p}{p_0} \ll 1. \tag{98b}$$

Thus, (97) is the asymptotic expression of the generation linewidth of a strongly-nonlinear ($|\nu| \gg 1$) auto-oscillator in the intermediate temperature interval. For larger temperatures or smaller oscillation powers p_0 (i.e., in the near-threshold region) expansion (92) is not valid anymore, and one has to analyze the full equations (74) or (78), which can be done only using numerical methods.

It is not clear, *a priori*, which of the above derived approximate expressions for the "nonlinear" auto-oscillator linewidth (95) or (97) is a better approximation for the description of the experimentally measured linewidth magnitudes in strongly nonlinear spin-torque oscillators. On one hand, many of the experiments, where the STO linewidth was measured, were done at a room temperature $T \simeq 300$ K, which is higher than the upper limit of applicability for the low-temperature expression (95). On the other hand, in many cases the experimentally observed linewidths of spin-torque oscillators are rather narrow $\Delta\omega < \Gamma_G$, so that the conditions of applicability of the other asymptotic expression (97) are, also, violated. As it will be demonstrated below, the direct comparison of the currently available experimental data with the above developed theory, in general, demonstrates that the low-temperature expression (95) gives a much better account of the existing experimental data. However, additional measurements of the generation linewidth of STO performed in a wide range of temperatures are necessary to clarify this important question and to, possibly, see in the experiment the cross-over from the linear to a square root dependence on temperature, when the temperature is increased.

4) Near-Threshold Region: The region near the generation threshold $\Gamma_+(0) \simeq \Gamma_-(0)$ (or, for a spin-torque oscillator, $I \simeq I_{th}$) is the most difficult region for the theoretical analysis. In this region there are no obvious linearization schemes by which (74) can be simplified, and one needs to employ various numerical methods to evaluate the auto-oscillator linewidth near the generation threshold.

The distortions of the power spectrum $\mathcal{S}(\Omega)$ in the vicinity of the generation threshold have been studied in [143] using analytical and numerical methods based on a nonstationary Fokker-Planck equation (78) for the probability distribution function $\mathcal{P}(t, p, \phi)$. The results of this analysis show that, in the near-threshold region, the power spectrum $\mathcal{S}(\Omega)$ does not have a simple Lorentzian form (86a), but rather can be represented as a sum of partial Lorentzians

$$\mathcal{S}(\Omega) = \sum_j \frac{2\Delta\omega_j \overline{p}_j}{(\Omega - \overline{\omega}_j)^2 + \Delta\omega_j^2}.$$

For a "linear" auto-oscillator ($\omega(p) = \omega_0 = \text{const}$) the central frequencies $\overline{\omega}_j$ for all the partial Lorentzians are equal $\overline{\omega}_j = \omega_0$, and the overall power spectrum $\mathcal{S}(\Omega)$ remains symmetric with respect to the central frequency ω_0.

In contrast, for a "nonlinear" auto-oscillator ($\omega(p) \neq \text{const}$) all the frequencies $\overline{\omega}_j$ are distinct and, in general, distributed nonsymmetrically around the zero-temperature frequency $\omega_g = \omega(p_0)$. As a result, the power spectrum $\mathcal{S}(\Omega)$ becomes asymmetric in the near-threshold region. Such asymmetry of the power spectrum has been recently observed in experiments with spin-torque oscillators based on metallic GMR nano-pillar [108].

Another specific feature of the behavior of a nonlinear auto-oscillator linewidth in the near-threshold region is an apparent broadening of this linewidth. Like in the case of a usual inhomogeneous linewidth broadening, the sum of several partial Lorentzians with different central frequencies $\overline{\omega}_j$ results in the power spectrum $\mathcal{S}(\Omega)$ that has the overall width that is larger,

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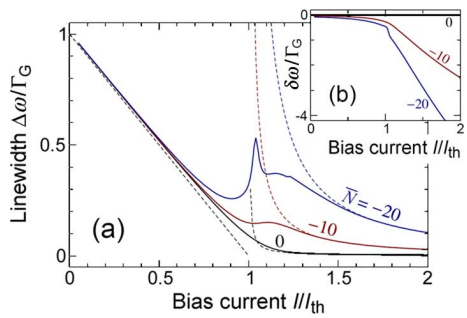

Fig. 20. Generation linewidth $\Delta\omega$ [panel (a)] and frequency shift $\delta\omega = \overline{\omega} - \omega_0$ [panel (b)] as a function of the bias current I obtained from single-Lorentzian fits to spectra computed for different values of the frequency nonlinearity $\overline{N} = N/\Gamma_G$ (from [143]). Noise power $\eta = 10^{-3}$, nonlinear damping coefficient $Q = 2$. Dashed lines in (a) show the below-threshold (88) and above-threshold (95) linewidth asymptotics. Used values of the parameters are typical for in-plane magnetized spin-torque oscillators (note that the nonlinear frequency shift N is negative).

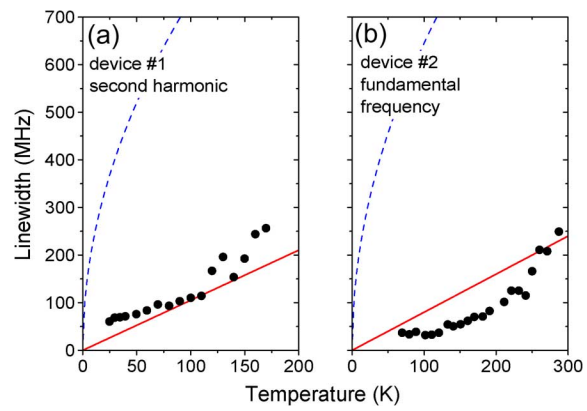

Fig. 21. Generation linewidth $2\Delta\omega$ of a spin-torque oscillator calculated from the approximate expressions (95) (solid lines) and (97) (dashed lines) in comparison with the temperature dependence of the linewidth in a nanopillar device #1 (a) and device #2 (b) measured in [107] (black dots). Panel (a) shows the linewidth measured at the *second* harmonic of the signal $2\Delta\omega_2 = 8\Delta\omega$.

than the partial linewidths $\Delta\omega_j$ of individual Lorentzians. This effect is illustrated by Fig. 20(a), where the linewidth $\Delta\omega$, obtained from a single-Lorentzian fit to the power spectrum $\mathcal{S}(\Omega)$, is shown as a function of the bias current I for a spin-torque oscillator. Good agreement is found between the numerically and analytically calculated linewidths far below and far above the generation threshold in all cases, as indicated by the dashed lines representing the linewidth values calculated using (88) and (95). In general, the overall linewidth $\Delta\omega$ decreases with the increasing current, except in the region near the threshold, where a local maximum of the linewidth can appear in the case of a strong frequency nonlinearity $|\nu| \gg 1$. Depending on the magnitude of the nonlinear frequency shift $|N|$, this maximum can be several times larger than the generation linewidth in both below- and above-threshold regimes [143]. The experimental measurements of the power spectra of current-induced microwave oscillations in STO, also, show the line narrowing followed by line broadening near the generation threshold with the increase of the bias current (see Fig. 3(c) in [108]).

Fig. 20(b) shows the variation of the generation frequency $\overline{\omega}$, obtained from the single-Lorentzian fit to the power spectrum $\mathcal{S}(\Omega)$, with the bias current I [143]. This variation is relatively slow below the generation threshold $\overline{\omega} \approx \omega_0$ and exhibits a fast quasi-linear change with current above the generation threshold due to the nonlinear frequency shift, in agreement with (29).

5) Comparison With Experiments: The above presented theory of the generation linewidth of nonlinear auto-oscillators successfully explains a number of qualitative features, experimentally observed in the linewidth measurements on spin-torque oscillators. First of all, the inverse dependence of the generation linewidth (95) on the energy of oscillations $\mathcal{E}(p_0)$ explains the general linewidth narrowing with the increase in the bias current and oscillation amplitude observed in, e.g., Fig. 4 in [109]. Note, also, that the energy $\mathcal{E}(p_0)$ is proportional to the effective volume $V_{\rm eff}$ of the auto-oscillator, which is much larger for a spin-torque auto-oscillator, based on a magnetic nano-contact, than for a similar device based on a nano-pillar. Thus, the linewidth expression (95) provides a natural explanation for a well-known experimental fact (see [13], [17], [18]) that the auto-oscillation linewidths associated with devices based on magnetic nano-pillars are, in general, several times broader that those in the devices based on magnetic nano-contacts.

In Fig. 21 we show the comparison of the generation linewidth calculated using two asymptotic expressions (95) and (97) with the results of experimental measurements of the temperature dependence of the linewidth of a spin-torque oscillator performed on the nano-pillar devices #1 [Fig. 21(a)] and #2 [Fig. 21(b)] in [107] (see [107, Fig. 2]). The geometrical parameters of the nano-pillar devices were taken from [107] and it was assumed that the excited magnetization oscillation mode is pinned at the pillar lateral boundaries (see [144] for details), the Gilbert damping parameter $\alpha_G = 0.01$, the nonlinear damping parameter $Q = 3$, and the polarization efficiency $\varepsilon = 0.4$ were assumed to be the same for both devices [113]. As it is clear from Fig. 21, the simple analytical expression (95) obtained in the low-temperature limit gives a reasonably good estimate of the observed linewidths at different temperatures for *both* nano-pillar devices with *the same* parameters. At the same time, the second asymptotic expression (97) obtained in the relatively high-temperature limit significantly overestimates the generation linewidth. Assuming that the parameters of the two devices are slightly different (which is possible due to different nano-patterning and different thicknesses of the "free" magnetic layer), one can obtain much better quantitative agreement with the experiment [107] using the low-temperature expression (95). Although it is difficult to determine from the experimental data presented in Fig. 21 what is the real temperature dependence of the generation linewidth, it is clear that the low-temperature asymptotic expression (95) gives a much better estimate of the generation linewidth magnitude.

In Fig. 22 we compared the theoretical (95) and experimental [109] (see Fig. 6 in [109]) dependences of the full generation linewidth $2\Delta\omega$ on the out-of-plane magnetization angle θ_0 of nano-contact spin-torque oscillator. All the parameters of the nano-contact device Fig. 22 were taken from [109], while the current $I = 9$ mA and magnetic field $H_0 = 9$ kOe correspond to the center of the experimentally studied region, and the nonlinearity parameter of positive damping was again chosen to be

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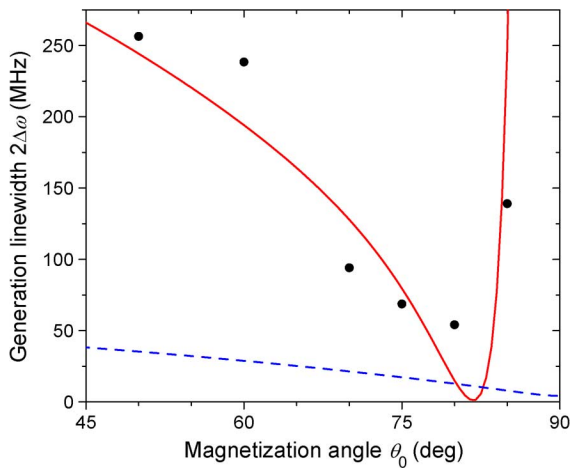

Fig. 22. Generation linewidth $2\Delta\omega$ of a spin-torque auto-oscillator calculated from (95) (solid line) in comparison with the angular dependence of the nano-contact STO linewidth measured at a room temperature in [109] (black dots) (from [113]). The dashed line represents the classical result for a "linear" auto-oscillator linewidth calculated from (91) and multiplied by 10.

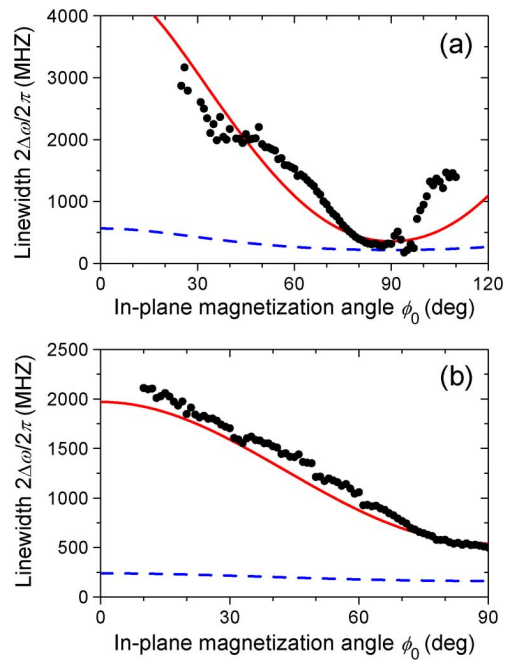

Fig. 23. Dependence of the generation linewidth $2\Delta\omega$ (95) on the in-plane magnetization angle ϕ_0 for two spin-torque oscillators based on anisotropic magnetic nano-pillars: (a) oscillator with a synthetic "fixed" layer Py-IrMn; (b) oscillator with a thick Py "fixed" layer (after [145]). Black dots—experimental data measured at room temperature from [145], solid lines—calculation using the expression (95) for a "nonlinear" auto-oscillator, dashed lines—calculation using the classical "linear" expression (91) multiplied by 10. All the calculations were done for the parameters of the experiment [145]: saturation magnetization $4\pi M_0 = 8\,\mathrm{kOe}$, bias magnetic field $H_0 = 1.08\,\mathrm{kOe}$ (a) and $H_0 = 1.2\,\mathrm{kOe}$ (b), in-plane anisotropy field $H_A = 0.2\,\mathrm{kOe}$ (a) and $H_A = 0.1\,\mathrm{kOe}$ (b), Gilbert damping constant $\alpha_G = 0.015$, nonlinear damping coefficient $Q = 3$, "free" layer thickness $L = 4\,\mathrm{nm}$, shape of the nano-pillar—elliptical with the sizes 150 nm × 50 nm (a) and 130 nm × 70 nm (b), spin-polarization efficiency $\varepsilon = 0.32$ (a) and $\varepsilon = 0.375$ (b), bias current $I = 5\,\mathrm{mA}$, temperature $T = 300\,\mathrm{K}$.

equal to $Q = 3$ [113]. It is clear from Fig. 22, that the linewidth dependence on the bias field orientation calculated using the low-temperature asymptotic expression (95) is in good quantitative agreement with the experimental results from [109]. In contrast, the classical result for a "linear" auto-oscillator (91) (see dashed line in Fig. 22, which shows classical result (91), multiplied by 10) predicts a much narrower lines and a monotonous decrease of the linewidth as a function of the out-of-plane magnetic field angle θ_0. The "nonlinear" result (95) gives a reasonable qualitative and quantitative description of the experimentally observed behavior of the auto-oscillator linewidth, and, in particular, it predicts the linewidth minimum around $\theta_0 \approx 80°$, which is clearly seen in the experimental data. This linewidth minimum occurs at the "linear" magnetization angle $\theta_{0,\mathrm{lin}}$, for which the nonlinear frequency shift coefficient N vanishes [see Fig. 6(a)], and, therefore, at this magnetization angle the spin-torque oscillator behaves as a classical "linear" oscillator with the generation linewidth determined by (91).

For an STO based on an anisotropic nano-pillar one may, also, expect a nontrivial dependence of the full generation linewidth $2\Delta\omega$ on the in-plane magnetization angle ϕ_0, since in this case the nonlinear frequency shift coefficient N significantly varies with ϕ_0 (see Fig. 6(b)). Recent experiments [145] performed using a spin-torque oscillator based on a nano-pillar geometry where the "free" layer was magnetized by the bias magnetic field oriented in the "free" layer plane, indeed, demonstrated strong variation of the oscillation linewidth with the variation of the in-plane magnetization angle ϕ_0. This behavior of the generation linewidth finds a natural explanation in the framework of the asymptotic linewidth expression (95). Fig. 23 shows the comparison of the experimental data from [145] with the angular dependence of the linewidth $2\Delta\omega$ calculated from the asymptotic equation (95) using the general expression (105a) for the nonlinear frequency shift coefficient in an anisotropic magnetic film and taking all the other parameters of the spin-torque oscillators from [145]. It is clear from Fig. 23, that the linewidth corresponding to the magnetization along the hard in-plane axis $(\phi_0 = 90°)$ of the nano-pillar "free" layer is much smaller than the linewidth in the case of the magnetization along the easy axis $(\phi_0 = 0)$. It is, also, clear, that the linewidth expression (95) gives a good qualitative and a reasonable quantitative description of the experimental data for both nano-oscillator devices used in the experiments [145].

We would, also, like to note that, similar to the case of an spin-torque oscillator with an out-of-plane magnetized "free" layer, in the case of an STO based on an anisotropic in-plane magnetized nano-pillar it is possible to choose the direction and magnitude of the bias magnetic field in such a way that the nonlinear frequency shift coefficient N vanishes [see Fig. 6(b)]. Under these conditions the linewidth of the anisotropic STO will have a minimum possible value, corresponding to the case of a "linear" auto-oscillator (91).

VIII. Conclusion

The above presented comparison of the analytical results following from the universal nonlinear auto-oscillator model (5) with results of laboratory experiments performed on current-driven magnetic nano-contacts or nano-pillars demonstrate that this model can give an adequate description of most of the salient characteristics of real spin-torque nano-oscillators.

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The main advantage of the model (5) is the fact that it gives a clear and simple physical picture of the nonlinear dynamical processes taking place in current-driven spin-torque oscillators both in the autonomous auto-oscillation regime and under the influence of periodic or/and stochastic external signals. The developed analytical approach puts the theory of STO on equal footing with the theory of conventional auto-oscillators, and, at the same time, stresses two characteristic features, that make STO different: i) strong dependence of the generated frequency on the power of the generated signal (strong nonlinear frequency shift) characterized by the normalized nonlinearity parameter $|\nu| \gg 1$ (33); ii) strong influence of the thermal noise on the process of microwave generation in STO, resulting from the fact that in a nano-sized STO the generated microwave power (of the order of 1 nW) is much smaller than in most conventional auto-oscillators and can be comparable to the power of the thermal noise at a room temperature in a rather wide range of supercritical bias currents ($I - I_{\rm th} \sim 0.2 I_{\rm th}$).

Another important advantage of the proposed analytical approach is that it formulates the language in which the theoretical description of STO could be written, and identifies the critical parameters of STO and natural variables in which the theoretical description of STO has the simplest possible form. For example, it follows from the developed theory of STO phase-locking (see Section VI) that the nonlinear renormalized phase of an STO (50) is the natural variable to describe the phase-locking of nonlinear auto-oscillators, and that the difference of the STO oscillation frequencies $\delta = \omega_{\rm g,2} - \omega_{\rm g,1}$ (rather than the difference of the STO anisotropies used in [104] and [105]) should be used as a simplest possible parameter describing the difference between the individual STO in a phase-locked array. Thus, we believe, that the proposed analytical model (5) can form a basis for the development of an engineering theory of STO and STO arrays, that is necessary for the development of practical nano-sized active spin-torque devices.

Of course, it would have been unrealistic to expect that an approximate theory based on a single ordinary differential equation of the first order (5) would be able to give exact quantitative description of *all* the experiments performed on spin-torque oscillators of different geometry for any magnitude and direction of the bias magnetic field and bias direct current. Nevertheless, it is clear from the above presented comparison of our approximate theory with various experiments performed in different STO geometries, that this theory gives sufficient accuracy for the practical development of STO and, in many cases, the accuracy of our auto-oscillator model (5) is better, than the experimental uncertainties in the values of the input parameters characterizing STO (like static magnetization and effective anisotropy of the "free" layer, magnitude of spin-polarization efficiency ε, and geometrical parameters of magnetic nano-contacts and nano-pillars) and in the values of experimentally measured STO characteristics (like generated power, frequency, and linewidth). Therefore, we believe, that at least in a certain range of supercritical bias currents ($\zeta = I/I_{\rm th} \leq 1.5$), in which only one spin wave mode is excited in an STO, the theory will be useful for practical calculations of STO characteristics and optimization of the STO design.

The universal auto-oscillator model (5) contains three functions of the oscillator power p: the oscillation frequency $\omega(p)$, and the positive $\Gamma_+(p)$ and negative $\Gamma_-(p)$ dissipative functions. These three functions contain all the information about the auto-oscillator and, if these functions are known, the model (5) allows us to predict all the practically important properties of the auto-oscillator. At the same time, the problem of the exact determination of these three functions for a particular geometry of a practical STO is rather complicated, and cannot be solved analytically. The approximate analytic expressions (19), obtained as Taylor expansions of these functions in the case of relatively small generated power $p \ll 1$, can be used as a first approximation, and, as we have shown above, could, in many cases, provide sufficient accuracy for many of the practical calculations. If a better accuracy is needed, there is a possibility to combine the analytical approach presented above with advanced micromagnetic numerical simulations, and extract the functions $\omega(p)$, $\Gamma_+(p)$, and $\Gamma_-(p)$ from the results of such simulations performed in the realistic geometry of a *single* STO. When the functions $\omega(p)$, $\Gamma_+(p)$, and $\Gamma_-(p)$ are determined numerically, they can be used in the model (5) for a more precise calculations of the STO working characteristics.

The same numerical modeling performed for a *single* STO can yield spatially inhomogeneous profiles of the spin wave modes excited in an STO of a particular geometry, and if these profiles are known, it will be possible to calculate analytically the coupling frequencies $\Omega_{j,j'}$ and phases $\beta_{j,j'}$, which determine the interaction between the individual oscillators in an STO array. An example of an analytic formula which allows one to calculate the coupling frequency between two nano-contact oscillators coupled by propagating spin waves, when the spatial profiles of these wave modes are known, is given in [102] (see equation (5) in [102]). Obviously, the use of the analytic coupled-oscillator model (71), in the situation when all the parameters of this model are determined from the micromagnetic simulations on a single STO, will lead to considerable savings in computational time and capacity, and will provide improved accuracy for the practical calculations of microwave generators based on STO arrays.

Another way to determine the realistic parameters of the universal model (5) is to perform a series of systematic laboratory experiments in a particular STO geometry. For example, systematic experimental measurements of the effect of frequency modulation in a spin-torque oscillator, where the variation of the bias current is done at several different modulation frequencies, will allow one to determine the important model parameter $\Gamma_{\rm p}$ (27b) using the above derived (37). Analogously, a direct measurement of coupling between two STO performed in a sub-critical regime, using, for example, the spin-torque FMR technique [36], will allow one to determine the coupling parameters of the model (71), and, therefore, to solve the problem of phase-locking in an array of STO.

The above developed analytic approach in the case when the model parameters are known can, also, be used to predict the behavior of an individual auto-oscillator (or an oscillator array) under the influence of composite external signals consisting of both deterministic and stochastic components. A good example of such a use of our universal model (5) has been demonstrated

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recently in [132]. The authors of [132] studied theoretically and experimentally the process of phase-locking of a spin-torque auto-oscillator to an external periodic signal in the presence of thermal noise. For the theoretical description of this effect they used (5) and, as a result of a series of well-staged and systematic experiments, were able to directly confirm (see Fig. 4b in [132]) the linear relation between the magnitude of a phase-locking range Δ and the normalized nonlinear frequency shift coefficient ν, which follows from (54) (see, also, [98, Eq. (6)]) in the limit $|\nu| \gg 1$. Working in the framework of the model (5), the authors of [132] were, also, able to calculate the dependence of the forced oscillator frequency on the free-running oscillator frequency (dependence analogous to (48b) shown in Fig. 10) in the presence of thermal noise (see [132, Fig. 3]). Thus, the model (5) in combination with well-targeted experiments allows one to obtain novel nontrivial results and, at the same time, cross-check the magnitudes of the integral parameters of the model.

Another important feature of the developed nonlinear auto-oscillator theory is that this theory is absolutely *generic*, and, therefore, can be used for the description of nonlinear auto-oscillators of any physical nature. In particular, this theory can be applied to both fully metallic spin-torque oscillators based on the effect of giant magnetoresistance (GMR) and to the auto-oscillators with a dielectric barrier layer between two metallic layers, which use the effect of tunneling magnetoresistance (TMR). Our preliminary calculations based on (5) show that a single nano-sized spin-torque oscillator based on the TMR effect with the MgO tunnel barrier can provide several microwatts of the output microwave power, which is three orders of magnitude higher than the typical output power of a metallic GMR nano-oscillator. This difference is mainly related to the much higher magnetoresistance ratio in a TMR contact compared to the fully metallic GMR contact.

We have demonstrated in [5] that the analytic model (5) works well not only for the description of the current-induced oscillations of magnetization taking place near the minimum of the magnetic energy of the "free" layer, like in most experiments performed with the in-plane orientation of spin polarization (see, e.g., [13]), but, also, in the case when these oscillations are taking place near the *maximum* of the "free" layer magnetic energy, like in the experiments with the perpendicular-to-plane orientation of the spin-polarization direction [45] (for details see [5] and [63]).

The general character of the qualitative results obtained above in the framework of the models (5) and (71) allows us to believe that these results can be successfully used in different branches of science, not related to spin-torque effect or/and to the generation of microwaves. For example, in our opinion, the above presented theory might be useful in the analysis of weakly connected neural networks [146], where individual auto-oscillators imitate the action if neurons in the human brain, and the phase-locking of several oscillators can be interpreted as a "learning" process.

In summary, we developed a general analytic approach to the problem of autonomous and nonautonomous dynamics of nonlinear auto-oscillators and arrays of such oscillators in the case when they are weakly connected to each other. We, also, presented a comprehensive review of analytical results describing characteristics, obtained in the framework of this approach for a particular case of nano-sized spin-torque oscillators, and compared these theoretical results to the available results of laboratory experiments. We believe that this simple, intuitive, and, at the same time, rather general analytic theory will find many applications in science and technology.

Appendix I
Parameters of the Nonlinear Oscillator Model for Spin-Torque Oscillator

In this Appendix we present explicit expressions for the coefficients of the weakly-nonlinear expansion (19) for a spin-torque oscillator in the case, when the excited spin wave mode is a uniform magnetization precession [5]. We consider an anisotropic "free" magnetic layer with an easy-axis in-plane magnetic anisotropy field H_A. Note, that in the case of a spin-torque oscillator, based on magnetic nano-pillar [Fig. 1(b)] of a noncircular shape, the magnetic anisotropy field H_A is a sum of crystallographic anisotropy field and effective field of the shape anisotropy.

We use coordinate system, in which the "free" magnetic layer lies in the $x-y$ plane, axis z is normal to the film plane, and the easy axis of the in-plane anisotropy is along the x axis. The effective magnetic field $\boldsymbol{H}_{\rm eff}$ in this case has the explicit form

$$\boldsymbol{H}_{\rm eff} = \boldsymbol{H}_0 - 4\pi(\boldsymbol{M}\cdot\widehat{\mathbf{e}}_z)\widehat{\mathbf{e}}_z + \frac{H_A}{M_0}(\boldsymbol{M}\cdot\widehat{\mathbf{e}}_x)\widehat{\mathbf{e}}_x \tag{99}$$

where $\boldsymbol{H}_0$ is the external bias magnetic field, $\boldsymbol{M}$ is the dynamic magnetization, and $M_0 = |\boldsymbol{M}|$ is the modulus of the saturation magnetization (length of the magnetization vector $\boldsymbol{M}$) of the "free" layer.

We will consider a general case of an obliquely magnetized "free" layer, when the external magnetic field $\boldsymbol{H}_0$ can be written as

$$\boldsymbol{H}_0 = (\cos\theta_0\cos\phi_0\widehat{\mathbf{e}}_x + \cos\theta_0\sin\phi_0\widehat{\mathbf{e}}_y + \sin\theta_0\widehat{\mathbf{e}}_z)H_0 \tag{100}$$

where H_0 is the field magnitude, θ_0 is the *out-of-plane* angle of the bias field, and ϕ_0 is the angle that the *in-plane* projection of the bias field makes with the easy anisotropy axis x.

Due to the magneto-dipolar and anisotropy contributions to the effective magnetic field $\boldsymbol{H}_{\rm eff}$, the equilibrium orientation of the magnetization $\boldsymbol{M}_{\rm eq}$ will be different from the direction of the applied field $\boldsymbol{H}_0$

$$\boldsymbol{M}_{\rm eq} = (\cos\theta\cos\phi\widehat{\mathbf{e}}_x + \cos\theta\sin\phi\widehat{\mathbf{e}}_y + \sin\theta\widehat{\mathbf{e}}_z)M_0. \tag{101}$$

Here, θ and ϕ are the *internal* magnetization angles, determined from the conditions

$$H\sin\theta = H_0\sin\theta_0 - 4\pi M_0\sin\theta \tag{102a}$$

$$H\cos\theta\cos\phi = H_0\cos\theta_0\cos\phi_0 + H_A\cos\theta\cos\phi \tag{102b}$$

$$H\cos\theta\sin\phi = H_0\cos\theta_0\sin\phi_0 \tag{102c}$$

where H is the magnitude of the *internal* magnetic field.

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It is convenient to introduce three characteristic frequencies, connected with the three contributions to the effective field $\boldsymbol{H}_{\rm eff}$:

$$\omega_H = \gamma H, \quad \omega_M = 4\pi\gamma M_0, \quad \omega_A = \gamma H_A. \tag{103}$$

Then, the linear coefficients of the small-amplitude expansion (19) are given by

$$\omega_0^2 = (\omega_H - \sin^2\phi\omega_A)(\omega_H + \cos^2\theta\omega_M) - \sin^2\theta\cos^2\phi\omega_H\omega_A \tag{104a}$$

$$\Gamma_{\rm G} = \alpha_{\rm G}\left[\omega_H + \frac{1}{2}\cos^2\theta\omega_M - \frac{1}{2}(\sin^2\theta\cos^2\phi + \sin^2\phi)\omega_A\right] \tag{104b}$$

$$\sigma = \sigma_0\cos\gamma_0. \tag{104c}$$

Here, $\alpha_{\rm G}$ is the Gilbert damping constant, σ_0 is given by (4b), and γ_0 is the angle between the equilibrium orientations of the magnetization vectors of the "free" and "fixed" magnetic layers. The simple cos-like dependence of the parameter σ on the angle γ_0 follows from our assumption that the spin-polarization efficiency ε does not depend on the angle γ between the magnetizations of the "free" and "fixed" layers [see discussion below (4b)]. If such angular dependence need to be taken into account, one should use in (104c) value of ε, corresponding to equilibrium orientation $\gamma = \gamma_0$.

The nonlinear coefficients of the weakly-nonlinear expansion (19) have much more complicated form, but still can be written analytically as

$$N = \frac{2\omega_0}{\boldsymbol{\mathcal{A}}}\left(\boldsymbol{\mathcal{T}} - 3\frac{|\boldsymbol{\mathcal{W}}_1|^2 + |\boldsymbol{\mathcal{W}}_2|^2}{\omega_0}\right) \tag{105a}$$

$$Q = 2 - 3\left(\frac{\omega_0}{\boldsymbol{\mathcal{A}}}\right)^2 + 2\frac{\boldsymbol{\mathcal{U}}}{\boldsymbol{\mathcal{A}}} + 24\frac{{\rm Re}(\boldsymbol{\mathcal{B}}^*\boldsymbol{\mathcal{V}}^2)}{\omega_0^2\boldsymbol{\mathcal{A}}} + 2\left[3 - \left(\frac{\omega_0}{\boldsymbol{\mathcal{A}}}\right)^2\right] \times \left[q_1\left(\frac{\omega_0}{\omega_M}\right)^2 - \left(\frac{|\boldsymbol{\mathcal{V}}|}{\omega_0}\right)^2\right]. \tag{105b}$$

Here q_1 is the first coefficient in the expansion of the damping function $\alpha(\xi)$ and all the other coefficients are determined by the following equations:

$$\boldsymbol{\mathcal{A}} = \omega_H + \frac{1}{2}\cos^2\theta\omega_M - \frac{1}{2}(\sin^2\theta\cos^2\phi + \sin^2\phi)\omega_A \tag{106a}$$

$$\boldsymbol{\mathcal{B}} = \frac{1}{2}\cos^2\theta\omega_M - \frac{1}{2}(\sin\theta\cos\phi - i\sin\phi)^2\omega_A \tag{106b}$$

$$\boldsymbol{\mathcal{V}} = \cos\theta\left[\sin\theta\omega_M + \cos\phi(\sin\theta\cos\phi - i\sin\phi)\omega_A\right] \tag{106c}$$

$$\boldsymbol{\mathcal{U}} = \frac{1}{4}\left[(1 - 3\cos 2\theta)(\omega_M + \omega_A/2) - 3\cos^2\theta\cos 2\phi\omega_A\right] \tag{106d}$$

$$u = \sqrt{\frac{\boldsymbol{\mathcal{A}} + \omega_0}{2\omega_0}}, \qquad v = \frac{\boldsymbol{\mathcal{B}}^*}{|\boldsymbol{\mathcal{B}}|}\sqrt{\frac{\boldsymbol{\mathcal{A}} - \omega_0}{2\omega_0}} \tag{106e}$$

$$\boldsymbol{\mathcal{W}}_1 = \frac{3}{2}\left(u^2 + |v|^2\right)(u\boldsymbol{\mathcal{V}} - v^*\boldsymbol{\mathcal{V}}^*) - \frac{1}{2}(u\boldsymbol{\mathcal{V}} + v^*\boldsymbol{\mathcal{V}}^*) \tag{106f}$$

$$\boldsymbol{\mathcal{W}}_2 = -uv^*(u\boldsymbol{\mathcal{V}} - v^*\boldsymbol{\mathcal{V}}^*) \tag{106g}$$

$$\boldsymbol{\mathcal{T}} = \frac{1}{2}\left[3\left(u^2 + |v|^2\right)^2 - 1\right]\boldsymbol{\mathcal{U}} + 3u\left(u^2 + |v|^2\right){\rm Re}(v\boldsymbol{\mathcal{B}}). \tag{106h}$$

Note, that the direct measurements of the nonlinear properties of magnetic damping (in particular, the direct determination of the expansion coefficient q_1) are quite complicated. Therefore, the parameter q_1 can be considered as a phenomenological parameter, whose value can be determined from the comparison of theoretical and experimental characteristics of an auto-oscillator. In the nonlinear auto-oscillator model the parameter q_1 enters only into the expression for the nonlinear damping coefficient Q. Therefore, it is natural to consider the coefficient Q (instead of q_1) as a basic phenomenological parameter. The comparison of predictions of the nonlinear oscillator model with the experimental results on the current-induced spin wave excitations show, that the values of this coefficient typically lie in the range $3 \geq Q \geq 0$. We would like to stress, that the coefficient Q is *the only* phenomenological parameter of the nonlinear auto-oscillator model (5) which is difficult (but still possible) to measure by standard magnetic techniques.

The expressions for the frequency ω_0, linear damping rate $\Gamma_{\rm G}$, and nonlinear frequency shift coefficient N become significantly simpler in the case, when the external magnetic field is aligned along one of the three main axis (x, y, or z).

Thus, for the magnetization oriented along the easy in-plane axis x ($\theta_0 = \theta = 0$, $\phi_0 = \phi = 0$, $H = H_0 + H_A$) one has

$$\omega_0 = \sqrt{\omega_H(\omega_H + \omega_M)} \tag{107a}$$

$$\Gamma_{\rm G} = \alpha_{\rm G}(\omega_H + \omega_M/2) \tag{107b}$$

$$N = -\frac{\omega_H\omega_M(\omega_H + \omega_M/4) + \omega_A\left(2\omega_0^2 + 3\omega_M^2/4\right)}{\omega_0(\omega_H + \omega_M/2)}. \tag{107c}$$

This magnetization orientation is stable for any value of the external magnetic field H_0.

For the magnetization oriented along the hard in-plane axis y ($\theta_0 = \theta = 0$, $\phi_0 = \phi = \pi/2$, $H = H_0$) the expressions for the coefficients become

$$\omega_0 = \sqrt{(\omega_H - \omega_A)(\omega_H + \omega_M)} \tag{108a}$$

$$\Gamma_{\rm G} = \alpha_{\rm G}(\omega_H + \omega_M/2 - \omega_A/2) \tag{108b}$$

$$N = -\frac{1}{\omega_0(\omega_H + \omega_M/2)}\left[(\omega_H - 4\omega_A)\omega_M(\omega_H + \omega_M/4) - \omega_A\left(\omega_0^2 - 3\omega_H(\omega_M - \omega_A/2)/2\right)\right]. \tag{108c}$$

This magnetization state is stable for external bias magnetic field $H_0 > H_A$.

Finally, for the normal-to-the-plane magnetization direction z ($\theta_0 = \theta = \pi/2$, $H = H_0 - 4\pi M_0$, and angles ϕ_0 and ϕ are undefined in this case) one obtains

$$\omega_0 = \sqrt{\omega_H(\omega_H - \omega_A)} \tag{109a}$$

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$$\Gamma_{\rm G} = \alpha_{\rm G}(\omega_H - \omega_A/2) \tag{109b}$$

$$N = \frac{2\omega_0^2(\omega_M + \omega_A/2) + 3\omega_A^2(\omega_H + \omega_M)/4}{\omega_0(\omega_H - \omega_A/2)}. \tag{109c}$$

This state of magnetization is stable for $H_0 > 4\pi M_0 + H_A$.

With a little modification, the expressions for the linear frequency ω_0 and other parameters of the nonlinear oscillator model (5), presented above, can be used for the description of a number of different quasi-linear critically excited nonuniform spin wave modes. To do this, one has to take into account the "exchange" renormalization of the magnetic field $H \to H + \Delta H_{\rm ex}$, where $\Delta H_{\rm ex} = 4\pi M_0 \lambda_{\rm ex}^2 k_{\rm c}^2$ is the effective exchange field for a spin wave mode having a wavenumber $k_{\rm c}$ ($\lambda_{\rm ex} = \sqrt{A/2\pi M_0^2}$ is the exchange length, where A is the exchange stiffness). Another parameter that should be renormalized in a spatially nonuniform case is the volume occupied by the excited spin wave mode $V = LS$ that enters in the denominator of (4b) for the spin-polarization efficiency σ_0. For instance, in the case of a Slonczewski's propagating spin wave mode [90], excited in a circular magnetic nano-contact of the radius $R_{\rm c}$, the wavenumber $k_{\rm c}$ is approximately equal to $1.2/R_{\rm c}$, while the effective volume $V_{\rm eff}$ is given by the expression (59).

Appendix II
Spin Wave Modes Excited by Spin-Polarized Current

The expressions for the coefficients of a weakly-nonlinear expansion (19) presented in the Appendix I were obtained in the macrospin approximation, i.e., assuming a spatially-uniform profile of the excited spin wave mode. These expressions, after a certain modification, can be used for the estimation of parameters of a spin-torque oscillator in the geometries where spatially nonuniform spin wave modes are excited by spin-polarized current. In a thin magnetic layer, that is magnetized to saturation by the external bias magnetic field, the linear frequency of any excited spin wave mode is always close to the frequency of the ferromagnetic resonance (FMR) ω_0. The nonlinearity of spin wave modes in a magnetized magnetic layer is caused, for the most part, by the magnetostatic interaction and the fact that the length of the total magnetization vector is conserved [72]. Therefore, for a given magnitude and direction of the bias magnetic field, this nonlinearity is approximately the same for any spin wave mode. Therefore, the macrospin approximation is often used as a simplifying assumption in the theoretical investigations of current-induced magnetization dynamics in thin magnetic layers.

In this Appendix we give a short overview of different types of *spatially nonuniform* spin wave modes, that can be excited by spin-polarized current in different geometries. We would like to stress, that all these modes are the eigen-modes of the corresponding magnetic structure, i.e., spin-polarized current, usually, does not create additional spin wave modes, but rather stabilizes one of the existing eigenmodes of a magnetic nano-structure. It is, also, worth mentioning, that the spin-torque effect provides a very effective method of spin wave excitation and modes having very large amplitudes (i.e., very large precession angles) can be excited by spin-polarized current. Some of the excited modes could be strongly nonlinear, and it would be practically impossible to excite them by any other means (e.g., using the excitation by a microwave magnetic field). However, the existence of such strongly nonlinear spin wave modes can be verified by numerical solution of the conservative Landau-Lifshitz equation.

In the case of a spin-torque auto-oscillator, based on a magnetic nano-pillar (see Fig. 1(b)), the area of the "free" layer is finite and, consequently, the spin wave spectrum of a pillar is discrete. The frequencies and spatial profiles of linear spin waves in this geometry can be calculated using the theory developed in [147]. Spin-polarized current will excite, generally, the mode with the smallest damping rate $\Gamma_{\rm G}$, i.e., with the smallest frequency (since, for the Gilbert damping the damping rate increases with the increase of frequency). Due to the strong dipolar pinning of microwave magnetization at the lateral edges of magnetic nano-elements with large aspect ratio [144], this lowest spin wave mode can have a finite wave number $k_{\rm c} \sim 1/R_{\rm c}$, where $R_{\rm c}$ is the radius of the nano-pillar. The excitation of such discrete modes can be studied in the framework of the nonlinear oscillator model (5) with the renormalized magnetic field $H \to H + \Delta H_{\rm ex}$, as it was explained in Appendix I. One should, also, take into account the fact, that the effective volume of the mode that is pinned at the nano-pillar lateral edges is roughly 2 times smaller than the nominal volume of the "free" layer of a pillar.

In the nano-pillar geometry spin-polarized current can, also, excite strongly nonlinear soliton-like spin wave excitations [97]. The excitation of such nonlinear modes is, usually, sub-critical, i.e., these modes can have a finite amplitude at the threshold of their excitation (see [148] for details of sub-critical instability) and, as it was discussed in Section IV-C, application of the nonlinear oscillator model (5) for their description is limited. We would like to stress that the nonlinear soliton-like modes of a magnetic nano-pillar, observed in [97], are not specific to the situations where the spin-polarized current is involved. The spin wave modes that are qualitatively similar have been studied theoretically in [149] in the framework of a conservative Landau-Lifshitz equation. This example illustrates again the general statement that spin-polarized current does not, usually, create new modes, but rather stabilizes eigen-modes that are already present in the magnetic system.

In the case of a spin-torque oscillator, based on a magnetic nano-contact (see Fig. 1(a)), the situation is more complicated, since there are no natural lateral boundaries in the "free" magnetic layer, and the spectrum of spin wave modes in such a layer is continuous. Therefore, this case requires a more detailed investigation.

The current-induced excitation of spin waves in a perpendicularly magnetized nano-contact was studied for the first time in [90] in a linear approximation. It was shown that the excited mode is a propagating cylindrical spin wave mode (Slonczewski's mode) with the wave number $k_{\rm c} \approx 1.2/R_{\rm c}$, where $R_{\rm c}$ is the radius of the nano-contact. The weakly-nonlinear analysis of this geometry [91] confirmed the results of the linear theory, and showed that the nonlinear properties of the excited mode are the same as one could expect from the macrospin analysis. In particular, the frequency of the excited spin wave mode increases with the bias current and oscillation amplitude,

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which corresponds to the positive nonlinear frequency shift coefficient N—which is the usual situation for a normally-magnetized magnetic layer (see, e.g., [72]). The same Slonczewski's-like propagating spin wave mode can be excited in the case on an oblique magnetization of the "free" layer [93]. Since this mode is critically-unstable (i.e., it has a vanishingly-small amplitude at the threshold of its excitation), it can be well described using the nonlinear oscillator model (5) with the coefficients presented in Appendix I, taking into account the renormalization of the bias magnetic field and effective volume of the excited mode.

The Slonczewski's propagating spin wave mode can, also, be considered as an eigen-mode of a conservative magnetic system. However, since the "free" layer is infinite, i.e., the magnetic system is *open*, one should use in this case a quasi-normal mode expansion [150]. The main difference between the quasi-normal modes of an open system and conventional normal modes of a closed system is that the eigen-frequencies of the quasi-normal modes are complex (which corresponds to the decay of the energy of the mode with time) even without the account of any physical dissipation mechanism. The imaginary part of an eigen-frequency of a quasi-normal mode describes energy radiation due to wave propagation, i.e., an additional source of damping, absent in the case of a closed system. For typical parameters of magnetic nano-contacts this type of damping is, in many cases, more important than the direct energy losses due to the magnetic dissipation.

Another type of a spin wave mode, that can be excited in a nano-contact geometry, is a self-localized strongly nonlinear standing spin wave "bullet" [92]–[95]. This mode exists only for relatively small magnetization angles, when the nonlinear frequency shift coefficient N is negative. In that case the mode nonlinearity, determined by the magneto-dipole interaction, and the mode dispersion, determined by the exchange interaction, can compensate each other and, as a result of this competition, a soliton-like standing spin wave mode is formed [92]. Due to the self-localized character of this mode, its radiation losses vanish, and it has a lower excitation threshold than the propagating Slonczewski's mode. The characteristic feature of this mode is that it is evanescent and its frequency lies *below* the continuous spectrum of linear spin waves, i.e., below the FMR frequency. Numerical simulations of the Landau-Lifshitz-Gilbert-Slonczewski equation showed, also, the possibility of current-induced excitations of more complicated strongly-nonlinear spin wave modes in a nano-contact geometry [94].

The spin wave modes considered above are the magnetic excitations on the background of a spatially-uniform equilibrium magnetization distribution (saturated magnetic state). Spin-polarized current can, also, excite spin wave modes in spatially-nonuniform (unsaturated) magnetic states. A good example of such an excitation is the current-induced excitation of a gyrotropic mode in a magnetic nano-pillar in a vortex ground state [50]–[52].

An interesting and unusual example of a magnetic excitation that is not only excited by spin-polarized current, but is, actually, *created* by this current is presented in [54], where the excitation of a "current-induced-vortex" in a magnetic nano-contacts was observed experimentally and described theoretically. In the experiment [54] the Oersted magnetic field of the bias current creates a vortex-like structure in a magnetically unsaturated "free" layer. The spin-transfer torque, exerted by the same current, excites a gyrotropic-like motion of this vortex. Thus, in this case bias current plays two roles simultaneously: it changes the ground state of the magnetic system (by the Oersted field) and destabilizes it (by the spin-transfer-torque effect). Probably, a similar type of a current-created and current-excited magnetic excitation was observed in the earlier work [53].

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